

2 Graphs

The interest in the study of graph theory has increased due to its applicability in so many fields like artificial intelligence, electrical engineering, transportation system, scheduling problems, economics, chemistry and operations research.

2.1 DEFINITIONS AND BASIC CONCEPTS

Definition 2.1. A graph $G=(V, E)$ is a mathematical structure consisting of two finite sets V and E . The elements of V are called **vertices (or nodes)** and the elements of E are called **edges**. Each edge is associated with a set consisting of **either one or two vertices** called its **endpoints**.

The correspondence from edges to endpoints is called **edge-endpoint function**. This function is generally denoted by γ . Due to this function, some authors denote graph by $G=(V, E, \gamma)$.

Definition 2.2. A graph consisting of one vertex and no edges is called a **trivial graph**.

Definition 2.3. A graph whose vertex and edge sets are empty is called a **null graph**.

Definition 2.4. An edge with just one endpoint is called a **loop** or a **self-loop**.

Thus, a loop is an edge that joins a single endpoint to itself.

Definition 2.5. An edge that is not a self-loop is called a **proper edge**.

Definition 2.6. If two or more edges of a graph G have the same vertices, then these edges are said to be **parallel** or **multi-edges**.

Definition 2.7. Two vertices that are connected by an edge are called **adjacent**.

Definition 2.8. An endpoint of a loop is said to be **adjacent to itself**.

Definition 2.9. An edge is said to be **incident** on each of its endpoints.

Definition 2.10. Two edges incident on the same endpoint are called **adjacent edges**.

Definition 2.11. The number of edges in a graph G which are incident on a vertex is called the **degree of that vertex**.

Definition 2.12. A vertex of degree zero is called an **isolated vertex**.

Thus, a vertex on which no edges are incident is called **isolated**.

Definition 2.13. A graph without multiple edges (**parallel edges**) and loops is called **simple graph**.

Notations: In pictorial representations of a graph, the vertices will be denoted by dots and edges by line segments.

EXAMPLE 2.1

(a) Let

$$V = \{1, 2, 3, 4\} \quad \text{and} \quad E = \{e_1, e_2, e_3, e_4, e_5\}.$$

Let γ be defined by

$$\begin{aligned} \gamma(e_1) &= \gamma(e_5) = \{1, 2\}, & \gamma(e_2) &= \{4, 3\}, \\ \gamma(e_3) &= \{1, 3\}, & \gamma(e_4) &= \{2, 4\}. \end{aligned}$$

2.2 ■ Engineering Mathematics-II

We note that both edges e_1 and e_5 have same end-points $\{1, 2\}$. The endpoints of e_2 are $\{4, 3\}$, the endpoints of e_3 are $\{1, 3\}$ and endpoints of e_4 are $\{2, 4\}$. Thus the graph is as shown in the Figure 2.1.

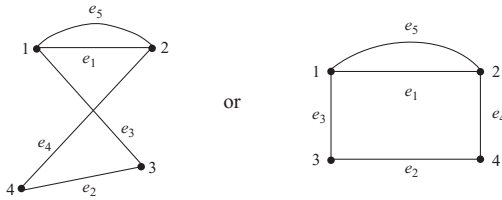


Figure 2.1

- (b) For the graph pictured in the Figure 2.2,
- Write down the degree of the vertices A, B, D .
 - Which of the edges are parallel?
 - Which vertex is isolated?
 - Point out whether it is a simple graph or a multi-graph.
 - Point out one pair of adjacent vertices and one pair of adjacent edges.

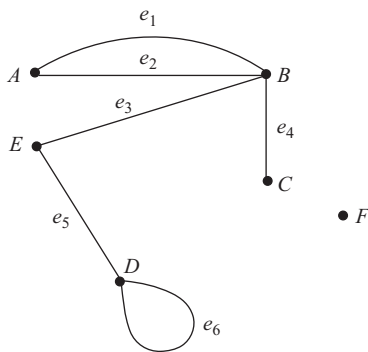


Figure 2.2

In this graph,

- (i) We notice that

Degree of the vertex $A=2$

Degree of the vertex $B=4$

Degree of the vertex $D=3$
(because loop e_6 has degree 2).

The vertex D has degree 3 (because e_6 is a loop and so has degree 2)

- The edges e_1 and e_2 are parallel
- The vertex F is isolated because no edge is incident on F
- It is a multi-graph because it has multi-edges and a loop.
- A and B are adjacent vertices because they are connected by the edge e_2 or e_1 whereas A and E are not adjacent.

The edges e_2 and e_3 are adjacent edges because they are incident on the same vertex B .

- (c) Consider the graph with the vertices A, B, C, D and E pictured in Figure 2.3.

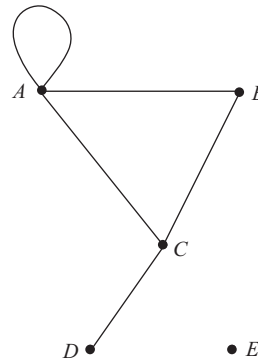


Figure 2.3

In this graph, we note that

Number of edges = 5

Degree of vertex $A=4$

Degree of vertex $B=2$

Degree of vertex $C=3$

Degree of vertex $D=1$

Degree of vertex $E=0$

Sum of the degree of vertices
 $= 4 + 2 + 3 + 1 + 0 = 10$.

Thus, we observe that

$$\sum_{i=1}^5 \deg(v_i) = 2e,$$

where $\deg(v_i)$ denotes the degree of vertex v_i and e denotes the number of edges.

Euler's Theorem 2.1. (The First Theorem of Graph Theory)

The sum of the degrees of the vertices of a graph G is equal to twice the number of edges in G .

Thus, total degree of a graph is even.

Proof: Each edge in a graph contributes a count of 1 to the degree of two vertices (endpoints of the edge), that is, each edge contributes 2 to the degree sum. Therefore, the sum of degrees of the vertices is equal to twice the number of edges.

Corollary 2.1. There can be only an even number of vertices of odd degree in a given graph G .

Proof: We know, by the fundamental theorem, that

$$\sum_{i=1}^n \deg(v_i) = 2 \times \text{number of edges}.$$

Thus the right-hand side is an even number. Hence to make the left-hand side an even number there can be only even number of vertices of odd degree.

Remarks 2.1.

- (i) A vertex of degree d is also called a d -**valent vertex**.
- (ii) The degree (or valence) of a vertex v in a graph G is the number of proper edges incident on v plus twice the number of self-loops.

Theorem 2.2. A nontrivial simple graph G must have at least one pair of vertices whose degrees are equal.

Proof: Let the graph G has n vertices. Then there appear to be n possible degree values, namely $0, 1, \dots, n-1$. But there cannot be both a vertex of degree 0 and a vertex of degree $n-1$ because if there is a vertex of degree 0 then each of the remaining $n-1$ vertices is adjacent to at most $n-2$ other vertices. Hence the n vertices of G can realize at most $n-1$ possible values for their degrees. Hence, the pigeonhole principle implies that at least two of the vertices have equal degree.

EXAMPLE 2.2

Is there a graph with eight vertices of degree 2, 2, 3, 6, 5, 7, 8, 4?

Solution. The answer is No. In such a graph, there will be three vertices of odd degree which is impossible.

We can also argue as follows: Total degree of the graph (if possible) is equal to $2+2+3+6+5+7+8+4=37$, which is odd. This contradicts the first theorem of graph theory, according to which, total degree of a graph is always even.

2.2 SPECIAL GRAPHS

Definition 2.14. A graph G is said to **simple** if it has no parallel edges or loops. In a simple graph, an edge with endpoints v and w is denoted by $\{v, w\}$.

Definition 2.15. For each integer $n \geq 1$, let D_n denote the graph with n **vertices** and **no edges**. Then D_n is called the **discrete graph on n vertices**.

For example, we have



Definition 2.16. Let $n \geq 1$ be an integer. Then a simple graph with n vertices in which there is an edge between each pair of distinct vertices is called the **complete graph on n vertices**. It is denoted by K_n .

For example, the complete graphs K_2 , K_3 and K_4 are shown in Figure 2.4.

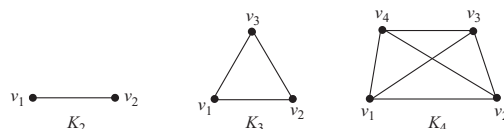


Figure 2.4

Definition 2.17. If each vertex of a graph G has the same degree as every other vertex, then G is called a **regular graph**.

A k -**regular graph** is a regular graph whose common degree is k .

For example, consider K_3 . The degree of each vertex in K_3 is 2. Hence K_3 is regular. Similarly, K_4 is regular. Also the graph shown in Figure 2.5 is regular because degree of each vertex here is 2.

But this graph is not complete because v_2 and v_4 have not been connected through an edge. Similarly, v_1 and v_3 are not connected by any edge.

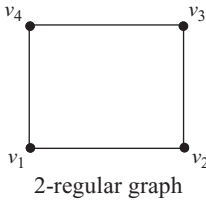


Figure 2.5

Thus, a complete graph is always regular but a regular graph need not be complete.

EXAMPLE 2.3

The oxygen molecule O_2 , made up of two oxygen atoms linked by a double bond can be represented by the regular graph shown in Figure 2.6.

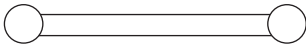


Figure 2.6

Definition 2.18. Let $n \geq 1$ be an integer. Then a graph L_n with n vertices $\{v_1, v_2, \dots, v_n\}$ and with edges $\{v_i, v_{i+1}\}$ for $1 \leq i < n$ is called a **linear graph on n vertices**.

For example, the linear graphs L_2 and L_4 are shown in Figure 2.7.

It is also called **path graph** denoted by P_n .

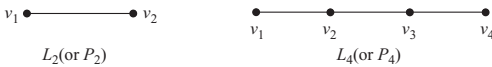


Figure 2.7

Definition 2.19. A **bipartite graph** G is a graph whose vertex set V can be partitioned into two subsets U and W , such that each edge of G has one endpoint in U and one endpoint in W .

The pair (U, W) is called a **vertex bipartition of G** and U and W are called the bipartition subsets. Obviously, a bipartite graph cannot have any self-loop.

EXAMPLE 2.4

If vertices in U are solid vertices and vertices in W are hollow vertices, then the graphs shown in Figure 2.8 are bipartite graphs.

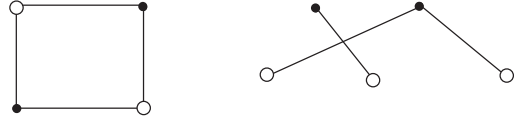


Figure 2.8

EXAMPLE 2.5

The smallest possible simple graph that is not bipartite is the complete graph K_3 shown in Figure 2.9.

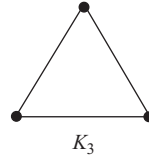


Figure 2.9

Definition 2.20. A **complete bipartite graph** G is a simple graph whose vertex set V can be partitioned into two subsets $U = \{v_1, v_2, \dots, v_m\}$ and $W = \{w_1, w_2, \dots, w_n\}$ such that for all i, k in $\{1, 2, \dots, m\}$ and j, l in $\{1, 2, \dots, n\}$

- (i) There is an edge from each vertex v_i to each vertex w_j .
- (ii) There is not an edge from any vertex v_i to any other vertex v_k .
- (iii) There is not an edge from any vertex w_j to any other vertex w_l .

A complete bipartite graph on (m, n) vertices is denoted by $K_{m,n}$.

EXAMPLE 2.6

The complete bipartite graphs $K_{3,2}$ and $K_{3,4}$ are shown in Figure 2.10.

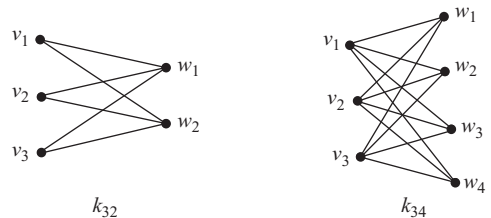


Figure 2.10

Definition 2.21. A **cycle graph** is a single vertex with a self-loop or a simple connected graph C with $|V_c| = |E_c|$, that can be drawn so that all of its vertices and edges lie on a circle.

An n -vertex cycle graph is denoted by C_n . The cycle graphs C_1 , C_2 , and C_4 are shown in Figure 2.11.

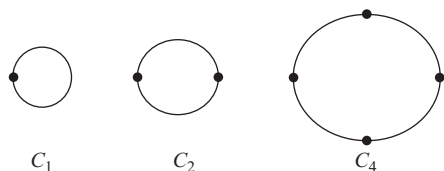


Figure 2.11

Definition 2.22. The **Petersen graph** is the 3-regular graph shown in Figure 2.12.

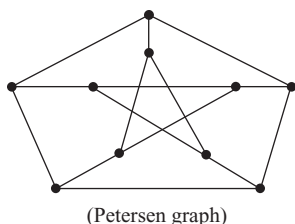


Figure 2.12

This graph is frequently used to establish theorems and to test conjectures.

Definition 2.23. The **hypercube graph** Q_n is the n -regular graph whose vertex set is the set of bit string of length n and such that there is an edge between two vertices if and only if they differ in exactly one bit.

For example, 8-vertex cube graph is a hypercube graph Q_3 shown in Figure 2.13.

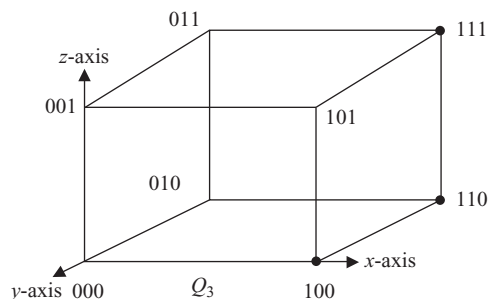


Figure 2.13

Definition 2.24. A graph consisting of two concentric n -cycles in which each of the n pairs of the corresponding vertices is joined by an edge is called the **circular ladder graph** CL_n .

For example, the graphs CL_2 and CL_4 are shown in the Figure 2.14.

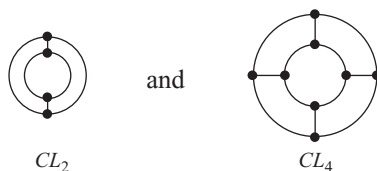


Figure 2.14

Definition 2.25. A graph consisting of a single vertex having n loops is called a **bouquet** and is denoted by B_n .

For example, the graphs shown in Figure 2.15 represent bouquet B_2 and bouquet B_4 .

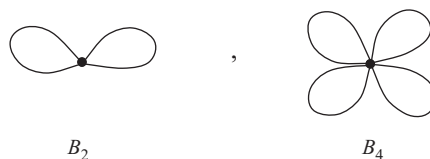


Figure 2.15

Definition 2.26. A graph consisting of two vertices and n edges joining them is called a **dipole**. It is denoted by D_n .

For example, Figure 2.16 represents D_2 , D_3 and D_4 .

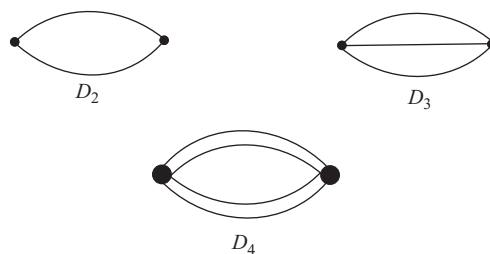


Figure 2.16

EXAMPLE 2.7

Draw all simple graphs with vertices $\{u, v, x, y\}$ and two edges, one of which is $\{u, v\}$.

Solution. In a simple graph, an edge corresponds to a subset of two vertices. Since there are four vertices,

we can have $4c_2=6$ such subsets in all and they are $\{u, v\}$, $\{u, x\}$, $\{u, y\}$, $\{v, x\}$, $\{v, y\}$ and $\{x, y\}$. But one edge of the graph is specified to be $\{u, v\}$. Hence the possible simple graphs are as shown in Figure 2.17.

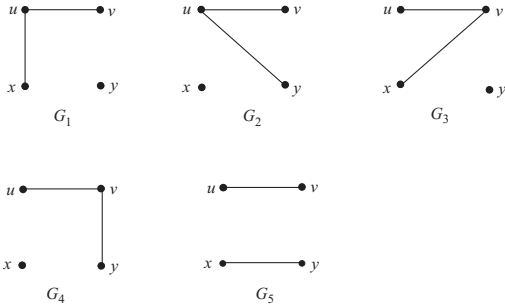


Figure 2.17

2.3 SUBGRAPHS

Definition 2.27. A graph H is said to be a subgraph of a graph G if and only if every vertex in H is also a vertex in G , every edge in H is also an edge in G and every edge in H has the same endpoints as in G .

We may also say that G is a supergraph of H . For example, the graphs (Figure 2.18).

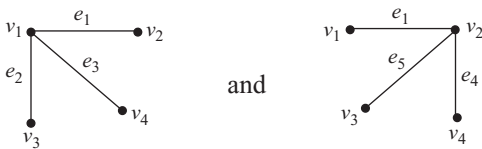


Figure 2.18

are subgraphs of the graph given in Figure 2.19.

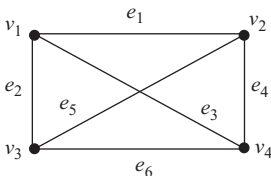


Figure 2.19

Similarly, the graph (Figure 2.20)

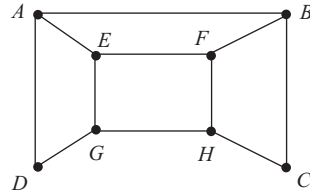


Figure 2.20

is a subgraph of the graph given in the Figure 2.21.

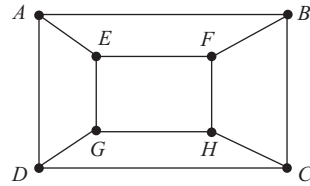


Figure 2.21

Definition 2.28. A subgraph H is said to be a **proper subgraph** of a graph G if vertex set V_H of H is a proper subset of the vertex set V_G of G or edge set E_H is a proper subset of the edge set E_G .

For example, the subgraphs in the above examples are proper subgraphs of the given graphs.

Definition 2.29. A subgraph H is said to **span** a graph G if $V_H = V_G$.

Thus H is a spanning subgraph of graph G if it contains all the vertices of G .

For example, the subgraph (Figure 2.22)

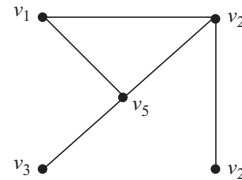


Figure 2.22

spans the graph give in Figure 2.23.

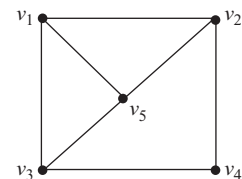


Figure 2.23

Definition 2.30. Let $G=(V, E)$ be a graph. Then the **complement of a subgraph** $G'=(V', E')$ with respect to the graph G is another subgraph $G''=(V'', E'')$ such that $E''=E-E'$ and V'' contains only the vertices with which the edges in E'' are incident.

For example, the subgraph (Figure 2.24)



Figure 2.24

is the complement of the subgraph (Figure 2.25).

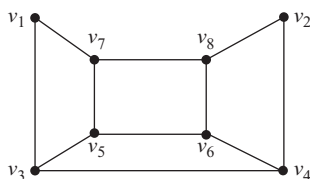


Figure 2.25

with respect to the graph G shown in Figure 2.26.

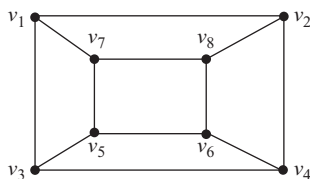


Figure 2.26

Definition 2.31. If G is a simple graph, the **complement of G (edge complement)**, denoted by G' or G^c is a graph such that

- (i) The vertex set of G' is identical to the vertex set of G , that is, $V_{G'}=V_G$.
- (ii) Two distinct vertices v and w of G' are **connected** by an edge if and only if v and w are **not connected** by an edge in G .

For example, consider the graph G shown in Figure 2.27.

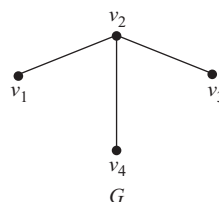


Figure 2.27

Then complement G' of G is the graph shown in Figure 2.28.

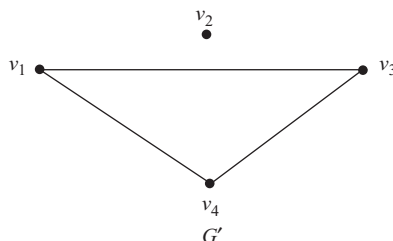


Figure 2.28

EXAMPLE 2.8

Find the complement of the graphs:

(a)

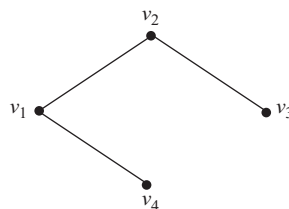
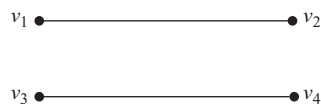


Figure 2.29

(b)



(c) Complete graph K_4 :

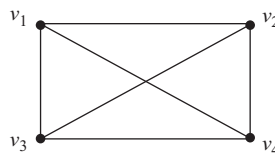


Figure 2.31

Solution.

(a)

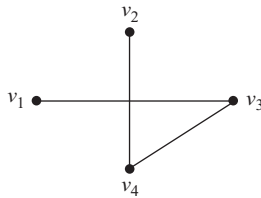


Figure 2.32

(b)

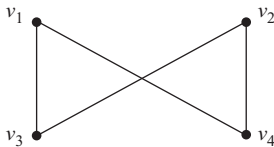


Figure 2.33

(c) Null graph.

EXAMPLE 2.9

Find the edge complement of the graph G shown in Figure 2.34.

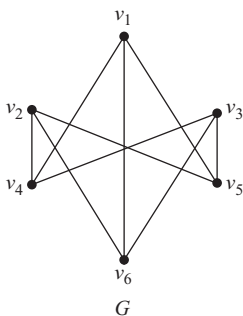


Figure 2.34

Solution. The edge complement of G is the following graph G^c (Figure 2.35).

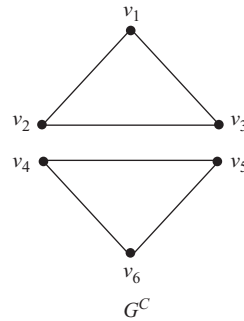


Figure 2.35

Definition 2.32. If a new vertex v is joined to each of the pre-existing vertices of a graph G , then the resulting graph is called the **join of G and v** or the **suspension of G from v** . It is denoted by $G + v$.

Thus, a graph obtained by joining a new vertex v to each of the vertices of a given graph G is called the **join of G and v** or the **suspension of G from v** . For example, consider the graph given below (Figure 2.36).

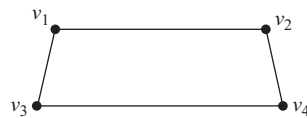


Figure 2.36

Let v be a vertex. Then the graph shown in the Figure 2.37 is the join of G to v .

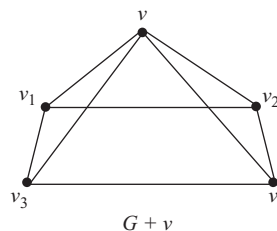


Figure 2.37

2.4 ISOMORPHISMS OF GRAPHS

We know that shape or length of an edge and its position in space are not part of specification of a

graph. For example, the Figures 2.38 represent the same graph.

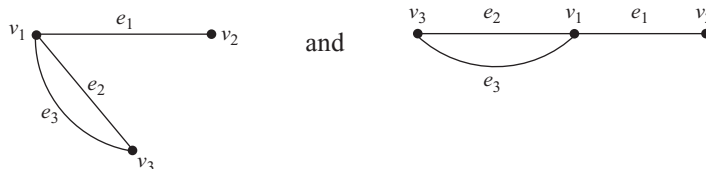


Figure 2.38

Definition 2.33. Let G and H be graphs with vertex sets $V(G)$ and $V(H)$ and edge sets $E(G)$ and $E(H)$, respectively. Then G is said to be **isomorphic to H** if there exist one-to-one correspondences $g: V(G) \rightarrow V(H)$ and $h: E(G) \rightarrow E(H)$ such that for all $v \in V(G)$ and $e \in E(G)$,

v is an endpoint of $e \Leftrightarrow g(v)$ is an endpoint of $h(e)$.

Definition 2.34. The property of mapping endpoints to endpoints is called **preserving incidence** or the **continuity rule** for graph mappings.

As a consequence of this property, a self-loop must map to a self-loop.

Thus, two isomorphic graphs are same except for the labelling of their vertices and edges. For example, the graphs shown in the Figure 2.39 are isomorphic.

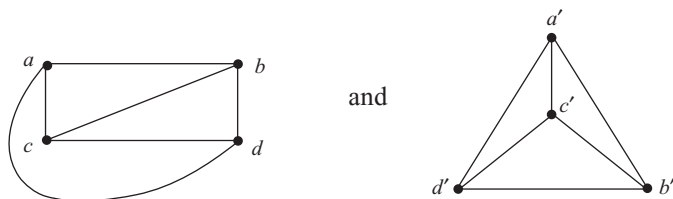


Figure 2.39

Similarly, the hypercube graph Q_3 and circular ladder CL_4 shown in Figure 2.40 are isomorphic.

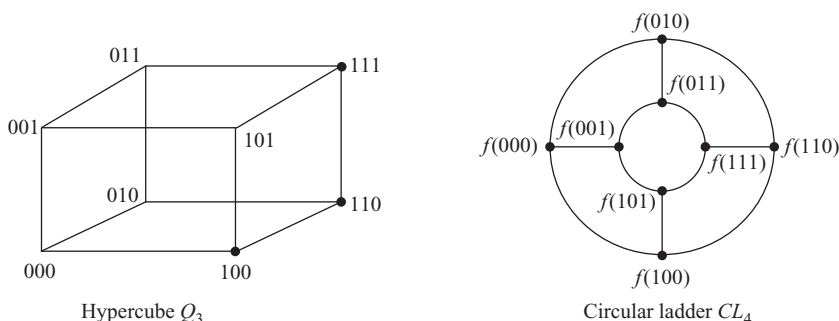


Figure 2.40

EXAMPLE 2.10

Show that the graphs shown in Figure 2.41 are isomorphic.

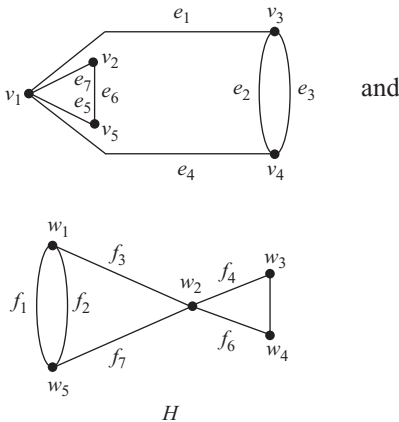


Figure 2.41

Solution. To solve this problem, we have to find $g: V(G) \rightarrow V(H)$ and $h: E(G) \rightarrow E(H)$ such that for all $v \in V(G)$ and $e \in E(G)$,

v is an endpoint of $e \Leftrightarrow g(v)$ is an endpoint of $h(e)$.

Since e_2 and e_3 are parallel (have the same endpoints), $h(e_2)$ and $h(e_3)$ must also be parallel. Thus we have

$$h(e_2)=f_1 \text{ and } h(e_3)=f_2 \quad \text{or} \quad h(e_2)=f_2 \text{ and } h(e_3)=f_1.$$

Also the endpoints of e_2 and e_3 must correspond to the endpoints of f_1 and f_2 and so

$$g(v_3)=w_1 \text{ and } g(v_4)=w_5 \quad \text{or} \quad g(v_3)=w_5 \text{ and } g(v_4)=w_1.$$

Further, we note that v_1 is the endpoint of four distinct edges e_1, e_7, e_5 and e_4 and so $g(v_1)$ should be the endpoint of four distinct edges. We observe that w_2 is the vertex having four edges and so $g(v_1)=w_2$. If $g(v_3)=w_1$, then since v_1 and v_3 are endpoints of e_1 in G , $g(v_1)=w_2$ and $g(v_3)=w_1$ must be endpoints of $h(e_1)$ in H . This implies that $h(e_1)=f_3$.

Continuing this way, we can find g and h to define the isomorphism between G and H .

One such pair of functions (of course there exist several) is shown in Figure 2.42.

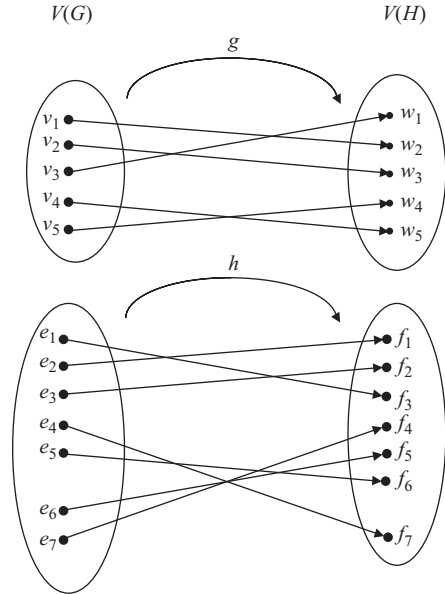


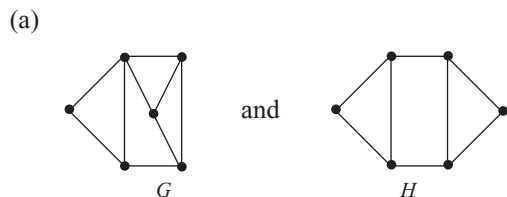
Figure 2.42

Remark 2.2. Each of the following properties of a graph is invariant under graph isomorphism, where n, m and k are all non-negative integers:

1. Has n vertices
 2. Has m edges
 3. Has a vertex of degree k
 4. Has m vertices of degree k
 5. Has a circuit of length k
 6. Has a simple circuit of length k
 7. Has m simple circuits of length k ,
 8. Is connected
 9. Has an Euler circuit
 10. Has a Hamiltonian circuit
- } to be studied later on in this chapter

EXAMPLE 2.11

Examine for isomorphism



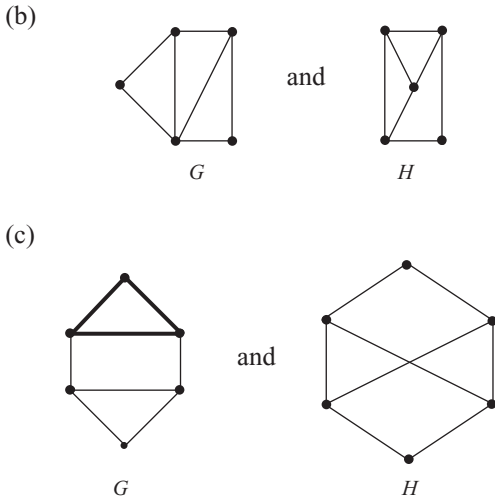


Figure 2.43

Solution. We note in Figure 2.43 that

- (a) G has nine edges whereas H has only eight. Hence, G is not isomorphic to H .
- (b) G has a vertex v of degree 4, whereas H has no vertex of degree 4. Hence, G is not isomorphic to H .
- (c) G has a simple circuit of length 3 (**marked dark**) whereas H does not have. Hence, G is not isomorphic to H .

2.5 WALKS, PATHS AND CIRCUITS

Definition 2.35. In a graph G , a **walk** from vertex v_0 to vertex v_n is a finite alternating sequence $\{v_0, e_1, v_1, e_2, \dots, v_{n-1}, e_n, v_n\}$ of vertices and edges such that v_{i-1} and v_i are the endpoints of e_i .

The **trivial walk** from a vertex v to v consists of the single vertex v .

Definition 2.36. In a graph G , a **path** from the vertex v_0 to the vertex v_n is a walk from v_0 to v_n that does not contain a repeated edge.

Thus a **path** from v_0 to v_n is a walk of the form

$$\{v_0, e_1, v_1, e_2, v_2, \dots, v_{n-1}, e_n, v_n\},$$

where all the edges e_i are distinct.

Definition 2.37. In a graph, a **simple path** from v_0 to v_n is a path that does not contain a repeated vertex.

Thus a simple path is a walk of the form

$$\{v_0, e_1, v_1, e_2, v_2, \dots, v_{i-1}, e_n, v_n\},$$

where all the e_i and all the v_i are distinct.

Definition 2.38. A walk in a graph G that starts and ends at the same vertex is called a **closed walk**.

Definition 2.39. A closed walk that does not contain a repeated edge is called a **circuit**.

Thus, a closed path is called a circuit (or a **cycle**) if it is a walk of the form

$$\{v_0, e_1, v_1, e_2, v_2, \dots, v_{n-1}, e_n, v_n\},$$

where $v_0 = v_n$ and all the e_i are distinct.

Definition 2.40. A **simple circuit** is a circuit that does not have any other repeated vertex except the first and the last.

Thus, a simple circuit is a walk of the form

$$\{v_0, e_1, v_1, e_2, \dots, v_{n-1}, e_n, v_n\},$$

where all the e_i are distinct and all the v_j are distinct except that $v_0 = v_n$.

EXAMPLE 2.12

In the graph given in the Figure 2.44, determine whether the following walks are paths, simple paths, closed walks, circuits, simple circuits or are just walks.

1. $v_1 e_2 v_2 e_3 v_3 e_4 v_4 e_5 v_2 e_2 v_1 e_1 v_0$
2. $v_5 v_4 v_2 v_1$
3. $v_2 v_3 v_4 v_5 v_2$
4. $v_4 v_2 v_3 v_4 v_5 v_2 v_4$
5. $v_2 v_1 v_5 v_2 v_3 v_4 v_2$
6. $v_0 v_5 v_2 v_3 v_4 v_2 v_1$

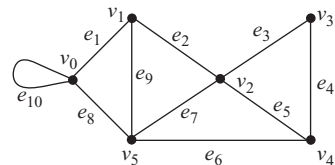


Figure 2.44

Solution

1. It is **just a walk** because it has repeated edges and repeated vertices.

2. It is a **simple path**, because here no vertex is repeated and no edge is repeated.
3. It is **closed walk** in which no edge is repeated and hence it is a **circuit**. It starts and ends at the same vertex, and therefore is a **simple circuit**.
4. It is a closed walk in which no edge is repeated but vertices v_2 and v_4 are repeated. Hence it is a **circuit**.
5. It is a closed walk in which no edge is repeated. Hence it is a circuit. Only one vertex is repeated twice, hence it is **not a simple circuit**.
6. It is a walk in which no edge is repeated but the vertex v_2 is repeated. Hence it is a **path**.

EXAMPLE 2.13

Consider the graph shown in Figure 2.45.

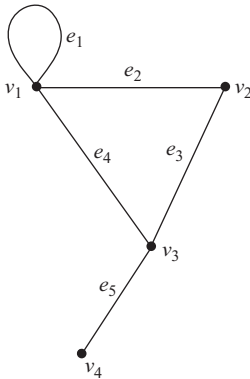


Figure 2.45

We note that e_3, e_5 is a path. The walk e_1, e_2, e_3, e_5 is a path but it is not a simple path because the vertex v_1 is repeated (e_1 being a self-loop). The walk e_2, e_3, e_4 is a circuit. The walk e_2, e_3, e_4, e_1 is a circuit but it is not simple circuit because vertex v_1 repeats twice (or we may write that v_1 is met twice).

Definition 2.41. In a graph the number of edges in the path $\{v_0, e_1, v_1, e_2, \dots, e_n, v_n\}$ from v_0 to v_n is called the **length of the path**.

Definition 2.42. A cycle with k -edges is called a **k -cycle** or **cycle of length k** .

For example, loop is a cycle of length 1. On the other hand, a pair of parallel edges e_1 and e_2 , shown in the Figure 2.46, is a cycle of length 2.

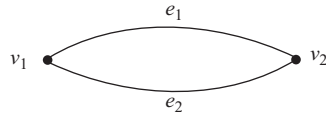


Figure 2.46

EXAMPLE 2.14

Consider the graph shown in Figure 2.47.

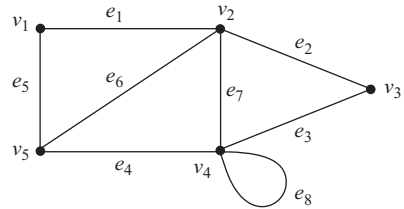


Figure 2.47

In this graph, we observe that

- (i) The path $v_4 e_8 v_4$ is a cycle of length 1.
- (ii) The path $v_5 e_4 v_4 e_8 v_4 e_3 v_3$ from v_5 to v_3 (Figure 2.48) is a path of length 3.

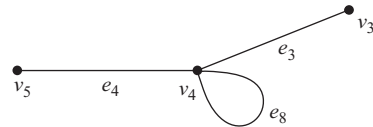


Figure 2.48

- (iii) The path $v_1 e_1 v_2 e_6 v_5 e_5 v_1$ (Figure 2.49) is a

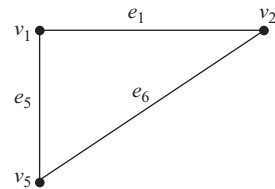


Figure 2.49

cycle of length 3, since it consists of three edges.

Definition 2.43. A graph is said to be **acyclic** if it contains no cycle.

For example, the graphs shown in Figure 2.50 are acyclic.

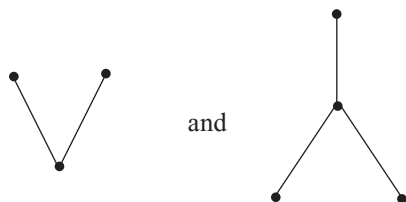


Figure 2.50

Theorem 2.3. If there is a path from vertex v_1 to v_2 in a graph with n vertices, then there does not exist a simple path of more than $n - 1$ edges from vertex v_1 to v_2 .

Proof: Suppose there is a path from v_1 to v_2 . Let $v_1, \dots, v_i, \dots, v_2$ be the sequence of vertices which the path meets between the vertices v_1 and v_2 . Let there be m edges in the simple path. Then there will be $m + 1$ vertices in the sequence. Therefore if $m > n - 1$, then there will be more than n vertices in the sequence. But the graph is with n vertices. Therefore some vertex, say v_k , appears more than once in the sequence. So the sequence of vertices shall be $v_1, \dots, v_i, \dots, v_k, \dots, v_k, \dots, v_2$. Deleting the edges in the path that lead v_k back to v_k we have a path from v_1 to v_2 that has fewer edges than the original one. This argument is repeated until we get a path that has $n - 1$ or fewer edges.

Definition 2.44. Two vertices v_1 and v_2 of a graph G are said to be **connected** if and only if there is a walk from v_1 to v_2 .

Definition 2.45. A graph G is said to be **connected** if and only if given any two vertices v_1 and v_2 in G , there is a walk from v_1 to v_2 .

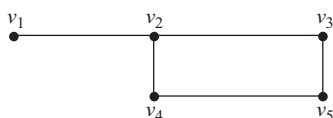
Thus, a graph G is connected if there exists a walk between every two vertices in the graph.

Definition 2.46. A graph which is not connected is called **disconnected graph**.

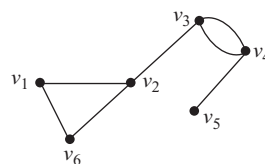
EXAMPLE 2.15

Which of the graphs shown below are connected?

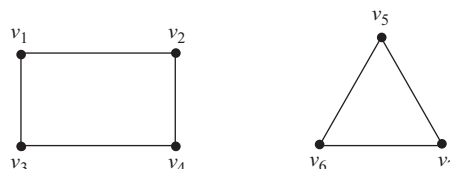
(a)



(b)



(c)



(d)

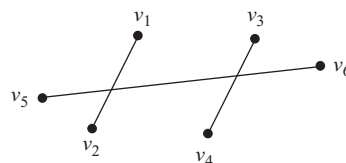


Figure 2.51

Solution. In Figure 2.51, we observe that

- Since every two vertices in the given graph are connected by a path, therefore the given graph is connected.
- This graph is also connected.
- The graph in (c) is disconnected. It has two connected components.
- In this graph, the edge (v_5, v_6) cross the edges (v_1, v_2) and (v_3, v_4) at points which are not vertices. Therefore the graph can be redrawn as shown in Figure 2.52.

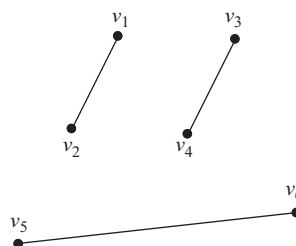


Figure 2.52

There is no path from v_2 to v_4 , etc. Hence the given graph is disconnected and has three connected components.

EXAMPLE 2.16

Which of the graphs shown in Figure 2.53 are connected?

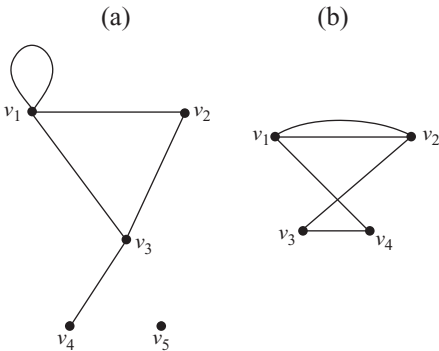


Figure 2.53

Solution. Graph (a) is not connected as there is no walk from any of v_1, v_2, v_3, v_4 to the vertex v_5 . The graph (b) is clearly connected.

Definition 2.47. If a graph G is disconnected, then the various connected pieces of G are called the **connected components of the graph**.

EXAMPLE 2.17

Find the connected components of the graphs given below (Figure 2.54).

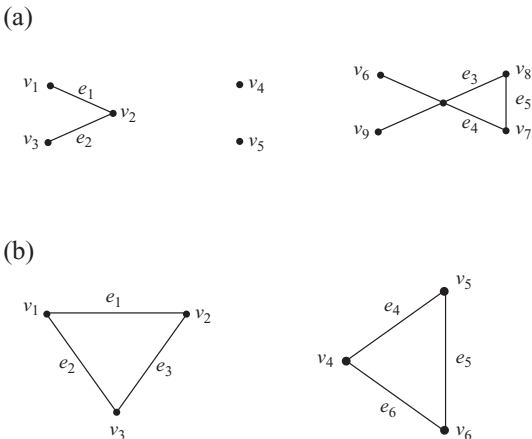


Figure 2.54

Solution

(a) The graph in (a) has four connected components (Figures 2.55 and 2.56)

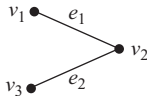


Figure 2.55

H_1 : with vertex set $\{v_1, v_2, v_3\}$ and edge set $\{e_1, e_2\}$
 H_2 : with vertex set $\{v_4\}$ and edge set ϕ
 H_3 : with vertex set $\{v_5\}$ and edge set ϕ
 H_4 : with vertex set $\{v_6, v_7, v_8, v_9\}$ and edge set $\{e_3, e_4, e_5\}$

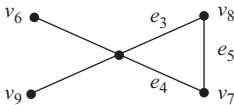


Figure 2.56

(b) The graph in (b) is disconnected and have two connected components (Figures 2.57(a), 2.57(b))

H_1 :

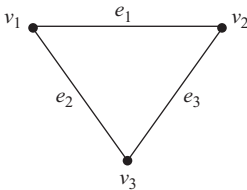


Figure 2.57(a)

with vertex set $\{v_1, v_2, v_3\}$ and edge set $\{e_1, e_2, e_3\}$

H_2 :

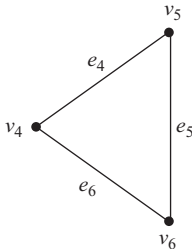


Figure 2.57(b)

with vertex set $\{v_4, v_5, v_6\}$ and edge set $\{e_4, e_5, e_6\}$.

EXAMPLE 2.18

Find the number of connected components in the graph shown below (Figure 2.58).

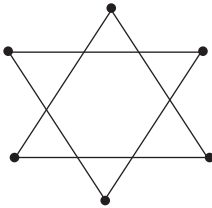


Figure 2.58

Solution. The connected components are shown in Figure 2.59.

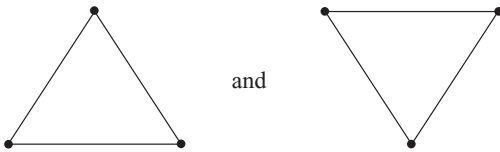


Figure 2.59

Remark 2.3. If a connected component has n vertices, then degree of any vertex cannot exceed $n - 1$.

2.6 EULERIAN PATHS AND CIRCUITS

Definition 2.48. A path in a graph G is called an **Euler path** if it includes **every edge exactly once**.

Definition 2.49. A circuit in a graph G is called an **Euler circuit** if it includes every edge exactly once. Thus, an Euler circuit (Eulerian trail) for a graph G is a sequence of adjacent vertices and edges in G that starts and ends at the same vertex, uses **every vertex of G at least once**, and uses **every edge of G exactly once**.

Definition 2.50. A graph is called **Eulerian graph** if there exists an Euler circuit for that graph.

EXAMPLE 2.19

Find which of the following are Euler paths and Euler circuits in the graph given below (Figure 2.60)

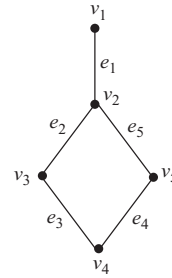


Figure 2.60

- (a) $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_2$
- (b) $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_2 e_1 v_1$
- (c) $v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_2$

Solution

- (a) The walk in (a) is an Euler path but it is not Euler circuit because it is not closed.
- (b) It is not an Euler circuit because the edge e_1 is covered twice.
- (c) It is neither an Euler path nor an Euler circuit because the vertex v_1 of G has not been used.

Theorem 2.4. If a graph has an Euler circuit, then every vertex of the graph has even degree.

Proof: Let G be a graph which has an Euler circuit. Let v be a vertex of G . We shall show that degree of v is even. By definition, Euler circuit contains every edge of graph G . Therefore the Euler circuit contains all edges incident on v . We start a journey beginning in the middle of one of the edges adjacent to the start of Euler circuit and continue around the Euler circuit to end in the middle of the starting edge. Since Euler circuit uses every edge exactly once, the edges incident on v occur in entry/exist pair and hence the degree of v is a multiple of 2. Therefore, the degree of v is even. This completes the proof of the theorem.

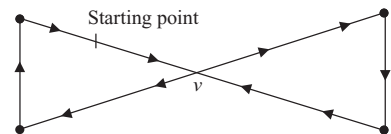


Figure 2.61

We know that **contrapositive of a conditional statement is logically equivalent to the statement**. Thus, Theorem 2.4 is equivalent to the following:

Theorem 2.5. If a vertex of a graph is not of even degree, then it does not have an Euler circuit.

Thus,

If some vertex of a graph has odd degree, then that graph does not have an Euler circuit.

EXAMPLE 2.20

Show that the graphs shown in Figure 2.62 do not have Euler circuits.

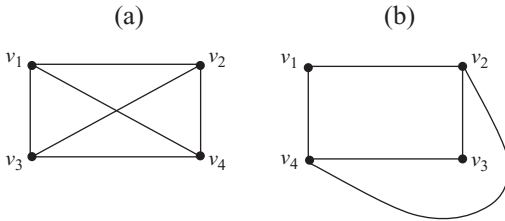


Figure 2.62

Solution. In graph (a), degree of each vertex is 3. Hence this **does not** have an Euler circuit.

In graph (b), we have

$$\deg(v_2)=2 \text{ and } \deg(v_3)=3.$$

Since there are vertices of odd degree in the given graph, therefore it **does not** have an Euler circuit.

Remark 2.4. The converse of Theorem 2.4 is not true. There exist graphs in which every vertex has even degree but the Euler circuits do not exist.

For example,

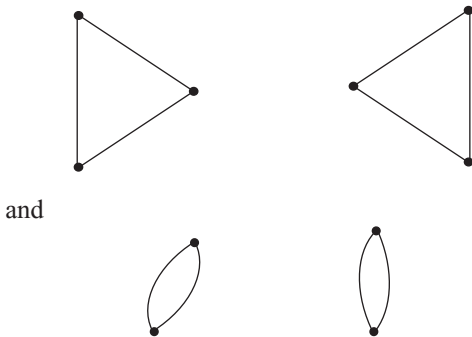


Figure 2.63

are graphs (Figure 2.63) in which each vertex has degree 2 but these graphs do not have Euler circuits since there is no path which uses each vertex at least once.

Theorem 2.6. If G is a connected graph and every vertex of G has even degree, then G has an Euler circuit.

Proof: Let every vertex of a connected graph G has even degree. If G consists of a single vertex v , the trivial walk from v to v is an Euler circuit. So suppose that G consists of more than one vertices. We start from any vertex v of G . Since the degree of each vertex of G is even, if we reach each vertex other than v by travelling on one edge, the same vertex can be reached by travelling on another previously unused edge. Thus a sequence of distinct adjacent edges can be produced indefinitely as long as v is not reached. Since the number of edges of the graph is finite (by definition of graph), the sequence of distinct edges will terminate. Thus the sequence must return to the starting vertex. We thus obtain a sequence of adjacent vertices and edges starting and ending at v without repeating any edge. Thus we get a circuit C .

If C contains every edge and vertex of G , then C is an Euler circuit.

If C does not contain every edge and vertex of G , remove all edges of C from G and also any vertices that become isolated when the edges of C are removed. Let the resulting subgraph be G' . We note that when we removed edges of C , an even number of edges from each vertex have been removed. Thus degree of each remaining vertex remains even.

Further, since G is connected, there must be at least one vertex common to both C and G' . Let it be w (in fact there are two such vertices). Pick any sequence of adjacent vertices and edges of G' starting and ending at w without repeating an edge. Let the resulting circuit be C' .

Join C and C' together to create a new circuit C'' . Now, we observe that if we start from v and follow C all the way to reach w and then follow C' all the way to reach back to w . Then continuing travelling along the untravelled edges of C , we reach v .

If C'' contains every edge and vertex of C , then C'' is an Euler circuit. If not, then we again repeat our process. Since the graph is finite, the process must terminate.

The process followed has been described in the graph G shown below (Figure 2.64).

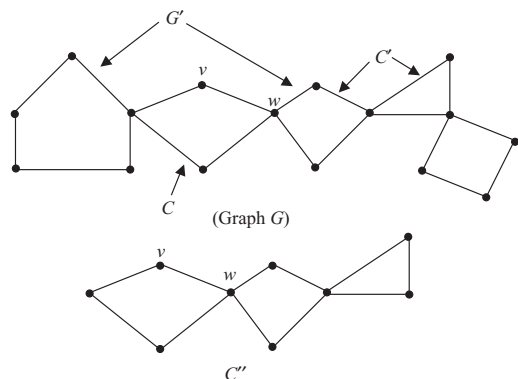


Figure 2.64

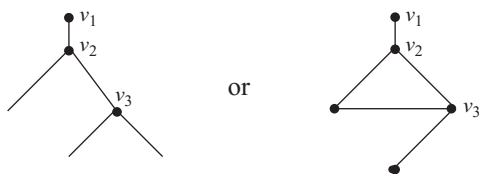
Theorems 2.4 and 2.6 taken together imply:

Theorem 2.7 (Euler's Theorem). A finite connected graph G has an Euler circuit if and only if every vertex of G has even degree.

Thus, finite connected graph is Eulerian if and only if each vertex has even degree.

Theorem 2.8. If a graph G has more than two vertices of odd degree, then there can be no Euler path in G .

Proof: Let v_1 , v_2 and v_3 be vertices of odd degree. Since each of these vertices had odd degree, any possible Euler path must leave (arrive at) each of v_1 , v_2 , v_3 with no way to return (or leave). One vertex of these three vertices may be the beginning of Euler path and another the end, but this leaves the third vertex at one end of an untravelled edge. Thus, there is no Euler path.



(Graphs having more than two vertices of odd degree).

Figure 2.65

Theorem 2.9. If G is a connected graph and has exactly two vertices of odd degree, then there is an Euler path in G . Further, any Euler path in G must begin at one vertex of odd degree and end at the other.

Proof: Let u and v be two vertices of odd degree in the given connected graph G (see Figure 2.66)

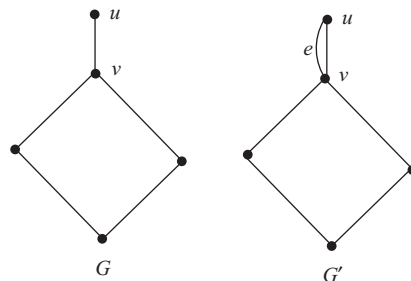


Figure 2.66

If we add the edge e to G , we get a connected graph G' all of whose vertices have even degree. Hence there will be an Euler circuit in G' . If we omit e from Euler circuit, we get an Euler path beginning at u (or v) and ending at v (or u).

EXAMPLE 2.21

Has the graph given in Figure 2.67 an Eulerian path?

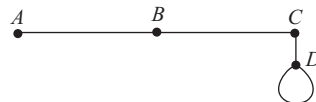


Figure 2.67

Solution. In the given graph,

$$\begin{aligned} \deg(A) &= 1, & \deg(B) &= 2, \\ \deg(C) &= 2, & \deg(D) &= 3. \end{aligned}$$

Thus the given connected graph has exactly two vertices of odd degree. Hence, it has an Eulerian path.

If it starts from A (vertex of odd degree), then it ends at D (vertex of odd degree). If it starts from D (vertex of odd degree), then it ends at A (vertex of odd degree).

But, on the other hand, let the graph be as shown below (Figure 2.68)

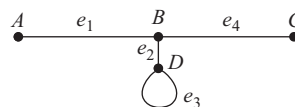


Figure 2.68

Then $\deg(A)=1$, $\deg(B)=3$, $\deg(C)=1$, degree of $D=3$ and so we have four vertices of odd degree. Hence it does not have an Euler path.

EXAMPLE 2.22

Does the graph given in Figure 2.69 possess an Euler circuit?

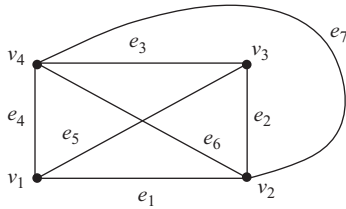


Figure 2.69

Solution. The given graph is connected. Further

$$\begin{aligned}\deg(v_1) &= 3, & \deg(v_2) &= 4, \\ \deg(v_3) &= 3, & \deg(v_4) &= 4.\end{aligned}$$

Since this connected graph has vertices with odd degree, it cannot have Euler circuit. But this graph has Euler path, since it has exactly two vertices of odd degree. For example, $v_3 e_2 v_2 e_7 v_4 e_6 v_2 e_1 v_1 e_4 v_4 e_3 v_3 e_5 v_1$ is an Euler path.

EXAMPLE 2.23

Consider the graph given below (Figure 2.70):



Figure 2.70

Here, $\deg(v_1)=4$, $\deg(v_2)=4$, $\deg(v_3)=2$, $\deg(v_4)=2$. Thus degree of each vertex is even. But the graph is not Eulerian since it is **not connected**.

EXAMPLE 2.24

Is it possible to trace the graph in Figure 2.71 without lifting the pencil?

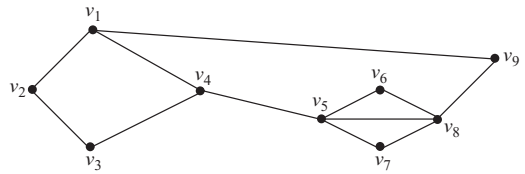


Figure 2.71

Solution. The problem is equivalent to say that “Is it possible for this graph to have an Eulerian circuit?” We observe that

$$\begin{aligned}\deg(v_1) &= 3, & \deg(v_2) &= 2, & \deg(v_3) &= 2, & \deg(v_4) &= 3, \\ & & & & & & \deg(v_5) &= 4, \\ \deg(v_6) &= 2, & \deg(v_7) &= 2, & \deg(v_8) &= 4, & \deg(v_9) &= 2.\end{aligned}$$

Since the graph contains vertex of odd degree, therefore it cannot have Euler’s circuit and therefore can not be traced without lifting the pencil.

EXAMPLE 2.25

Find the Euler’s circuit for the graph given below (Figure 2.72):

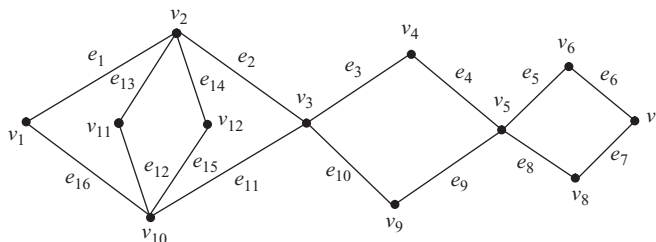


Figure 2.72

Solution. The given connected graph has 12 vertices and degree of each vertex is even. Hence, by Euler's theorem, this has an Euler's circuit. For example, we observe that $v_1 e_1 v_2 e_{13} v_{11} e_{12} v_{10} e_{15} v_{12} e_{14} v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_6 e_6 v_7 e_7 v_8 e_8 v_9 v_{10} v_3 e_{11} v_{10} e_{16} v_1$ is an Euler's circuit. In short we can represent it by $e_1, e_{13}, e_{12}, e_{15}, e_{14}, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10}, e_{11}, e_{16}$.

EXAMPLE 2.26 (The bridges of Königsberg)

The graph theory began in 1736 when Leonhard Euler solved the problem of seven bridges on Pregel River in the town of Königsberg in Prussia (now Kaliningrad in Russia). The two islands and seven bridges are shown in Figure 2.73.

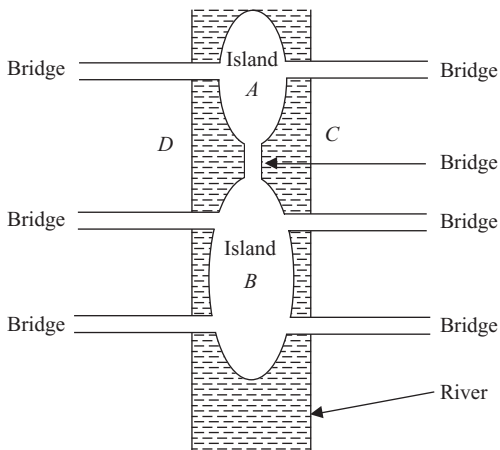


Figure 2.73

The people of Königsberg posed the following question to famous Swiss Mathematician Leonhard Euler:

“Beginning anywhere and ending any where, can a person walk through the town of Königsberg crossing all the seven bridges exactly once?”

Euler showed that such a walk is impossible. He replaced the islands A , B and the two sides (banks) C and D of the river by vertices and the bridges as edges of a graph. We note then that

$$\begin{aligned} \deg(A) &= 3, & \deg(B) &= 5, \\ \deg(C) &= 3, & \deg(D) &= 3. \end{aligned}$$

Thus the graph of the problem is as shown in Figure 2.74.

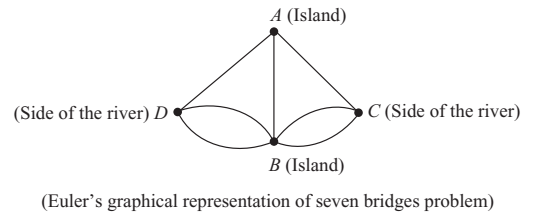


Figure 2.74

The problem then reduces to

“Is there any Euler's path in the above diagram?”

To find the answer, we note that there are more than two vertices having odd degree. Hence there exists no Euler path for this graph.

EXAMPLE 2.27

The floor plan shown in Figure 2.75 is for a house that is open for public viewing. Each room is connected to every room with which it has a common wall and to the outside along each wall. Is it possible to begin in a room or outside and take a walk which goes through each door exactly once?

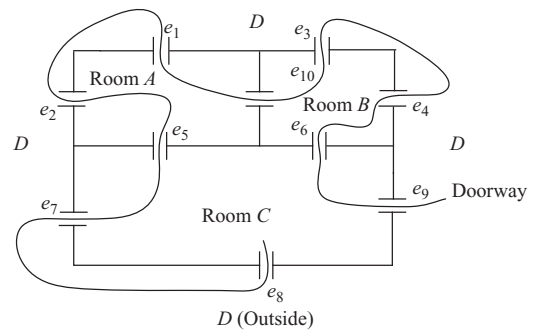


Figure 2.75

If each room and out side constitute a vertex and each door corresponds to an edge, then the floor plan converts into the graph. There are two edges e_1 and e_2 from A to D , two edges e_3 and e_4 from B to D , three edges e_7, e_8, e_9 from C to D , one edge e_{10} from A to B , one edge e_5 from A to C , one edge e_6 from B to C . Hence the graph is as shown below (Figure 2.76):

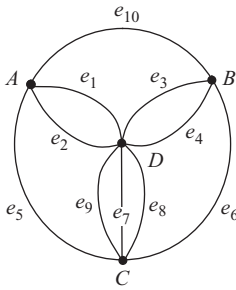


Figure 2.76

We note that

$$\deg(A) = \deg(B) = 4, \quad \deg(C) = 5, \quad \deg(D) = 7.$$

Since the graph is connected and degree of exactly two vertices C and D is odd, there exists an Euler's path and that path should start from one vertex of odd degree and end to the other vertex of odd degree. Therefore, either the path will begin from C and end at D or it will begin from D and end at C . An Euler's path is shown (which begins from C).

Definition 2.51. An edge in a connected graph is called a **bridge** or a **cut edge** if deleting that edge creates a disconnected graph.

For example, consider the graph shown below (Figure 2.77):

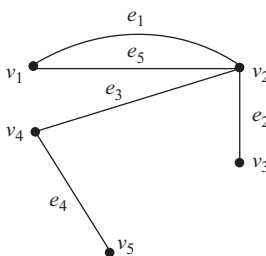


Figure 2.77

In this graph, if we remove the edge e_3 , then the graph breaks into two connected components given below (Figure 2.78):

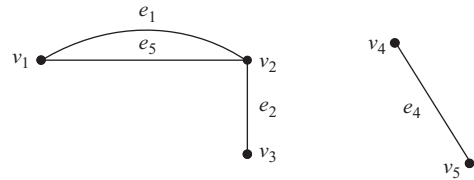


Figure 2.78

Hence the edge e_3 is a bridge in the given graph.

2.6.1 Methods for Finding Euler Circuit

Method 1. We know that if every vertex of a non-empty connected graph has even degree, then the graph has an Euler circuit. We shall make use of this result to find an Euler path in a given graph.

Consider the graph shown in Figure 2.79.

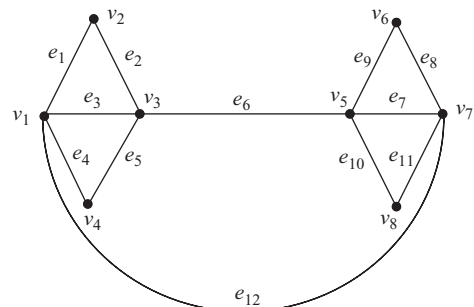


Figure 2.79

We note that

$$\deg(v_2) = \deg(v_4) = \deg(v_6) = \deg(v_8) = 2,$$

$$\deg(v_1) = \deg(v_3) = \deg(v_5) = \deg(v_7) = 4.$$

Hence all vertices have even degree. Also the given graph is connected. Hence the given graph has an Euler circuit. We start from the vertex v_1 and let C be

$$C: v_1 v_2 v_3 v_1.$$

Then C is not an Euler circuit for the given graph but C intersects the rest of the graph at v_1 and v_3 .

Let C' be

$$C': v_1 v_4 v_3 v_5 v_7 v_6 v_5 v_8 v_7 v_1.$$

(In case we start from v_3 , then C' will be $v_3 v_4 v_1 v_7 v_6 v_5 v_7 v_8 v_5$).

Path C' into C and obtain

$$C'': v_1 v_2 v_3 v_1 v_4 v_5 v_6 v_5 v_8 v_7 v_1,$$

that is,

$$C'': e_1 e_2 e_3 e_4 e_5 e_6 e_7 e_8 e_9 e_{10} e_{11} e_{12}.$$

(If we had started from v_2 , then C'' : $v_1 v_2 v_3 v_4 v_1 v_7 v_6 v_5 v_8 v_5 v_3 v_1$ **or** $e_1 e_2 e_5 e_4 e_{12} e_8 e_9 e_7 e_{11} e_{10} e_6 e_3$). In C'' all edges are covered exactly once. Also every vertex has been covered at least once. Hence C'' is an Euler circuit.

Method 2: Fleury's Algorithm: This algorithm is used to find Euler's circuit for a connected graph with no vertices of odd degree.

We shall illustrate the method with the help of the following example:

Let G be the connected graph as shown in the Figure 2.80.

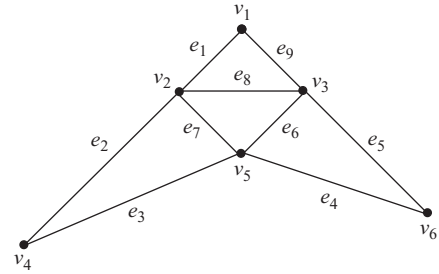


Figure 2.80

Step 1. We begin from any vertex, say v_1

Step 2. We then construct the following table:

Current Path	Next Edge	Reason
v_1	$\{v_1, v_2\}$	No edge from v_1 is a bridge, so choose any edge, say $\{v_1, v_2\}$
$v_1 v_2$	$\{v_2, v_4\}$	(v_2, v_5) a bridge, so choose $\{v_2, v_4\}$
$v_1 v_2 v_4$	$\{v_4, v_5\}$	only one edge $\{v_4, v_5\}$ from v_4 remains
$v_1 v_2 v_4 v_5$	$\{v_5, v_6\}$	Since $\{v_2, v_5\}$ and $\{v_5, v_3\}$ are bridges, choose $\{v_5, v_6\}$
$v_1 v_2 v_4 v_5 v_6$	$\{v_6, v_3\}$	only one edge $\{v_6, v_3\}$ remains
$v_1 v_2 v_4 v_5 v_6 v_3$	$\{v_3, v_5\}$	Since $\{v_1, v_3\}$ is a bridge, choose either $\{v_3, v_2\}$ or $\{v_3, v_5\}$
$v_1 v_2 v_4 v_5 v_6 v_3 v_5$	$\{v_5, v_2\}$	only one edge $\{v_5, v_2\}$ remains
$v_1 v_2 v_4 v_5 v_6 v_3 v_5 v_2$	$\{v_2, v_3\}$	only one edge $\{v_2, v_3\}$ remains
$v_1 v_2 v_4 v_5 v_6 v_3 v_5 v_2 v_3$	$\{v_3, v_1\}$	only one edge $\{v_3, v_1\}$ remains

Hence one possible Euler circuit is

$$C: v_1 v_2 v_4 v_5 v_6 v_3 v_5 v_2 v_3 v_1.$$

In term of edges, this Euler circuit can be expressed as

$$e_1 e_2 e_3 e_4 e_5 e_6 e_7 e_8 e_9.$$

2.7 HAMILTONIAN CIRCUITS

Definition 2.52. A **Hamiltonian path** for a graph G is a sequence of adjacent vertices and distinct

edges in which every vertex of G appears exactly once.

Definition 2.53. A **Hamiltonian circuit** for a graph G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once, except for the first and the last which are the same.

Definition 2.54. A graph is called **Hamiltonian** if it admits a Hamiltonian circuit.

EXAMPLE 2.28

The wooden graph shown in Figure 2.81 and constructed by William Hamilton in the shape of a regular dodecahedron is a Hamiltonian circuit.

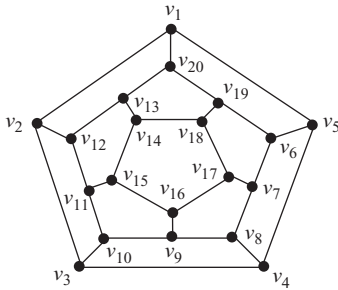


Figure 2.81

The Hamilton circuit is $v_1 v_2 v_3 \dots v_{18} v_{19} v_{20} v_1$.

EXAMPLE 2.29

A complete graph K_n has a Hamiltonian circuit. In particular, the graphs shown in Figure 2.82 are Hamiltonian.

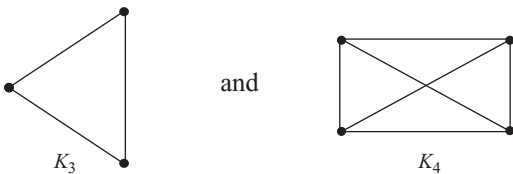


Figure 2.82

EXAMPLE 2.30

The graph shown below does not have a Hamiltonian circuit.

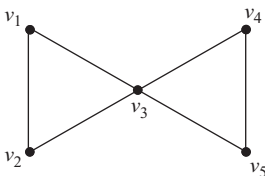


Figure 2.83

EXAMPLE 2.31

The graph shown in Figure 2.84 does not have a Hamiltonian circuit

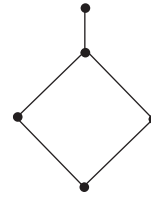


Figure 2.84

Remark 2.5. It is clear that **only connected graphs can have Hamiltonian circuit**. However, there is no simple criterion to tell us whether or not a given graph has Hamiltonian circuit. The following results give us some sufficient conditions for the existence of Hamiltonian circuit/path.

Theorem 2.10. Let G be a linear graph of n vertices. If the sum of the degrees for each pair of vertices in G is greater than or equal to $n - 1$, then there exists a Hamiltonian path in G .

Theorem 2.11. Let G be a connected graph with n vertices. If $n \geq 3$ and $\deg(v) \geq n$ for each vertex v in G , then G has a Hamiltonian circuit.

Theorem 2.12. Let G be a connected graph with n vertices and let u and v be two vertices of G that are not adjacent. If $\deg(u) + \deg(v) \geq n$, then G has a Hamiltonian circuit.

Corollary 2.2. Let G be a connected graph with n vertices. If each vertex has degree greater than or equal to $n/2$, then G has a Hamiltonian circuit.

Proof: It is given that degree of each vertex is greater than or equal to $n/2$. Hence the sum of the degree of any two vertices is greater than or equal to $n/2 + n/2 = n$. So, by the above theorem, the graph G has a Hamiltonian circuit.

Theorem 2.13. Let n be the number of vertices and m be the number of edges in a connected graph G . If $m \geq \frac{1}{2}(n^2 - 3n + 6)$, then G has a Hamiltonian circuit.

The following example shows that the above conditions are not necessary for the existence of Hamiltonian path.

EXAMPLE 2.32

Let G be the connected graph shown in the Figure 2.85.

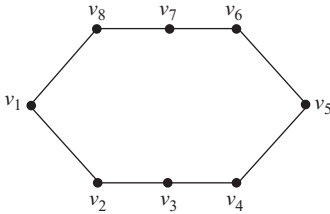


Figure 2.85

We note that

$$\begin{aligned} n &= \text{Number of vertices in } G = 8, \\ m &= \text{Number of edges in } G = 8, \\ \text{Degree of each vertex} &= 2. \end{aligned}$$

Thus, if u and v are non-adjacent vertices, then

$$\deg(u) + \deg(v) = 2 + 2 = 4 \not\geq 8.$$

Also,

$$\frac{1}{2}(n^2 - 3n + 6) = \frac{1}{2}(64 - 24 + 6) = 23.$$

Clearly $m \not\geq \frac{1}{2}(n^2 - 3n + 6)$. Therefore the above two theorems fail. But the given graph has Hamiltonian circuit. For example, $v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_1$ is an Hamiltonian circuit for the graph.

Proposition 2.1. Let G be a graph with at least two vertices. If G has a Hamiltonian circuit, then G has a subgraph H with the following properties:

1. H contains every vertex of G
2. H is connected
3. H has the same number of edges as vertices
4. Every vertex of H has degree 2

The contrapositive of this proposition is

“If a graph G with at least two vertices does not have a subgraph H satisfying (1) – (4), then G does not have a Hamiltonian circuit”.

Also we know that contrapositive of a statement is logically equivalent to the statement. Therefore the above result can be used to show non-existence of a Hamiltonian circuit.

EXAMPLE 2.33

Consider the graph G shown below (Figure 2.86):

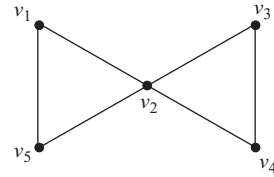


Figure 2.86

Let us verify whether this graph has a Hamiltonian circuit.

Suppose that G has a Hamiltonian circuit. Then it has a subgraph H such that

1. H contains every vertex of G , i.e., H has 5 vertices v_1, v_2, v_3, v_4, v_5
2. H is connected
3. H has 5 edges
4. Every vertex of H has degree 2

Since the degree of v_2 in G is 4 and every vertex of H has degree 2, two edges incident on v_2 must be removed from G to create H . But we note that the edge $\{v_1, v_2\}$ cannot be removed, because the removal of $\{v_1, v_2\}$ will decrease the degree of v_1 to 1, which is less than 2. Similarly the edge $\{v_2, v_3\}$, $\{v_2, v_4\}$ and $\{v_2, v_5\}$ cannot be removed. Hence the degree of v_2 in H must stand at 4 which contradicts the condition that every vertex in H has degree 2 in H . Therefore, no such subgraph H exists. Hence G does not have a Hamiltonian circuit.

EXAMPLE 2.34

Does the graph G given below (Figure 2.87) have Hamiltonian circuit?

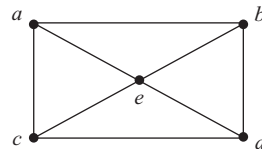


Figure 2.87

Solution. The given graph has

$$\begin{aligned} n &= \text{Number of vertices} = 5 \\ m &= \text{Number of edges} = 8 \end{aligned}$$

$$\deg(a) = \deg(b) = \deg(c) = \deg(d) = 3$$

$$\deg(e) = 4$$

We observe that

- (i) Degree of each vertex is greater than $n/2$
- (ii) The sum of degrees of any non-adjacent pair of vertices is greater than n
- (iii) $\frac{1}{2}(n^2 - 3n + 6) = \frac{1}{2}(25 - 15 + 6) = 8$.

Thus the condition $m \geq \frac{1}{2}(n^2 - 3n + 6)$ is satisfied.

- (iv) The sum of degrees of each pair of vertices in the given graph is greater than $n - 1 = 5 - 1 = 4$.

Thus four sufficiency conditions are satisfied (whereas one condition out of these four conditions is sufficient for the existence of Hamiltonian path/circuit). Hence the graph has a Hamiltonian circuit.

For example, the following circuits in G are Hamiltonian (Figure 2.88).

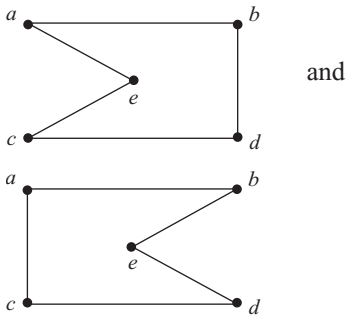


Figure 2.88

EXAMPLE 2.35

Does the graph shown below (Figure 2.89) has Hamiltonian circuit?

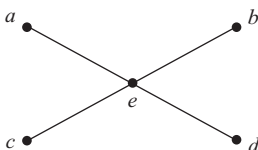


Figure 2.89

Solution.

Here

$$\text{Number of vertices } (n) = 5$$

$$\text{Number of edges } (m) = 4$$

$$\deg(a) = \deg(b) = \deg(c) = \deg(d) = 1$$

$$\deg(e) = 4$$

We note that

- (i) $\deg(a) = \deg(b) = \deg(c) = \deg(d) \not\geq \frac{5}{2}$
- (ii) $\deg(a) + \deg(b) = 2 \not\geq 5$, that is sum of any non-adjacent pair of vertices is not greater than 5
- (iii) $\frac{1}{2}(n^2 - 3n + 6) = \frac{1}{2}(25 - 15 + 6) = 8$

Therefore the condition $m \geq \frac{1}{2}(n^2 - 3n + 6)$ is **not satisfied**,

- (iv) $\deg(a) + \deg(b) = 2 \not\geq 4$, i.e., the condition that sum of degrees of each pair of vertices in the graph is not greater than or equal to $n - 1$.

Hence no sufficiency condition is satisfied. So we try the Proposition 2.1. Suppose that G has a Hamiltonian circuit. Then G should have a subgraph which contains every vertex of G , and number of vertices and number of edges in H should be same. Thus, H should have five vertices a, b, c, d, e and five edges. Since G has only four edges, H cannot have more than four edges. Hence no such subgraph is possible. Hence, the given graph does not have Hamiltonian circuit.

EXAMPLE 2.36

Does the graph shown below (Figure 2.90) possess a Hamiltonian circuit?

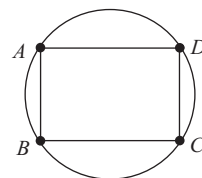


Figure 2.90

Solution. In the given graph

$$\text{Number of vertices } (n) = 4$$

$$\text{Number of edges } (m) = 8$$

$$\text{Degree of each vertex} = 4$$

Thus we see that

- (i) Degree of each vertex is greater than $n/2$
- (ii) Sum of degree of each pair of vertices is greater than $n-1$,
- (iii) $\frac{1}{2}(n^2-3n+6) = \frac{1}{2}(16-12+6) = 5$ and so $m \geq \frac{1}{2}(n^2-3n+6)$.

Hence the given graph has a Hamiltonian path. For example, $ABCD A$ is a Hamiltonian path in the given graph (see Figure 2.91).



Figure 2.91

EXAMPLE 2.37

Is the graph shown in Figure 2.92 a Hamiltonian?

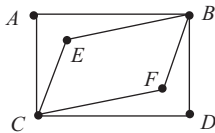


Figure 2.92

Solution. We note that

$$\deg(A) = \deg(D) = \deg(E) = \deg(F) = 2,$$

$$\deg(B) = \deg(C) = 4.$$

Further,

$$\text{Number of vertices } (n) = 6$$

$$\text{Number of edges } (m) = 8.$$

So,

$$\frac{1}{2}(n^2-3n+6) = \frac{1}{2}(36-18+6) = 12.$$

Thus the conditions

- (i) Degree of each vertex is greater than or equal to $n/2$ is not satisfied
- (ii) $m \geq \frac{1}{2}(n^2-3n+6)$ is not satisfied

- (iii) $\deg(A) + \deg(E) = 2 + 2 = 4$ and so the condition that “the sum of degrees of non-adjacent vertices is greater than or equal to n ” is not satisfied
- (iv) “The sum of degrees of any pair of vertices is greater than or equal to $n-1$ ” is not satisfied.

So suppose that G has a Hamiltonian circuit. Then it should have a connected subgraph H containing six vertices, six edges and degree of each vertex should be 2. To have degree of each vertex equal to 2, we should remove two edges from C and two edges from B .

For example, now, if we remove CE from C , then $\deg(E) \neq 2$. If we remove CF , then $\deg(F) \neq 2$. So, we cannot remove CE and CF . If we remove CD , then $\deg(D) \neq 2$. If we remove CA , then $\deg(A) \neq 2$. Hence no such subgraph H exists. So, G cannot have a Hamiltonian circuit.

Remark 2.6. Since the degree of each vertex in the above graph is even, it has Eulerian circuit. Thus the graph in EXAMPLE 2.37 is Eulerian but not Hamiltonian.

EXAMPLE 2.38

Is the graph given in Figure 2.93 Hamiltonian?

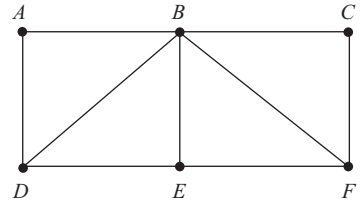


Figure 2.93

Solution. In this graph,

$$\text{Number of vertices } (n) = 6$$

$$\text{Number of edges } (m) = 9.$$

So,

$$\frac{1}{2}(n^2-3n+6) = \frac{1}{2}(36-18+6) = 12,$$

$$\deg(A) = \deg(C) = 2,$$

$$\deg(D) = \deg(F) = 3 = \deg(E),$$

$$\deg(B) = 5.$$

Thus, no sufficient condition is satisfied in this case.

But, the graph has Hamiltonian circuit $ADEFCBA$ shown in the Figure 2.94.

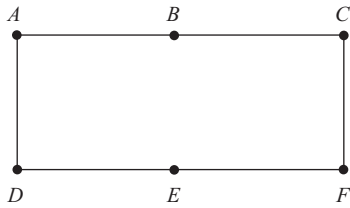


Figure 2.94

EXAMPLE 2.39

Show that the graph G shown in Figure 2.95 is not Hamiltonian.

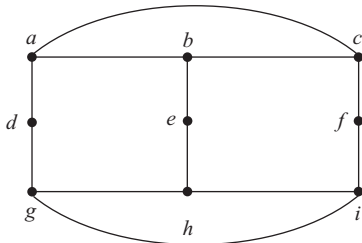


Figure 2.95

Solution. If G is Hamiltonian, then it has a subgraph H that

1. Contains every vertex of G
2. Is connected
3. Has the same number of edges as vertices
4. Is such that every vertex has degree 2

Thus, if such a graph exists, then it will have vertices $\{a, b, c, d, e, f, g, h, i\}$, will be connected, will have six edges and the degree of each vertex shall be 2. We note that $\deg(d) = \deg(e) = \deg(f) = 2$. The degree of the vertex h in G is 3, one edge incident on h must be deleted from G to create H . The edge $\{h, e\}$ is required since otherwise $\deg(e) \neq 2$. So we have to delete either $\{g, h\}$ or $\{h, i\}$. If we delete $\{h, i\}$ and retain $\{g, h\}$, then $\{g, i\}$ has to be deleted to keep g of degree 2. In such a case, there is only one edge $\{i, f\}$ on i and so there is a contradiction to (4). So we cannot remove $\{h, i\}$. Similarly, we see that $\{g, h\}$ cannot be deleted.

Hence no such subgraph H exists and so G does not have a Hamiltonian circuit.

Definition 2.55. A **weighted graph** is a graph for which each edge or each vertex or both is (are) labelled with a numerical value, called its **weight**.

For example, if vertices in a graph denote recreational sites of a town and weights of edges denote the distances in kilometers between the sites, then the graph shown in Figure 2.96 is a weighted graph.

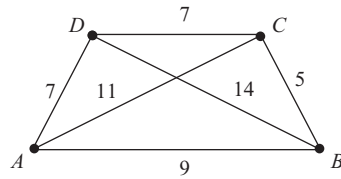


Figure 2.96

Definition 2.56. The **weight of an edge** (v_i, v_j) is called **distance between the vertices** v_i and v_j .

Definition 2.57. A vertex u is a **nearest neighbour** of vertex v in a graph if u and v are adjacent and no other vertex is joined to v by an edge of lesser weight than (u, v) .

For example, in the above example, B is the nearest neighbour of C , whereas A and C are both nearest neighbours of the vertex D . **Thus, nearest neighbour of a set of vertices is not unique.**

Definition 2.58. A vertex u is a nearest neighbour of a set of vertices $\{v_1, v_2, \dots, v_n\}$ in a graph if u is adjacent to some member v_i of the set and no other vertex adjacent to member of the set is joined by an edge of lesser weight than (u, v_i) .

In the above example, if we have set of vertices as $\{B, D\}$, C is the nearest neighbour of $\{B, D\}$ because the edge (C, B) has weight 5 and no other vertex adjacent to $\{B, D\}$ is linked by an edge of lesser weight than (C, B) .

Definition 2.59. The **length of a path** in a graph is the sum of lengths of edges in the path.

Definition 2.60. Let $G = (V, E)$ be a graph and let l_{ij} denote the length of edge (v_i, v_j) in G . Then a **shortest path** from v_i to v_k is a path such that the sum $l_{i_2} + l_{i_2 i_3} + \dots + l_{i_{k-1} i_k}$ of lengths of its edges is **minimum**, that is, total edge weight is minimum.

2.7.1 Travelling Salesperson Problem

This problem requires the determination of a **shortest Hamiltonian circuit** in a given graph of cities

and lines of transportation to minimize the total fare for a travelling person who wants to make a tour of n cities visiting each city exactly once before returning home.

The weighted graph model for this problem consists of vertices representing cities and edges with weight as distances (fares) between the cities. The salesman starts and ends his journey at the same city and visits each of $n-1$ cities once and only once. We want to find minimum total distance.

We discuss the case of five cities and so consider the weighted graph shown in the Figure 2.97.

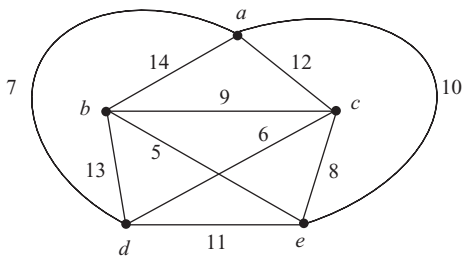


Figure 2.97

We shall use **nearest neighbour** algorithm to solve the problem:

Algorithm: Nearest neighbour (closest insertion)

Input: A weighted complete graph G .

Output: A sequence of labelled vertices that forms a Hamiltonian cycle.

Start at any vertex v .

Initialize $l(v)=0$.

Initialize $i=0$.

While there are unlabelled vertices

$i:=i+1$.

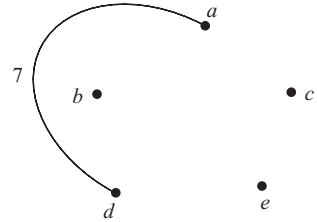
Traverse the cheapest edge that join v to an unlabelled vertex, say w

Set $l(w)=i$.

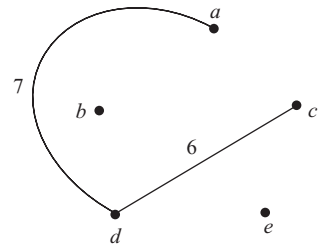
$v:=w$.

For the present example,

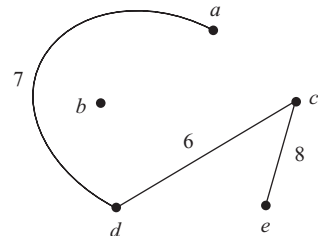
- (i) Let us choose a as the starting vertex. Then d is the nearest vertex and then (a, d) is the corresponding edge. Thus we have the figure



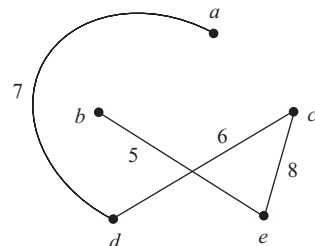
- (ii) From d , the nearest vertex is c , so we have a path shown below:



- (iii) From c , the nearest vertex is e . So we have the path as shown below:



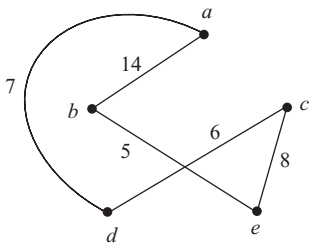
- (iv) From e , the nearest vertex is b and so we have the path



- (v) Now, from b , the only vertex to be covered is a to form **Hamiltonian circuit**. Thus we have a

Hamiltonian circuit as given below. The length of this Hamiltonian circuit is

$7+6+8+5+14=40.$



However, this is not Hamiltonian circuit of minimal length.

The total distance of a minimum Hamiltonian circuit (Figure 2.98) is 37.

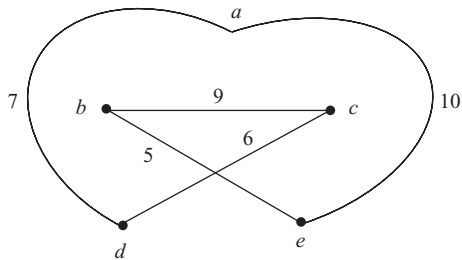


Figure 2.98

Total length = $7+6+9+5+10=37.$

Remark 2.7. Unless otherwise stated, try to start from a vertex of largest weight.

EXAMPLE 2.40

Find a Hamiltonian circuit of minimal weight for the graph given below (Figure 2.99)

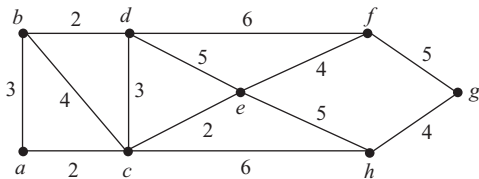


Figure 2.99

Solution. Starting from c and applying nearest neighbour method, we have the required Hamiltonian circuit as $c a b d e f g h c$ with total length as

$2+3+2+5+4+5+4+6=31.$

EXAMPLE 2.41

Find a Hamiltonian circuit of minimal weight for the graph shown below (Figure 2.100)

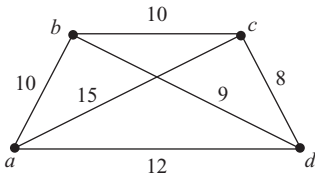


Figure 2.100

Solution. Starting from the point a and using nearest neighbour method, we have the required Hamiltonian circuit as $a b c d a$ with total length as

$10+10+8+12=40.$

Definition 2.61. A k -factor of a graph is a spanning subgraph of the graph with the degree of its vertices being k .

Consider the graph shown in Figure 2.101.

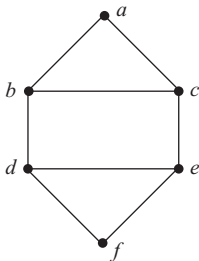


Figure 2.101

Then the graph shown in Figure 2.102 shows a 1-factor of the given graph.

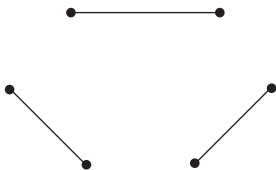


Figure 2.102

Also then, (see Figure 2.103)



Figure 2.103

is a 2-factor of the given graph.

2.8 MATRIX REPRESENTATION OF GRAPHS

A graph can be represented inside a computer by using the adjacency matrix or the incidence matrix of the graph.

Definition 2.62. Let G be a graph with n ordered vertices $v_1, v_2, \dots, v_n$. Then the **adjacency matrix** of G is the $n \times n$ matrix $A(G) = (a_{ij})$ over the set of non-negative integers such that

a_{ij} = the number of edges connecting v_i and v_j for all $i, j = 1, 2, \dots, n$.

We note that if G has no loop, then there is no edge joining v_i to v_i , $i = 1, 2, \dots, n$. Therefore, in this case, all the entries on the main diagonal will be 0.

Further, if G has no parallel edge, then the entries of $A(G)$ are either 0 or 1.

It may be noted that adjacent matrix of a graph is symmetric.

Conversely, given a $n \times n$ symmetric matrix $A(G) = (a_{ij})$ over the set of non-negative integers, we can associate with it a graph G , whose adjacency matrix is $A(G)$, by letting G have n vertices and joining v_i to vertex v_j by a_{ij} edges.

EXAMPLE 2.42

Find the adjacency matrix of the graph shown below (Figure 2.104):

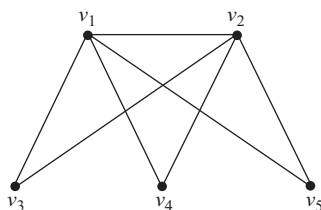


Figure 2.104

Solution. The adjacency matrix $A(G) = (a_{ij})$ is the matrix such that

a_{ij} = Number of edges connecting v_i and v_j .

So, for the given graph, the adjacency matrix is

$$A(G) = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

EXAMPLE 2.43

Find the adjacency matrix of the graph given below:

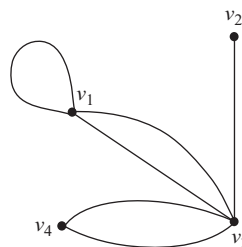


Figure 2.105

Solution. We note that there is a loop at v_1 and parallel edges between v_1, v_3 , and v_3, v_4 . So the adjacency matrix $A(G)$ is given by the following 4×4 matrix:

$$A(G) = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 1 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{bmatrix}.$$

EXAMPLE 2.44

Find the graph that has the following adjacency matrix:

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 0 \end{bmatrix}.$$

Solution. We note that there is a loop at v_1 and a loop at v_3 . There are parallel edges between v_1, v_2 ; v_1, v_4 ; v_2, v_1 ; v_2, v_3 ; v_3, v_2 ; v_4, v_1 . Thus the graph is as shown in Figure 2.106.

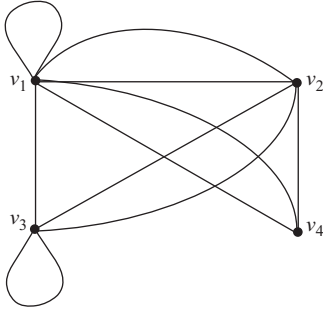


Figure 2.106

The following theorem is stated without proof.

Theorem 2.14. Let G be a graph with n vertices $v_1, v_2, \dots, v_n$ and let $A(G)$ denote the matrix of G . If $B = (b_{ij})$ is the matrix

$$B = A + A^2 + \dots + A^{n-1},$$

then the graph G is a connected graph if and only if $b_{ij} \neq 0$ for $i \neq j$, that is, B has no zero entry off the main diagonal.

EXAMPLE 2.45

A graph G has the following adjacency matrix. Verify whether it is connected.

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

Solution. We have

$$A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 1 & 1 & 0 & 0 & 2 \end{bmatrix},$$

$$A^3 = \begin{bmatrix} 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 2 & 1 & 0 & 0 & 3 \\ 1 & 2 & 0 & 0 & 3 \\ 0 & 0 & 3 & 3 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 2 & 1 & 0 & 0 & 3 \\ 1 & 2 & 0 & 0 & 3 \\ 0 & 0 & 5 & 4 & 0 \\ 0 & 0 & 4 & 5 & 0 \\ 3 & 3 & 0 & 0 & 6 \end{bmatrix}.$$

Therefore,

$$B = A + A^2 + A^3 + A^4 = \begin{bmatrix} 3 & 1 & 3 & 1 & 4 \\ 1 & 3 & 1 & 3 & 4 \\ 3 & 1 & 7 & 5 & 4 \\ 1 & 3 & 5 & 7 & 4 \\ 4 & 4 & 4 & 4 & 8 \end{bmatrix}.$$

Since B has no zero entry off the main diagonal, the graph is connected.

Definition 2.63. Suppose a graph G has n vertices $v_1, v_2, \dots, v_n$ and t edges $e_1, e_2, \dots, e_t$. The **incidence matrix** $B(G)$ of G is the $n \times t$ matrix $B(G) = (b_{ij})$, where

b_{ij} = the number of times that the vertex v_i is incident with the edge e_j ,

that is,

$$b_{ij} = \begin{cases} 0 & \text{if } v_i \text{ is not end of } e_j \\ 1 & \text{if } v_i \text{ is an end of the non-loop } e_j \\ 2 & \text{if } v_i \text{ is an end of the loop } e_j \end{cases}$$

2.9 PLANAR GRAPHS

Definition 2.64. A graph which can be drawn in the plane so that its edges do not cross is said to be **planar**.

For example, the graph shown in Figure 2.107 is planar:

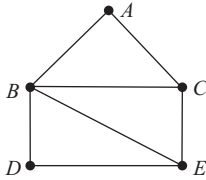


Figure 2.107

Also the complete graph K_4 shown below (Figure 2.108) is planar.

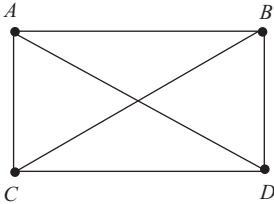


Figure 2.108

In fact, it can be redrawn as shown below in Figure 2.109 so that no edges cross.

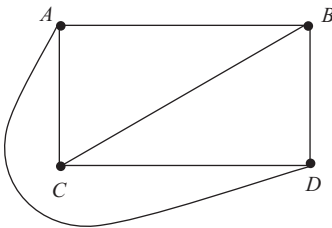


Figure 2.109

But the map K_5 is not planar because in this case, the edges cross each others (see Figure 2.110).

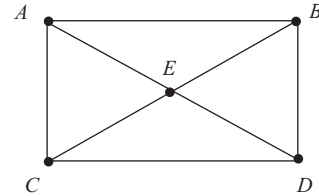


Figure 2.110

Definition 2.65. An area of the plane that is bounded by edges of the planar graph and is not further subdivided into subareas is called a **region** or **face** of the planar graph.

A face is characterized by the cycle that forms its boundary.

Definition 2.66. A region is said to be **finite** if its area is finite and **infinite** if its area is infinite. Clearly a planar graph has exactly one infinite region.

For example, consider the graphs shown in Figure 2.111.

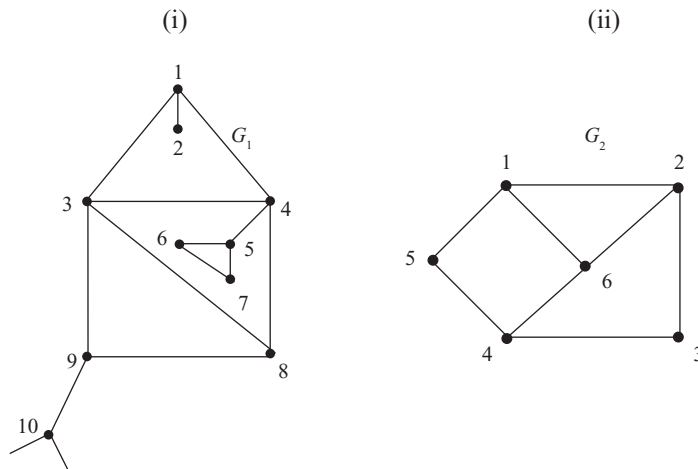
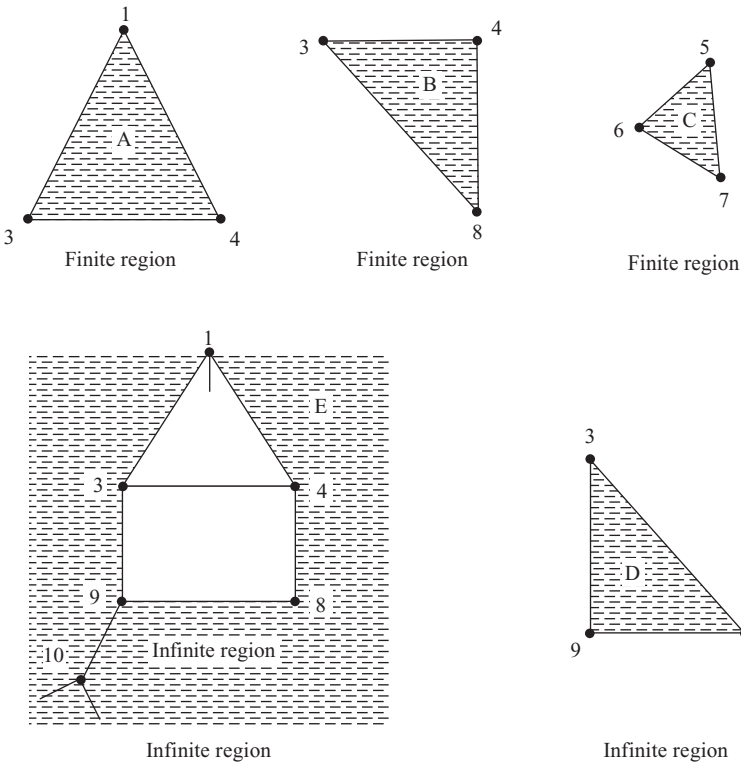


Figure 2.111

We note that in the graph G_1 , there are five regions **A**, **B**, **C**, **D**, **E** as shown below:



In graph G_2 , there are four region **A**, **B**, **C**, **D** (Figure 2.112)

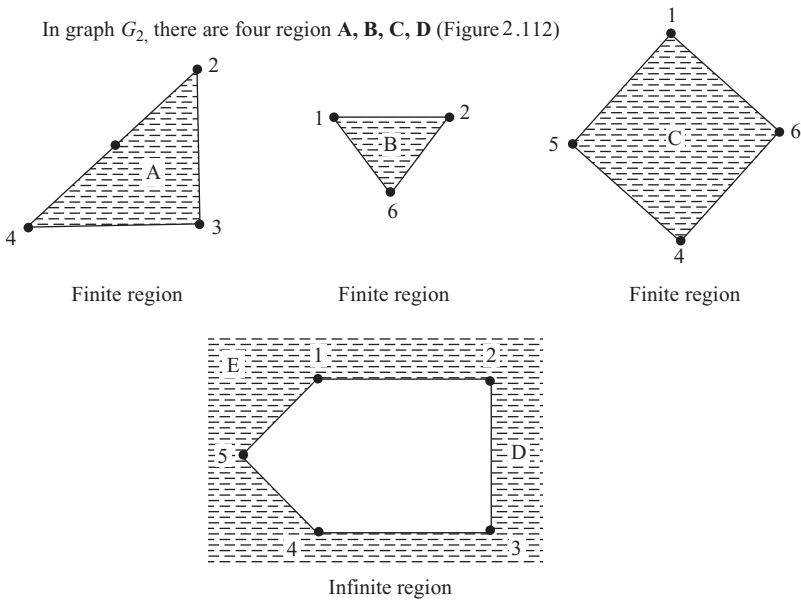


Figure 2.112

Definition 2.67. Let f be a face (region) in a planar graph. The length of the cycle (or closed walk) which borders f is called the **degree of the region f** . It is denoted by $\deg(f)$.

In a planar graph we note that **each edge either borders two regions or is contained in a region and will occur twice in any walk along the border of the region**. Thus we have the following:

Theorem 2.15. The sum of the degrees of the regions of a map is equal to twice the number of edges.

For example, in the graph G_2 , discussed above, we have

$$\deg(A)=4, \deg(B)=3, \deg(C)=4, \deg(d)=5.$$

The sum of degrees of all regions $= 4+3+4+5=16$.

$$\text{Number of edges in } G_2=8.$$

Hence the sum of degrees of regions is twice the number of edges.

Theorem 2.16. (Euler's Formula for Connected Planar Graphs) If G is a connected planar graph with e edges, v vertices and r regions, then

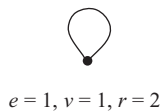
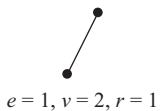
$$v - e + r = 2.$$

Proof: We shall use induction on the number of edges. Suppose that $e=0$. Then the graph G consists of a single vertex, say P . Thus, G is as shown below:



and we have $e=0$, $v=1$, $r=1$. Thus $1-0+1=2$ and the formula holds in this case.

Suppose that $e=1$. Then the graph G is one of the two graphs shown below:



We see that, in either case, the formula holds.

Suppose that the formula holds for connected planar graph with n edges. We shall prove that this holds for graph with $n+1$ edges. So, let G be the graph with $n+1$ edges. Suppose first that G contains no cycles. Choose a vertex v_1 and trace a path starting at v_1 . Ultimately, we will reach a vertex a with degree 1, that we cannot leave (see Figure 2.113).

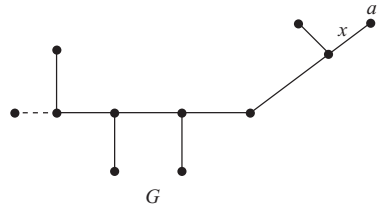


Figure 2.113

We delete “ a ” and the edge x incident on “ a ” from the graph G . The resulting graph G' (Figure 2.114) has n edges and so by induction hypothesis, the formula holds for G' . Since G has one more edge than G' , one more vertex than G' and the same number of faces as G' , it follows that the formula $v-e+r=2$ holds also for G .

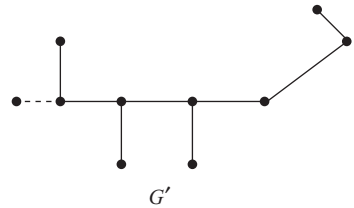


Figure 2.114

Now suppose that G contains a cycle. Let x be an edge in a cycle as shown in Figure 2.115.

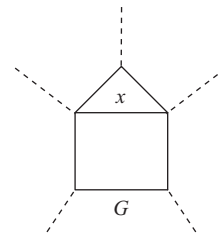


Figure 2.115

Now the edge x is part of a boundary for two faces. We delete the edge x but no vertices to obtain the graph G' as shown in Figure 2.116.

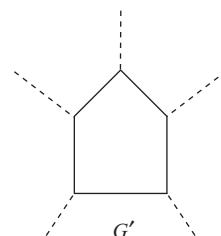


Figure 2.116

Thus G' has n edges and so by induction hypothesis the formula holds. Since G has one more face (region) than G' , one more edge than G' and the same number of vertices as G' , it follows that the formula $v-e+r=2$ also holds for G . Hence, by mathematical induction, the theorem is true.

Remark 2.8. Planarity of a graph is not affected if

- (i) An edge is divided into two edges by the insertion of new vertex of degree 2 (Figure 2.117).

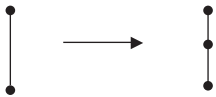


Figure 2.117

- (ii) Two edges that are incident with a vertex of degree 2 are combined as a single edge by the removal of that vertex (Figure 2.118).

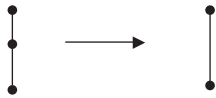


Figure 2.118

Definition 2.68. Two graphs G_1 and G_2 are said to be **isomorphic to within vertices of degree 2 (or homeomorphic)** if they are isomorphic or if they can be transformed into isomorphic graphs by repeated insertion and/or removal of vertices of degree 2.

Definition 2.69. The repeated insertion/removal of vertices of degree 2 is called **sequence of series reduction**.

For example, the graphs shown in Figure 2.119 are isomorphic to within vertices of degree 2.

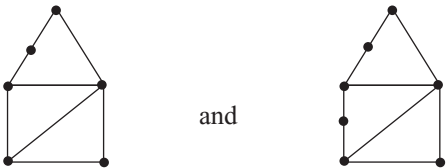


Figure 2.119

If we define a relation R on the set of graphs by $G_1 R G_2$ if G_1 and G_2 are homeomorphic, then R is an equivalence relation. Each equivalence class consists of a set of mutually homeomorphic graphs.

EXAMPLE 2.46

Show that the graph $K_{3,3}$ given in Figure 2.120 below, is not planar.

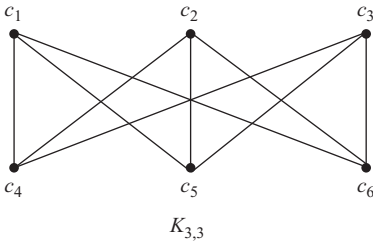


Figure 2.120

A problem based on this example can be stated as “Three cities c_1, c_2 and c_3 are to be directly connected by express ways to each of three cities c_4, c_5 and c_6 . Can this road system be designed so that the express ways do not cross?” This example shows that it cannot be done.

Solution. Suppose that $K_{3,3}$ is planar. Since every cycle in $K_{3,3}$ has at least four edges, each face (region) is bounded by at least four edges. Thus the number of edges that bound regions is at least $4r$. Also, in a planar graph each edge belongs to at most two bounding cycles. Therefore,

$$2e \geq 4r \text{ (sums of degrees of region is equal to twice the number of edges)}$$

But, by Euler’s formula for planar graph,

$$r = e - v + 2.$$

Hence,

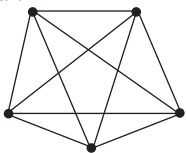
$$2e \geq 4(e - v + 2). \tag{1}$$

In case of $K_{3,3}$ we have $e=9, v=6$ and so (1) yields

$$18 \geq 4(9 - 6 + 2) = 20,$$

which is a contradiction. Therefore $K_{3,3}$ is not planar.

Remark 2.9. By a argument similar to the above example, we can show that the graph K_5 (Figure 2.121) is not planar.



(Non-planar graph K_5)

Figure 2.121

We observe that if a graph contains $K_{3,3}$ or K_5 as a subgraph, then it cannot be planar.

The following theorem, which we state without proof, gives necessary and sufficient condition for a graph to be planar.

Kuratowski's Theorem 2.17. A graph G is planar if and only if G does not contain a **subgraph** homeomorphic to $K_{3,3}$ or K_5 .

The complete graph K_5 and the complete bipartite graph $K_{3,3}$ are called the **Kuratowski graphs**.

EXAMPLE 2.47

Using Kuratowski's theorem show that the graph G , shown in Figure 2.122, is not planar.

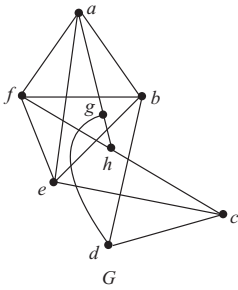
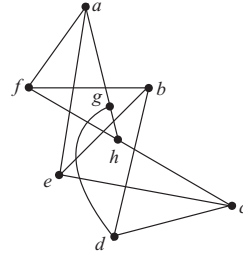


Figure 2.122

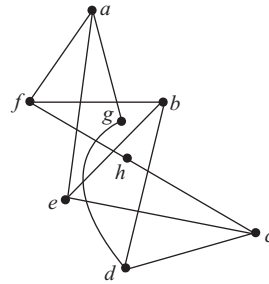
Solution. Let us try to find $K_{3,3}$ in the graph G . We know that in $K_{3,3}$, each vertex has degree 3. But we note that in G , the degree of a, b, f and e each is 4. So we eliminate the edges (a, b) and (f, e) so that all vertices have degree 3. If we eliminate one more edge, we will obtain two vertices of degree 2 and

we can then carry out series reduction. The resulting graph will have nine edges.

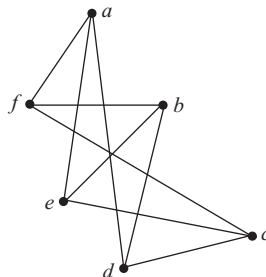
Also we know that $K_{3,3}$ has nine edges. So this approach seems promising. Using trial and error, we find that the edge (g, h) should be removed. Then g and h have degree 2.



(Graph obtained by deleting edges (a, b) and (f, e)).



(Graph obtained by eliminating the edge (g, h)). Performing series reduction now, we obtain an isomorphic copy of $K_{3,3}$ (Figure 2.123).



(Isomorphic copy of $K_{3,3}$, obtained by series reduction)

Figure 2.123

Hence, by Kurtowski's theorem, the given graph G is not planar.

EXAMPLE 2.48

If the graph given in Figure 2.124 is planar, redraw it so that no edges cross, otherwise find a subgraph homeomorphic to either K_5 or $K_{3,3}$.

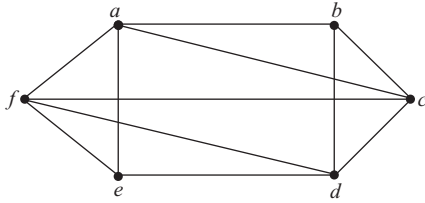


Figure 2.124

Solution. The given graph can be redrawn as shown in the Figure 2.125.

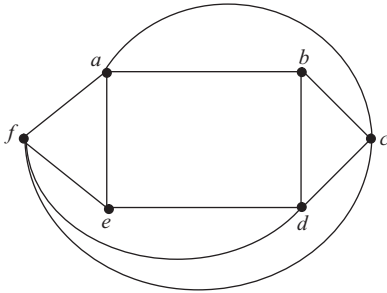


Figure 2.125

We then observe that no edges cross. Hence the given graph is planar.

Theorem 2.18. Let G be a connected planar graph with v vertices, e edges, where $v \geq 3$. Then $e \leq 3v - 6$ or $6 \leq 3v - e$.

(Note that the theorem is not true for K_1 , where $v=1$ and $e=0$ and is not true for K_2 , where $v=2$ and $e=1$).

Proof: Let r be the number of regions in a planar representation of the graph G . The sum of the degrees of the regions is equal to $2e$. But each region has degree greater than or equal to 3. Hence $2e \geq 3r$, that is, $r \leq \frac{2e}{3}$. But, by Euler's formula,

$$v - e + r = 2.$$

So,

$$2 = v - e + r \leq v - e + \frac{2e}{3} = v - \frac{e}{3}.$$

Hence,

$$6 \leq 3v - e \quad \text{or} \quad e \leq 3v - 6$$

Remark 2.10. Using above theorem, let us consider planarity of K_5 . Here $v=5$, $e=10$. Suppose K_5 is planar, then by the above theorem,

$$6 \leq 3(5) - 10 = 5,$$

which is impossible. Hence K_5 is non-planar.

Definition 2.70. Any graph homeomorphic either to K_5 or to $K_{3,3}$ is called **Kuratowski subgraph**.

EXAMPLE 2.49

Find Kuratowski subgraph of the following graph (Figure 2.126):

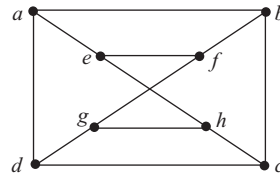


Figure 2.126

Solution. In the given graph,

Number of vertices = 8

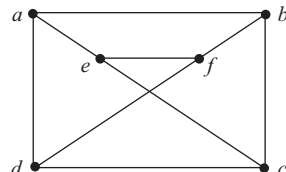
Number of edges = 12.

Firstly, we remove the edge (g, h) . Then $\deg(g)=2$, $\deg(h)=2$ and so the vertices g and h can be removed. Then, we have

Number of vertices = 6

Number of edges = 9

and the graph reduces to



which is homeomorphic to $K_{3,3}$ (Kuratowski graph) shown in Figure 2.127.

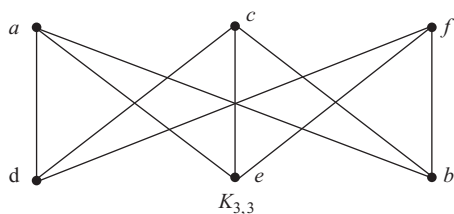


Figure 2.127

Thus the given graph is non-planar by Kuratowski theorem.

EXAMPLE 2.50

Find a subgraph homeomorphic to either K_5 or $K_{3,3}$ in the graph given below (Figure 2.128):

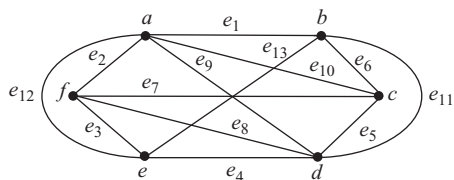


Figure 2.128

Solution.

Here,

Number of vertices = 6

Number of edges = 13

$\deg(a)=5$, $\deg(f)=4$, $\deg(c)=4$, $\deg(d)=5$

But, in $K_{3,3}$, the degree of each vertex is 3. We delete the edges (a, c) , (f, d) , (a, e) , (b, d) , then degree of each vertex a, f, c and d becomes 3. As such, the subgraph becomes as given below (Figure 2.129), which has six vertices each with degree 3 and has 9 edges.

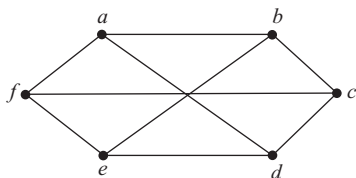


Figure 2.129

This subgraph is homeomorphic to $K_{3,3}$ as shown in Figure 2.130.

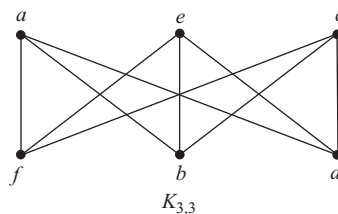


Figure 2.130

This also shows, by Kuratowski theorem, that the given graph is not planar.

2.10 COLOURING OF GRAPH

Definition 2.71. Let G be a graph. The assignment of colours to the vertices of G , one colour to each vertex, so that the adjacent vertices are assigned different colours is called **vertex colouring** or **colouring of the graph G** .

Definition 2.72. A graph G is n -colourable if there exists a colouring of G which uses n colours.

Definition 2.73. The minimum number of colours required to paint (colour) a graph G is called the **chromatic number of G** and is denoted by $\chi(G)$.

EXAMPLE 2.51

Find the chromatic number for the graph shown in the Figure 2.131.

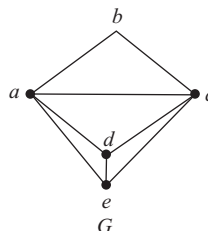


Figure 2.131

Solution. The triangle abc needs three colours. Suppose that we assign colours c_1, c_2, c_3 to a, b and c respectively. Since d is adjacent to a and c , d will have different colour than c_1 and c_3 . So we paint d by c_2 . Then e must be painted with a colour different from those of a, d and c , that is, we cannot colour e with c_1, c_2 or c_3 . Hence, we have to give e a fourth colour c_4 . Hence

$$\chi(G)=4.$$

Welsh–Powell algorithm to determine upper bound to the chromatic number of a given graph.

The input is a given graph G .

1. Order the vertices of G according to decreasing degree.
2. Assign the first colour, say c_1 , to the first vertex and then, in sequential order, assign c_1 to each vertex, which is not adjacent to a previous vertex assigned c_1 .
3. Repeat Step 2 with a second colour c_2 and the subsequence of the remaining non-painted vertices.
4. Repeat Step 3 with a third colour c_3 , then a fourth colour c_4 and so on until all vertices are coloured.
5. Exit.

EXAMPLE 2.52

Use Welsh-Powell algorithm to determine an upper bound to the chromatic number of the “Wheel” graph shown in Figure 2.132.

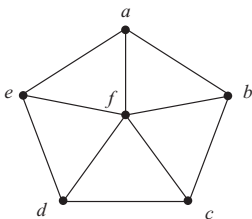


Figure 2.132

Solution. We note that

$$\deg(f)=5,$$

$$\deg(a)=\deg(b)=\deg(c)=\deg(d)=\deg(e)=3.$$

Step 1. Ordering the vertices according to decreasing degree yields

$$f, a, b, c, d, e$$

Step 2. Paint f with colour c_1 .

Step 3. Paint a, d with colour c_2 .

Step 4. Paint b, e with colour c_3 .

Step 5. Paint c with colour c_4 .

Hence $\chi(G) \leq 4$. Also since there is a triangle in the given graph so we have $\chi(G) \geq 3$. Thus $3 \leq \chi(G) \leq 4$. But we do not yet know exactly what the chromatic number is. We try to build a 3-colouring of G .

Let us start colouring the triangle abf with the colours c_1, c_2, c_3 , respectively. Since c is adjacent to the vertices b and f of colour c_2 and c_3 , respectively, c is forced to be coloured c_1 and then d is forced to be c_2 . However, now the adjacent vertices a and e cannot both have colour c_1 . Thus the graph cannot be 3-coloured. But using a fourth colour c_4 for e gives us 4-colouring of G . Hence, $\chi(G)=4$.

EXAMPLE 2.53

Use Welsh-Powell algorithm to colour the graph shown in Figure 2.133.

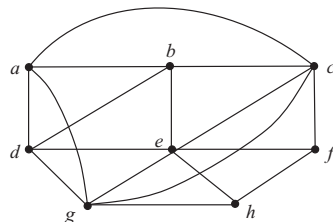


Figure 2.133

Solution. Ordering the vertices according to decreasing degrees, we get the sequence e, c, g, a, b, d, f, h .

Use the colour c_1 to colour (paint) e and a .

Use the colour c_2 to paint c, d and h .

Use third colour c_3 to paint vertices g, b and f .

Thus all the vertices are painted such that no adjacent vertices get the same colour. Hence $\chi(G)=3$.

Some rules for colouring a graph

1. $\chi(G) \leq |V|$, where $|V|$ is the number of vertices of G .
2. A triangle always requires three colours, that is, $\chi(K_3)=3$. Similarly $\chi(K_n)=n$, where K_n is the complete graph of n vertices.
3. If some subgraph of G requires k -colours, then $\chi(G) \geq k$.
4. If $\deg(v)=n$, then at most n colours are required to colour the vertices adjacent to v .
5. $\chi(G)=\max \{\chi(C): C \text{ is a connected component of } G\}$.
6. Every n -colourable graph has at least n vertices v such that $\deg(v) \geq n-1$.
7. $\chi(G) \leq 1 + \Delta(G)$, where $\Delta(G)$ is the largest degree of any vertex of G .
8. The following statements are equivalent:
 - (i) A graph G is 2-colourable
 - (ii) G is bipartite
 - (iii) Every cycle of G has even length
9. If $\delta(G)$ is the smallest degree of any vertex of G , then

$$\chi(G) \geq \frac{|v|}{|v| - \delta(G)}.$$

EXAMPLE 2.54

Find chromatic number of the graphs shown in Figure 2.134.

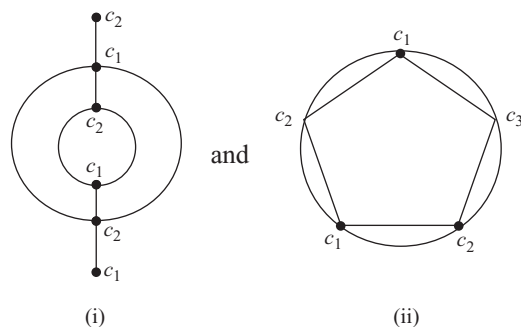


Figure 2.134

Solution. The graph (i) is 2-colourable whereas the graph (ii) is 3-colourable.

Theorem 2.19. Bipartite graph is 2-colourable unless G is edgeless.

Proof: A two colouring is obtained by assigning one colour to every vertex in one of the bipartition parts and another colour to every vertex in the other partition part as shown in Figure 2.135.

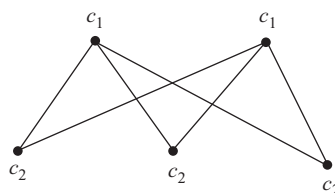


Figure 2.135

Corollary 2.3. Even cycle graphs C_{2n} have $\chi(C_{2n})=2$.

Proof: An even cycle graph is bipartite and therefore, by Theorem 2.19, its chromatic number is 2.

For example consider the graph shown in Figure 2.136.

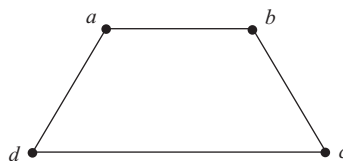


Figure 2.136

It is equivalent to the following bipartite graph shown in Figure 2.137.

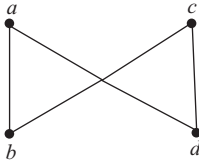


Figure 2.137

Therefore, it is 2-colourable.

Proposition 2.2. Odd cycle graph C_{2n+1} has $\chi(C_{2n+1})=3$.

Proof: Let $v_1, v_2, \dots, v_{2n}, v_{2n+1}$ be vertices of a cycle graph C_{2n+1} . If two colours were sufficient, then they would have to alternate around the cycle. Thus, the odd subscripted vertices would have to be one colour and the even subscripted vertices have to be second colour but vertex v_{2n+1} is adjacent to v_1 , and so according to this scheme two adjacent vertices v_1 and v_{2n+1} have the same colour. This is a contradiction and so C_{2n+1} is not 2-colourable but 3-colourable.

It follows, therefore, that “The chromatic number of a cycle is either two or three, depending on whether its length is even or odd.”

Definition 2.74. The **join** $G+H$ of the graphs G and H is obtained from the graph $G \cup H$ by adding an edge between each vertex of H .

For example, if we have a graph as shown below (Figure 2.138):

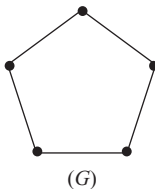


Figure 2.138

and a graph



then the join $G+K_1$ of the graphs G and K_1 is the graph as shown below (Figure 2.139):

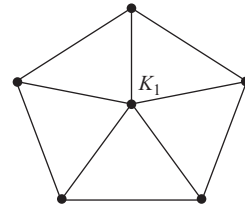


Figure 2.139

Regarding the chromatic number of the join $G+H$ of the graphs G and H we have the following:

Proposition 2.3. The join of a graph G and H has chromatic number

$$\chi(G+H)=\chi(G)+\chi(H).$$

Definition 2.75. The n vertex wheel graph W_n is called an **odd-order wheel** if n is odd and an **even-order wheel** if n is even.

Proposition 2.4. Odd-order wheel graph has

$$\chi(W_{2m+1})=3, \quad \text{for all } m \geq 1.$$

Proof: Using the fact that the wheel graph W_{2m+1} is the join $C_{2m}+K_1$, it follows that

$$\chi(W_{2m+1})=\chi(C_{2m})+\chi(K_1)=2+1=3.$$

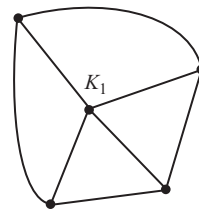


Figure 2.140 (W_5)

Proposition 2.5. Even-order wheel graph has $\chi(W_{2m})=4$.

Proof: Using the fact that $W_{2m}=C_{2m-1}+K_1$, it follows that

$$\chi(W_{2m})=\chi(C_{2m-1})+\chi(K_1)=3+1=4.$$

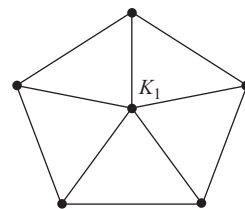


Figure 2.141 (W_6)

EXAMPLE 2.55

Find chromatic number of the graph shown in Figure 2.142:

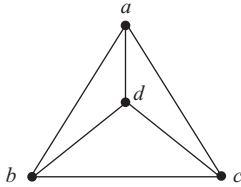


Figure 2.142

Solution. Here degree of each vertex is 3. Therefore, $\chi(G) \leq 1+3=4$. Since it has a triangle, $\chi(G) \geq 3$. Thus, $3 \leq \chi(G) \leq 4$.

We first colour the triangle abd with colours c_1, c_2, c_3 , respectively. Since c is adjacent to each of a, b and d , therefore a different colour c_4 have to be given to it. Hence the graph is 4-colourable and so $\chi(G)=4$.

Note: If we apply rule 9, then $\chi \geq \frac{4}{(4-3)} = 4$. Hence $\chi(G) = 4$.

Theorem 2.20 (Four-Colour Theorem of Appel and Haken)

Any planar graph is 4-colourable.

(Proof of the theorem is out of the scope of this book).

2.11 DIRECTED GRAPHS

Definition 2.76. A **directed graph** or **digraph** consists of two finite sets:

- (i) A set V of vertices (or nodes or points).
- (ii) A set E of directed edges (or arcs), where each edge is associated with an ordered pair (v, w) of vertices called its endpoints. If edge e is associated with the ordered pair (v, w) , then e is said to be **directed edge from v to w** .

The directed edges are indicated by arrows.

We say that edge $e=(v, w)$ is incident from v and is incident into w .

The vertex v is called **initial vertex** and the vertex w is called the **terminal vertex** of the directed edge (v, w) .

Definition 2.77. Let G be a directed graph. The **out-degree of a vertex v of G** is the number of edges beginning at v . It is denoted by $\text{outdeg}(v)$.

Definition 2.78. Let G be a directed graph. The **in-degree of a vertex v of G** is the number of edges ending at v . It is denoted by $\text{indeg}(v)$.

EXAMPLE 2.56

Consider the directed graph shown below (Figure 2.143):

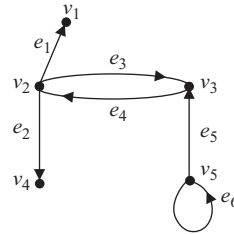


Figure 2.143

Here edge e_1 is (v_2, v_1) whereas e_6 is denoted by (v_5, v_5) and is called a loop. The indegree of v_2 is 1, outdegree of v_2 is 3.

Definition 2.79. A vertex with 0 indegree is called a **source**, whereas a vertex with 0 outdegree is called a **sink**.

For instance, in the above example, v_1 is a sink.

Definition 2.80. If the edges and/or vertices of a directed graph G are labelled with some type of data, then G is called a **labelled directed graph**.

Definition 2.81. Let G be a **directed graph** with ordered vertices $v_1, v_2, \dots, v_n$. The **adjacency matrix of G is the matrix $A=(a_{ij})$ over the set of non-negative integers such that**

a_{ij} = the number of arrows from v_i to v_j , $i, j=1, 2, \dots, n$.

EXAMPLE 2.57

Find the adjacency matrices for the graphs given below (Figure 2.144):

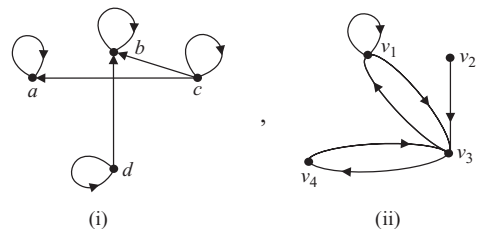


Figure 2.144

Solution. The edges in the directed graph are (a, a) , (b, b) , (c, c) , (d, d) , (c, a) , (c, b) and (d, b) . Therefore the adjacency matrix $A=(a_{ij})$ is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

(iii) The edges in the graph in (ii) are (v_2, v_3) , (v_1, v_1) , (v_1, v_3) , (v_3, v_1) , (v_3, v_4) , (v_4, v_3) . Hence the adjacency matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

EXAMPLE 2.58

Find the directed graph represented by the adjacency matrix:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solution. We observe that $a_{12}=1$, $a_{23}=1$, $a_{34}=1$, $a_{35}=1$, $a_{41}=1$, $a_{42}=1$. Hence the digraph is as shown in Figure 2.145.

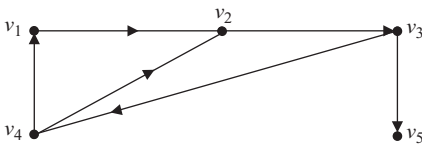


Figure 2.145

Definition 2.82. In a directed graph, if there is no more than one directed edge in a particular direction between a pair of vertices, then it is called **simple directed graph**.

For example, the graph shown in Figure 2.146 is a simple directed graph.

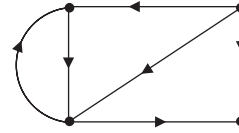


Figure 2.146

A directed graph which is not simple is called **directed multi-graph**.

Definition 2.83. A simple digraph is said to be **strongly connected** if for each pair of vertices v, w , there are path, from v to w and w to v .

For example, the graphs show in Figure 2.147 are strongly connected.

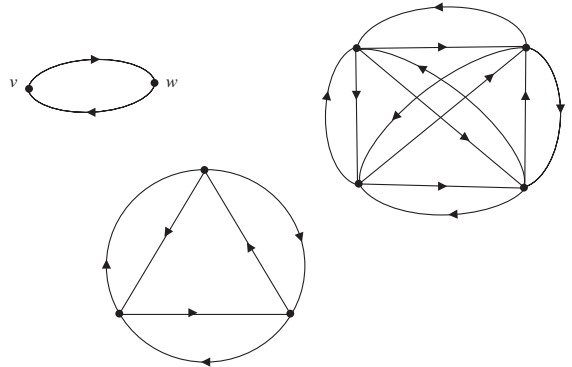


Figure 2.147

Definition 2.84. Let G be a simple digraph with n vertices. Then a $n \times n$ matrix $P=(p_{ij})$ such that

$$p_{ij} = \begin{cases} 1 & \text{if there is a path from } v_i \text{ to } v_j \\ 0 & \text{otherwise} \end{cases}$$

is called **path matrix** or **reachability matrix** of G .

The path matrix can be given in term of adjacency matrix:

$$P=A \vee A^{(2)} \vee A^{(3)} \vee \dots \vee A^{(n)}$$

and

$$A^{(2)}=A \odot A, \quad A^{(r)}=A^{(r-1)} \odot A,$$

where $\vee$ represents Boolean addition and $\odot$ denotes the Boolean matrix multiplication.

Warshall's algorithm for finding path matrix from adjacency matrix

We discuss this algorithm with the help of the following example.

EXAMPLE 2.59

Find the adjacency matrix and the path matrix for the digraph shown in Figure 2.148.

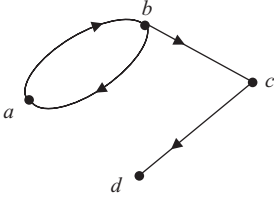


Figure 2.148

Solution. The adjacency matrix of this digraph is:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Then the path matrix $P=(p_{ij})$ is found as follows: We take

$$P_0 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = A \text{ itself.}$$

Now we find P_1 by consulting column 1 and row 1. We note that P_0 has 1 in location 2 of column 1 and location 2 of row 1. Thus the element p_{22} is replaced by 1 in P_0 . Thus we have

$$P_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

To find P_2 we consult column 2 and row 2 of P_1 . We note that P_1 has 1 in locations 1 and 2 of column 2 and locations 1, 2 and 3 of row 2. Thus, to obtain P_2 we should put 1 in positions p_{11} , p_{12} , p_{13} , p_{21} , p_{22} and p_{23} of matrix P_1 . We thus have

$$P_2 = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

We now consult column 3 and row 3 of P_2 . The matrix P_2 has 1 in locations 1 and 2 of column 3 and 1 in location 4 of row 3. Thus we shall obtain P_3 by putting 1 in positions p_{14} , p_{24} . Thus

$$P_3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Finally, we consult column 4 and row 4 of P_3 . We have 1 in location 1, 2, 3 of column 4 and there is no 1 in row 4. Hence $P_4=P_3$. Thus the path matrix is

$$P = (p_{ij}) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Remark 2.11. Given an adjacency matrix of a graph G , we first draw digraph of G and then apply definition of path matrix. We will obtain path matrix. Now, we state a result without proof.

Theorem 2.21. Let A be the adjacency matrix of a graph G with n vertices and let $B_n = A + A^2 + A^3 + \dots + A^n$. Then the path matrix P and B_n have the same non-zero entries.

Let G be a strongly connected directed graph. Then for any pair of vertices v and w in G , there is a path from v to w and from w to v . Accordingly, G is **strongly connected if and only if the path matrix P of G has no zero entries.**

Thus, in view of the above theorem, we have:

Theorem 2.22. Let A be adjacency matrix of a graph G with n vertices and let $B_n = A + A^2 + \dots + A^n$. Then G is strongly connected if and only if B_n has no zero entries.

EXAMPLE 2.60

A directed graph has the following adjacency matrix. Check whether it is strongly connected.

$$A(G) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution. To find the path matrix, we first consider column 1 and row 1 of

$$P_0 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = A(G).$$

In column 1, the entry 1 is located at 2 and 3 positions and 1 is at locations 2 and 3 in row 1. Thus put 1 at $p_{22}, p_{23}, p_{32}, p_{33}$ and get

$$P_1 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

We note that 1 is located in column 2 of P_1 at position 1, 2, 3, and 1 is located at position 1, 2, 3 of row 2. Thus put 1 at $p_{11}, p_{12}, p_{13}, p_{21}, p_{22}, p_{23}, p_{31}, p_{32}, p_{33}$, and get

$$P_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

The next step will not change the position. Hence,

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ (all non-zero entries).}$$

Therefore, G is strongly connected.
The graph of G is shown in Figure 2.149.

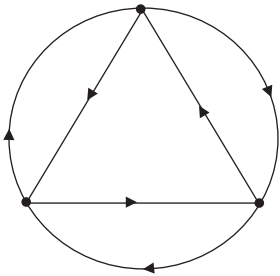


Figure 2.149

Remark 2.12. In the above example, we also note that

$$\begin{aligned} B_3 &= A + A^2 + A^3 \\ &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 5 & 5 \\ 5 & 4 & 5 \\ 5 & 5 & 4 \end{bmatrix} \text{ (all non-zero entries)} \end{aligned}$$

showing that G is strongly connected.

2.12 TREES

Definition 2.85. A graph is said to be a **tree** if it is a connected acyclic graph.

A **trivial tree** is a graph that consists of a single vertex. An **empty tree** is a tree that does not have any vertices or edges.

For example, the graphs shown in Figure 2.150 are all trees.

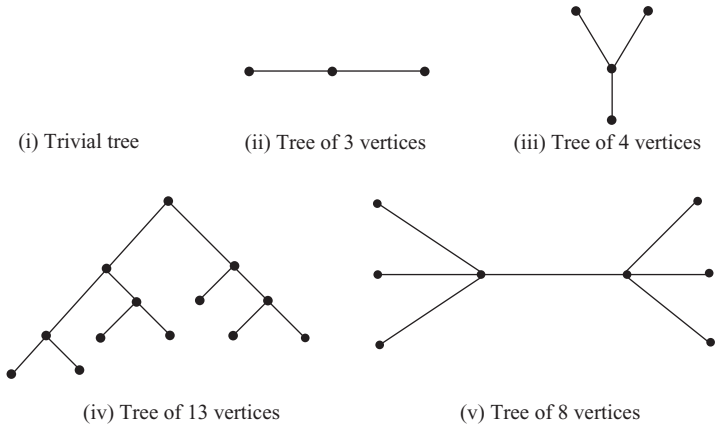


Figure 2.150

But the graphs shown in Figure 2.151 are not trees:

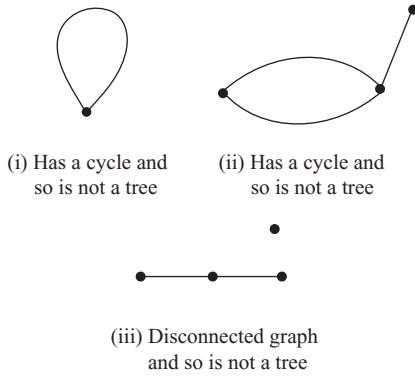


Figure 2.151

Definition 2.86. A collection of disjoint trees is called a **forest**.

Thus a graph is a forest if and only if it is circuit free.

Definition 2.87. A vertex of degree 1 in a tree is called a **leaf** or a **terminal node** or a **terminal vertex**.

Definition 2.88. A vertex of degree greater than 1 in a tree is called a **branch node** or **internal node** or **internal vertex**.

Consider the tree shown in Figure 2.152.

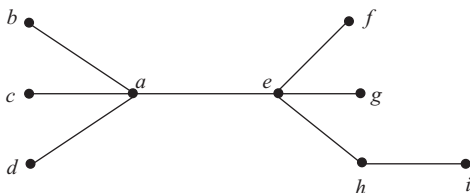


Figure 2.152

In this tree, the vertices b, c, d, f, g, i are leaves whereas the vertices a, e, h are branch nodes.

2.12.1 Characterization of Trees

We have the following interesting characterization of trees:

Lemma 2.1. A tree that has more than one vertex has at least one vertex of degree 1.

Proof: Let T be a particular but arbitrary chosen tree having more than one vertex (Figure 2.153).

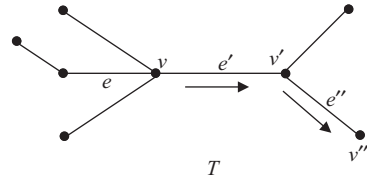


Figure 2.153

1. Choose a vertex v of T . Since T is connected and has at least two vertices, v is not isolated and there is an edge e incident on v .
2. If $\deg(v) > 1$, there is an edge $e' \neq e$ because there are, in such a case, at least two edges incident on v . Let v' be the vertex at the other end of e' . This is possible because e' is not a loop by the definition of a tree.
3. If $\deg(v') > 1$, then there are at least two edges incident on v' . Let e'' be the other edge different from e' and v'' be the vertex at other end of e'' . This is again possible because T is acyclic.
4. If $\deg(v'') > 1$, repeat the above process. Since the number of vertices of a tree is finite and T is circuit free, the process must terminate and we shall arrive at a vertex of degree 1.

Remark 2.13. In the proof of the Lemma 2.1, after finding a vertex of degree 1, if we return to v and move along a path outward from v starting with e , we shall reach to a vertex of degree 1 again. Thus it follows that “**Any tree that has more than one vertex has at least two vertices of degree 1**”.

Lemma 2.2. There is a unique path between every two vertices in a tree.

Proof: Suppose on the contrary that there are more than one path between any two vertices in a given tree T . Then T has a cycle which contradicts the definition of a tree because T is acyclic. Hence the lemma is proved.

Lemma 2.3. The number of vertices is one more than the number of edges in a tree.

Equivalently, we may state this lemma as

“For any positive integer n , a tree with n vertices has $n-1$ edges”.

Proof: We shall prove the lemma by mathematical induction.

Let T be a tree with **one** vertex. Then T has no edges, that is, T has 0 edge. But $0=1-1$. Hence the lemma is true for $n=1$.

Suppose that the lemma is true for $k > 1$. We shall show that it is then true also for $k+1$. Since the lemma is true for k , the tree has k vertices and $k-1$ edges. Let T be a tree with $k+1$ vertices. Since k is $+ve$, $k+1 \geq 2$ and so T has more than one vertex. Hence, by Lemma 2.1, T has a vertex v of degree 1. Also there is another vertex w and so there is an edge e connecting v and w . Define a subgraph T' of T so that

$$V(T') = V(T) - \{v\},$$

$$E(T') = E(T) - \{e\}.$$

Then number of vertices in $T' = (k+1) - 1 = k$ and since T is circuit free and T' has been obtained on removing one edge and one vertex, it follows that T' is acyclic. Also T' is connected. Hence T' is a tree having k vertices and therefore by induction hypothesis, the number of edges in T' is $k-1$.

But then

$$\begin{aligned} \text{Number of edges in } T &= \text{number of edges in } T' + 1 \\ &= k - 1 + 1 = k. \end{aligned}$$

Thus the Lemma is true for tree having $k+1$ vertices. Hence the lemma is true by mathematical induction.

Corollary 2.4. Let $C(G)$ denote the number of components of a graph. Then a forest G on n vertices has $n - C(G)$ edges.

Proof: Apply Lemma 2.3 to each component of the forest G .

Corollary 2.5. Any graph G on n vertices has at least $n - C(G)$ edges.

Proof: If G has cycle-edges, remove them one at a time until the resulting graph G^* is acyclic. Then G^* has $n - C(G^*)$ edges by Corollary 2.4. Since we have removed only circuit, $C(G^*) = C(G)$. Thus G^* has $n - C(G)$ edges. Hence G has at least $n - C(G)$ edges.

Lemma 2.4. A graph in which there is a unique path between every pair of vertices is a tree

(This lemma is converse of Lemma 2.2).

Proof: Since there is a path between every pair of points, therefore the graph is connected. Since a path between every pair of points is unique, there does not exist any circuit because existence of circuit implies existence of distinct paths between pair of vertices. Thus the graph is connected and acyclic and so is a tree.

Lemma 2.5. (Converse of Lemma 2.3)

A connected graph G with $e = v - 1$ is a tree

First Proof: The given graph is connected and $e = v - 1$. To prove that G is a tree, it is sufficient to show that G is acyclic. Suppose on the contrary that G has a cycle. Let m be the number of vertices in this cycle. Also, we know that **number of edges in a cycle is equal to number of vertices in that cycle**. Therefore number of edges in the present case is m . Since the graph is connected, every vertex of the graph which is not in cycle must be connected to the vertices in the cycle (see Figure 2.154).

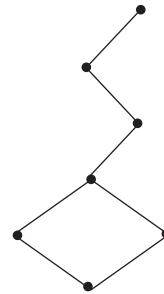


Figure 2.154

Now each edge of the graph that is not in the cycle can connect only one vertex to the vertices in the cycle. There are $v - m$ vertices that are not in the cycle. So the graph must contain at least $v - m$ edges that are not in the cycle. Thus we have $e \geq v - m + m = v$, which is a contradiction to our hypothesis. Hence there is no cycle and so the graph is a tree.

Second Proof: We shall show that a connected graph with v vertices and $v - 1$ edges is a tree. It is sufficient to show that G is acyclic. Suppose on the contrary

that G is not circuit free and has a nontrivial circuit C . If we remove one edge of C from the graph G , we obtain a graph G' which is connected (see Figure 2.155).

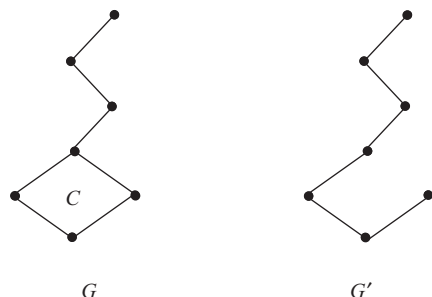


Figure 2.155

If G' still has a nontrivial circuit, we repeat the above process and remove one edge of that circuit obtaining a new connected graph. Continuing this process, we obtain a connected graph G^* which is circuit free. Hence G^* is a tree. Since no vertex has been removed, the tree G^* has v vertices. Therefore, by Lemma 2.3, G^* has $v-1$ edges. But at least one edge of G has been removed to form G^* . This means that G^* has not more than $v-1-1=v-2$ edges. Thus, we arrive at a contradiction. Hence our supposition is wrong and G has no cycle. Therefore G is connected and cycle free and so is a tree.

Lemma 2.6. A graph G with $e=v-1$, that has no circuit is a tree.

Proof: It is sufficient to show that G is connected. Suppose G is not connected and let $G', G'', \dots$ be the connected components of G . Since each of $G', G'', \dots$ is connected and has no cycle, they all are tree. Therefore, by Lemma 2.3,

$$\begin{aligned} e' &= v' - 1, \\ e'' &= v'' - 1, \\ &\dots \end{aligned}$$

where $e', e'', \dots$ are the number of edges and $v', v'', \dots$ are the number of vertices in $G', G'', \dots$, respectively. We have, on adding

$$e' + e'' + \dots = (v' - 1) + (v'' - 1) + \dots$$

Since

$$\begin{aligned} e &= e' + e'' + \dots, \\ v &= v' + v'' + \dots, \end{aligned}$$

we have

$$e < v - 1,$$

which contradicts our hypotheses. Hence G is connected. So G is connected and acyclic and is therefore a tree.

EXAMPLE 2.61

Construct a graph that has six vertices and five edges but is not a tree.

Solution. We have

$$\text{Number of vertices} = 6$$

$$\text{Number of edges} = 5$$

So the condition $e=v-1$ is satisfied. Therefore, to construct graph with six vertices and five edges that is not a tree, we should keep in mind that the graph should not be connected. The graph shown in Figure 2.156 has six vertices and five edges but is not connected. Hence it is not a tree.



Figure 2.156

Definition 2.89. A directed graph is said to be a **directed tree** if it becomes a tree when the direction of edges are ignored.

For example, the graph shown in Figure 2.157 is a directed tree.

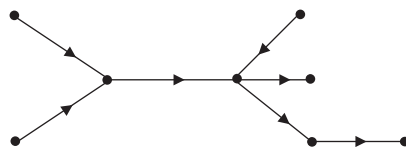


Figure 2.157

Definition 2.90. A directed tree is called a **rooted tree** if there is exactly one vertex whose incoming degree is 0 and the incoming degrees of all other vertices are 1.

The vertex with incoming degree 0 is called the **root** of the rooted tree.

A tree T with root v_0 will be denoted by (T, v_0) .

Definition 2.91. In a rooted tree, a vertex, whose outgoing degree is 0 is called a **leaf** or **terminal node**, whereas a vertex whose outgoing degree is non-zero is called a **branch node** or an **internal node**.

Definition 2.92. Let u be a branch node in a rooted tree. Then a vertex v is said to be **child (son or offspring)** of u if there is an edge from u to v . In this case, u is called **parent (father)** of v .

Definition 2.93. Two vertices in a rooted tree are said to be **siblings (brothers)** if they are both children of same parent.

Definition 2.94. A vertex v is said to be a **descendent** of a vertex u if there is a unique directed path from u to v .

In this case, u is called the **ancestor** of v .

Definition 2.95. The **level (or path length)** of a vertex u in a rooted tree is the number of edges along the unique path between u and the root.

Definition 2.96. The **height** of a rooted tree is the maximum level to any vertex of the tree.

As an example of these terms, consider the rooted tree shown in Figure 2.158:

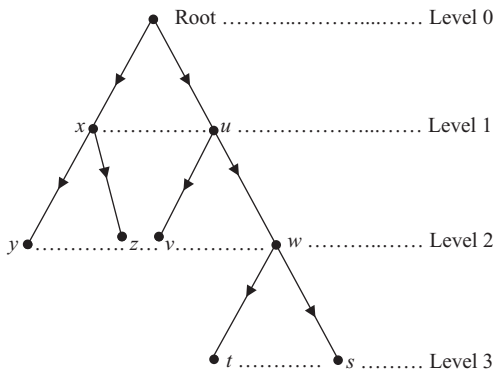


Figure 2.158

Here y is a child of x ; x is the parent of y and z . Thus y and z are siblings. The descendants of u are v , w , t and s . Levels of vertices are shown in the figure. The height of this rooted tree is 3.

Definition 2.97. Let u be a branch node in the tree $T=(V, E)$. Then the subgraph $T'=(V', E')$ of T such that the vertices set V' contains u and all of its descendants and E' contains all the edges in all directed paths emerging from u is called a **subtree** with u as the root.

Definition 2.98. Let u be a branch node. By a subtree of u , we mean a subtree that has child of u as root.

In the above example, we note that the Figure 2.159 is a subtree of T ,

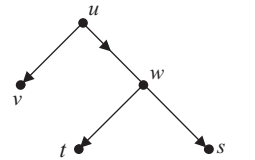


Figure 2.159

whereas the Figure 2.160 is a subtree of the branch node u .

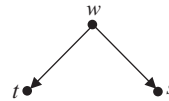


Figure 2.160

EXAMPLE 2.62

Let

$$V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$$

and let

$$E = \{(v_2, v_1), (v_2, v_3), (v_4, v_2), (v_4, v_5), (v_4, v_6), (v_6, v_7), (v_5, v_8)\}.$$

Show that (V, E) is rooted tree. Identify the root of this tree.

Solution. We note that

Incoming degree of $v_1=1$,

Incoming degree of $v_2=1$,

Incoming degree of $v_3=1$,

Incoming degree of $v_4=0$,

Incoming degree of $v_5=1$,

Incoming degree of $v_6=1$,

Incoming degree of $v_7=1$,

Incoming degree of $v_8=1$.

Since incoming degree of the vertex v_4 is 0, it follows that v_4 is root.

Further,

Outgoing degree of $v_1=0$,

Outgoing degree of $v_3=0$,

Outgoing degree of $v_7=0$,

Outgoing degree of $v_8=0$.

Therefore, v_1, v_3, v_7, v_8 are leaves. Also,

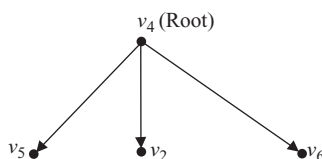
Outgoing degree of $v_2=2$,

Outgoing degree of $v_4=3$,

Outgoing degree of $v_5=1$,

Outgoing degree of $v_6=1$.

Now the root v_4 is connected to v_2, v_5 and v_6 . So, we have the following diagram:



Now v_2 is connected to v_1 and v_3 , v_5 is connected to v_8 , v_6 is connected to v_7 . Thus, we have the graph shown in Figure 2.161.

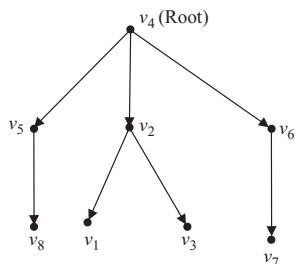


Figure 2.161

We thus have a connected acyclic graph and so (V, E) is a rooted tree with root v_4 .

EXAMPLE 2.63

Let F be the set of all female descendent of a lady v_0 . We define a relation T on F as follows: if $v_1, v_2 \in F$, then $v_1 T v_2$ if v_1 is mother of v_2 . Show that the relation T on F is a rooted tree with root v_0 . Is this relation an equivalence relation?

Solution. The graph of T is cycle free, because descendants cannot be parents of their parents. The graph is also connected. Hence T is rooted tree. We further note that

- (i) Since v_1 is not mother of itself, $(v_1, v_1) \notin T$.
- (ii) If $(v_1, v_2) \in T$, then v_1 is mother of v_2 . But then v_2 cannot be mother of v_1 . Hence $(v_2, v_1) \in T$ does not imply $(v_1, v_2) \in T$.
- (iii) If $(v_1, v_2) \in T, (v_2, v_3) \in T$, then

v_1 is mother of v_2 and v_2 is mother of v_3 .

This does not imply that v_1 is mother of v_3 , i.e., $(v_1, v_3) \notin T$. Thus T is irreflexive, asymmetric and is not transitive. Hence T is not an equivalence relation.

Consider the rooted trees T and T' shown in the Figure 2.162, where T is family tree of a man who has two sons, with the elder son having no child and younger having three children.

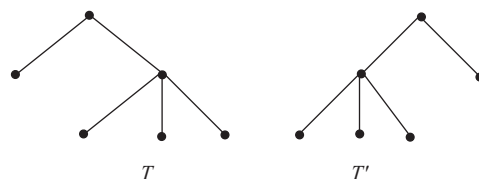


Figure 2.162

Although T' (as a graph) is isomorphic to T , it could be the family tree of another man whose elder son has three children and whose younger son has no child.

The above discussion motivates the following definitions:

Definition 2.99. A rooted tree in which the edges incident from each branch node are labelled with integers $1, 2, 3, \dots$ is called an **ordered tree**.

2.13 ISOMORPHISM OF TREES

Definition 2.100. Two ordered trees are said to be **isomorphic** if (i) there exists a one-to-one correspondence between their vertices and edges and that preserves the incident relation (ii) labels of the corresponding edges match.

In view of this definition, the ordered trees shown in Figure 2.163 are not isomorphic.

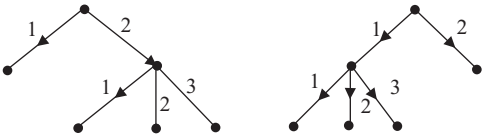


Figure 2.163

EXAMPLE 2.64

Show that the tree T_1 and T_2 shown in the diagram below are isomorphic.

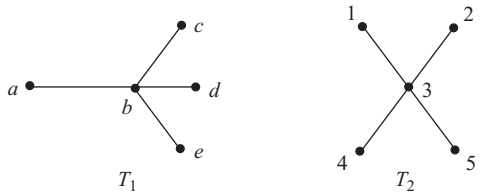


Figure 2.164

Solution. We observe that in the tree T_1 ,
 $\deg(b)=4$.

In the tree T_2 ,
 $\deg(3)=4$.

Further, $\deg(a)=\deg(1)=1$, $\deg(c)=\deg(2)$, $\deg(d)=\deg(4)=\deg(e)=1=\deg(5)$ and $\deg(b)=\deg(3)=4$. Thus we may define a function f from the vertices of T_1 to the vertices of T_2 by
 $f(a)=1$, $f(b)=3$, $f(c)=2$, $f(d)=4$, $f(e)=5$.

This is a one-to-one and onto function. Also, adjacency relation is preserved because if v_i and v_j are adjacent vertices in T_1 , then $f(v_i)$ and $f(v_j)$ are adjacent vertices in T_2 . Hence T_1 is isomorphic to T_2 .

EXAMPLE 2.65

Show that the tree T_1 and T_2 , shown in the Figure 2.165 are isomorphic

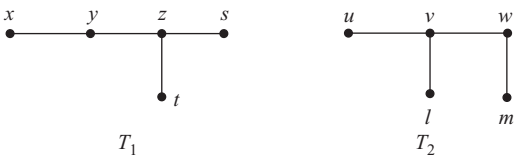


Figure 2.165

Solution. Let f be a function defined by

$$\begin{aligned} f(z) &= v, & f(y) &= w, & f(x) &= m, \\ f(s) &= u, & f(t) &= l. \end{aligned}$$

Then f is an one-one, onto mapping which preserves adjacency. Hence, T_1 and T_2 are isomorphic.

Remark 2.14. There are three non-isomorphic trees (Figure 2.166) with five vertices:

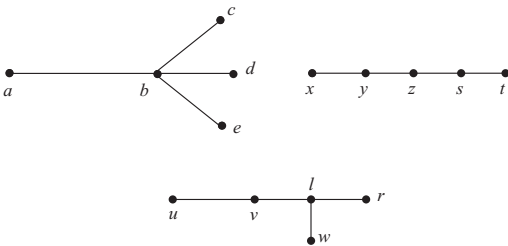


Figure 2.166

Definition 2.101. Let T_1 and T_2 be rooted trees with roots r_1 and r_2 , respectively. Then T_1 and T_2 are **isomorphic** if there exists a one-one, onto function f from the vertex set of T_1 to the vertex set of T_2 such that

- (i) Vertices v_i and v_j are adjacent in T_1 if and only if the vertices $f(v_i)$ and $f(v_j)$ are adjacent in T_2 ,
- (ii) $f(r_1)=r_2$.

The function f is then called an **isomorphism**.

EXAMPLE 2.66

Show that the trees T_1 and T_2 shown in Figure 2.167 are isomorphic.

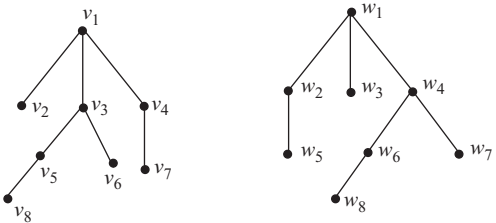


Figure 2.167

Solution. We observe that T_1 and T_2 are rooted trees. Define f : (vertex set of T_1) $\rightarrow$ (vertex set of T_2) by

$$\begin{aligned} f(v_1) &= w_1, & f(v_2) &= w_3, & f(v_3) &= w_4 \\ f(v_4) &= w_2, & f(v_5) &= w_6, & f(v_6) &= w_7 \\ f(v_7) &= w_5, & f(v_8) &= w_8. \end{aligned}$$

Then f is one-to-one and adjacency relation is preserved. Hence f is an isomorphism and so the rooted trees T_1 and T_2 are isomorphic

EXAMPLE 2.67

Show that the rooted trees shown in Figure 2.168 are not isomorphic:

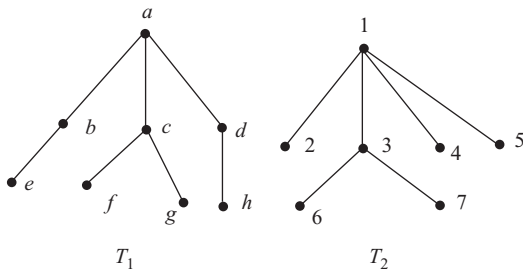


Figure 2.168

Solution. We observe that the degree of root in T_1 is 3, whereas the degree of root in T_2 is 4. Hence T_1 is not isomorphic to T_2 .

Definition 2.102. An ordered tree in which every branch node has at most n offspring is called a **n -ary tree** (or **n -tree**).

Definition 2.103. An n -ary tree is said to be **fully n -ary tree** (complete n -ary tree or **regular n -ary tree**) if every branch node has exactly n offspring.

Definition 2.104. An ordered tree in which every branch node has at most two offspring is called a **binary tree** (or **2-tree**).

Definition 2.105. A binary tree in which every branch node (internal vertex) has exactly two offspring is called a **fully binary tree**.

For example, the tree given in Figure 2.169 is a binary tree,

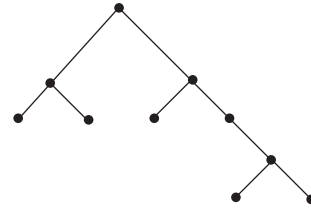


Figure 2.169

whereas the tree shown in Figure 2.170 is a fully binary tree.

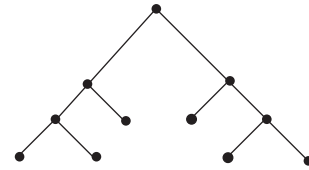


Figure 2.170

Definition 2.106. Let T_1 and T_2 be binary trees with roots r_1 and r_2 , respectively. Then T_1 and T_2 are **isomorphic** if there is a one-one, onto function f from the vertex set of T_1 to the vertex set of T_2 satisfying

- (i) Vertices v_i and v_j are adjacent in T_1 if and only if the vertices $f(v_i)$ and $f(v_j)$ are adjacent in T_2
- (ii) $f(r_1) = r_2$
- (iii) v is a left child of w in T_1 if and only if $f(v)$ is a left child of $f(w)$ in T_2
- (iv) v is a right child of w in T_1 if and only if $f(v)$ is a right child of $f(w)$ in T_2

The function f is then called an **isomorphism** between binary trees T_1 and T_2 .

EXAMPLE 2.68

Show that the trees given in Figure 2.171 are isomorphic.

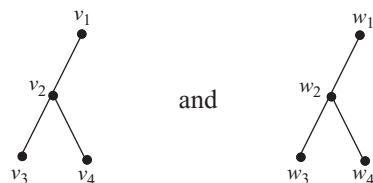


Figure 2.171

Solution. Define f by $f(v_i) = w_i$, $i = 1, 2, 3, 4$. Then f satisfies all the properties for isomorphism. Hence T_1 and T_2 are isomorphic.

EXAMPLE 2.69

Show that the trees given in Figure 2.172 are not isomorphic.

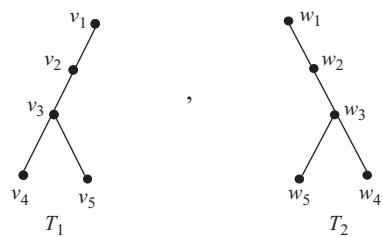


Figure 2.172

Solution. Since the root v_1 in T_1 has a left child but the root w_1 in T_2 has no left child, the binary trees are not isomorphic.

Definition 2.107. Let v be a branch node of a binary tree T . The **left subtree** of v is the binary tree whose root is the left child of v , whose vertices consists of the left child of v and all its descendents and whose edges consists of all those edges of T that connects the vertices of the left subtree together.

The **right subtree** can be defined analogously.

For example, the left subtree and the right subtree of v in the tree (Figure 2.173)

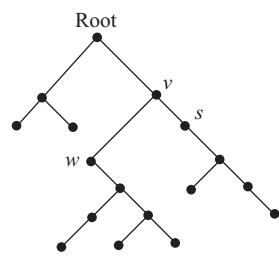


Figure 2.173

are, respectively the trees shown in Figure 2.174.

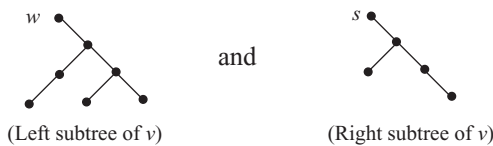


Figure 2.174

2.14 REPRESENTATION OF ALGEBRAIC EXPRESSIONS BY BINARY TREES

Binary trees are used in computer science to represent algebraic expressions involving parentheses. For example, the binary trees (Figure 2.175) represent the expressions, $a + b$, a/b and $b * (c * d)$, respectively.

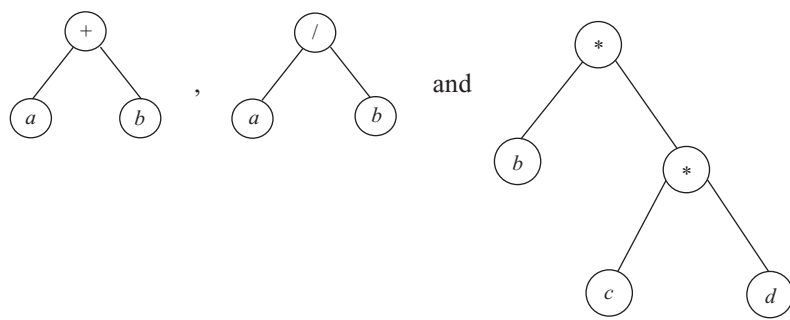


Figure 2.175

Thus, the **central operator acts as root of the tree.** (ii) $ab - (c/(d + e))$.

EXAMPLE 2.70

Draw a binary tree to represent

(i) $(2 - (3 \times x)) + ((x - 3) - (2 + x))$

Solution

(i) In this expression $+$ is the central operator. Therefore the root of tree is $+$. The binary tree is shown in Figure 2.176.

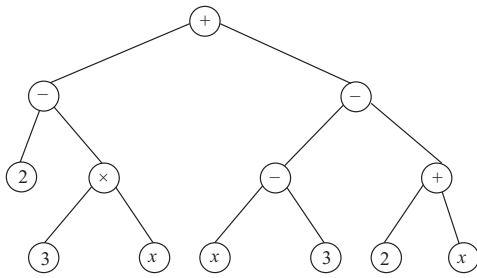


Figure 2.176

- (ii) Here the central operator is $-$. Therefore it is the root of the tree. We have the following binary tree (Figure 2.177) to represent this expression:

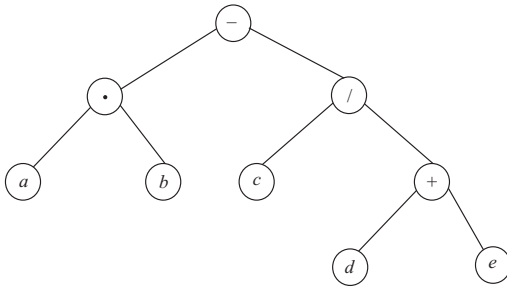


Figure 2.177

EXAMPLE 2.71

Consider a single elimination tournament with eight players in a wrestling game. How many rounds will be there to declare the champion and how many games are to be played?

Solution. In this example, the leaves of the tree shown in Figure 2.178 will represent player, the branch nodes will represent winners in the first round and the second round. Then the tree showing the tournament shall be as given below: The graph of single elimination tournament is a full binary tree. The champion shall be at the root.

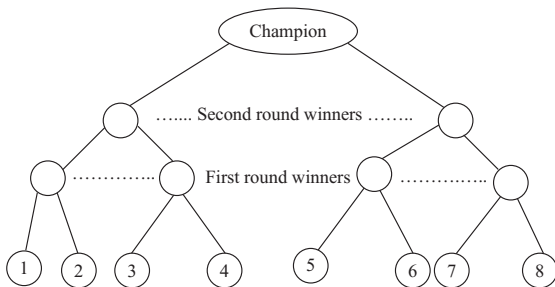


Figure 2.178

We note that there shall be three rounds to declare the champion. Let

i = Number of games played

$=$ number of internal vertices

p = Number of leaves = number of players.

Then, we see that

$$i = p - 1 = 7.$$

In general, for complete n -ary tree, we will have

$$(n - 1) i = p - 1.$$

To derive this formula we first prove the following result:

Theorem 2.23. If T is a full binary tree with i internal vertices, then T has $i + 1$ terminal vertices (leaves) and $2i + 1$ total vertices.

Proof: The vertices of T consist of the vertices that are children (of some parent) and the vertices that are not children (of any parent). There is a non-child (the root). Since there are i internal vertices, each having two children, there are $2i$ children. Thus the total number of vertices of T is $2i + 1$ and the number of terminal vertices is

$$(2i + 1) - i = i + 1.$$

This completes the Proof.

In the context of above example, we have

$$\begin{aligned} \text{Number of leaves} &= p = i + 1 \\ \Rightarrow i &= p - 1. \end{aligned}$$

Remark 2.15. In case of full n -ary tree, if i denotes the number of branch nodes, then total number of vertices of T is $ni + 1$ and the number of terminal vertices is

$$ni + 1 - i = i(n - 1) + 1.$$

If p is the number of terminal vertices, then

$$p = i(n - 1) + 1,$$

which yields

$$(n - 1) i = p - 1.$$

EXAMPLE 2.72

Find the minimum number of extension cords, each having four outlets, required to connect 22 bulbs to a single electric outlet.

Solution. Clearly, the graph of the problem is a regular quaternary tree with 22 leaves.

Let i denote the internal vertices and p denote the number of leaves, then using the expression $(n-1)i = p-1$, we have

$$(4-1)i = 22-1 \quad \text{or} \quad i = \frac{21}{3} = 7.$$

Thus seven extension cords, as shown in Figure 2.179, are required.

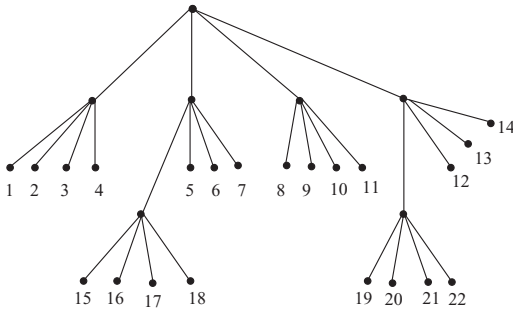


Figure 2.179

EXAMPLE 2.73

A computing machine has been given instruction which computes the sum of three numbers. How many times the addition instruction will be executed to perform the sum of 11 numbers?

Solution. In this problem, we will have a regular ternary tree with 11 leaves. Thus $n=3$, $p=11$ and so the relation $(n-1)i = p-1$ yields

$$i = \frac{p-1}{n-1} = \frac{10}{2} = 5.$$

Theorem 2.24. If a binary tree of height h has p leaves, then

$$\log_2 p \leq h \quad \text{or} \quad p \leq 2^h.$$

Proof: It is sufficient to prove that

$$p \leq 2^h, \quad (1)$$

because then taking logarithm to the base 2 of both sides, we will get

$$\log_2 p \leq h.$$

We shall prove the inequality (1) by induction on h .

If $h=0$, then the binary tree consists of a single vertex and $p=1$ and so (1) is satisfied.

Now suppose that (1) holds for a binary tree whose height is less than h . We shall prove that it holds for a tree of height h . So let T be a tree with height $h > 0$ with p leaves. We consider the following two cases:

- (a) Suppose that the root of T has only one child (Figure 2.180).

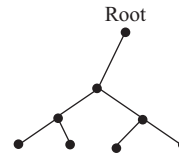


Figure 2.180

Eliminating the root and the edge incident on the root, we get a tree T' of height $h-1$ in which number of leaves is same as in T . Using induction hypothesis on T' , we get $p \leq 2^{h-1}$. Since $2^{h-1} < 2^h$, we have $p \leq 2^h$ and the inequality holds in this case.

- (b) Suppose the root of T has offspring's v and w (Figure 2.181).

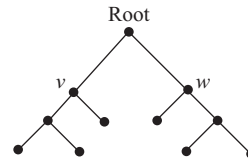


Figure 2.181

Let T_1 be the subtree rooted at v whose height is h_1 and has p_1 leaves. Similarly, let T_2 be the subtree rooted at w whose height is h_2 and has p_2 leaves. Using mathematical induction, to these trees, we have $p_1 \leq 2^{h_1}$ and $p_2 \leq 2^{h_2}$. But the leaves of T consist of the leaves of T_1 and T_2 . Hence,

$$\begin{aligned} p &= p_1 + p_2 \leq 2^{h_1} + 2^{h_2} \\ &\leq 2^{h-1} + 2^{h-1} = 2^{h-1} (1+1) = 2^h. \end{aligned}$$

Hence, by mathematical induction, the result holds.

It can be proved on the same lines that an **n -ary tree of height h has at most n^h leaves.**

EXAMPLE 2.74

Find the number of terminal vertices in a regular binary tree of height 4.

Solution. Let p be the number of terminal vertices. We are given that height (h) of the tree is 4. So, we must have $p \leq 2^4 = 16$. Since the tree is regular, we have $p=16$.

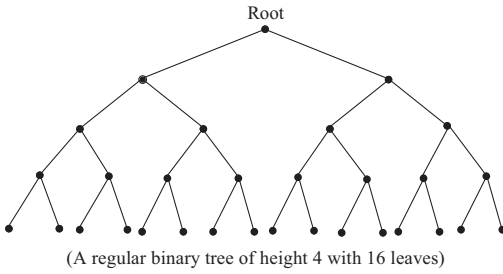


Figure 2.182

EXAMPLE 2.75

Does there exist a full binary tree with 12 internal vertices and 15 leaves?

Solution. We know that if i is the number of branch nodes in a full binary tree, then the number of leaves is $i+1$. Therefore for a tree with 12 branch nodes, the number of leaves should be 13 and not 15. Hence such tree does not exist.

EXAMPLE 2.76

Is there a binary tree with height 6 and 65 leaves?

Solution. No, since

$$p \leq 2^h = 2^6 = 64.$$

We now state some results without proof on n -ary tree.

Theorem 2.25. The number b_n of different binary trees on n vertices is given by

$$b_n = \frac{1}{n+1} \binom{2n}{n}.$$

Theorem 2.26. A regular n -ary tree of height h has at least $n+(n-1)(h-1)$ leaves.

Theorem 2.27. If in a rooted regular n -ary tree,

I = Sum of the path length of all branch nodes

E = Sum of the path length of all leaves

i = Number of branch nodes

then

$$E = (n-1)I + ni$$

Thus, in particular, for a full binary tree,

$$E = I + 2i.$$

EXAMPLE 2.77

How many binary trees are possible on three vertices?

Solution. We know that the number of different trees on n vertices is given by

$$b_n = \frac{1}{n+1} \binom{2n}{n}.$$

So we have

$$b_3 = \frac{1}{3+1} \binom{6}{3} = \frac{1}{4} \left(\frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 3 \times 2 \times 1} \right) = 5 \text{ trees}$$

and those are shown in Figure 2.183.

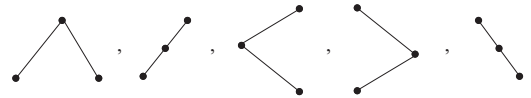


Figure 2.183

2.15 SPANNING TREE OF A GRAPH

Definition 2.108. Let G be a graph, then a subgraph of G which is a tree is called **tree of the graph**.

Definition 2.109. A **spanning tree** for a graph G is a subgraph of G that contains every vertex of G and is a tree.

Thus,

“A **spanning tree** for a graph G is a spanning subgraph of G which is a tree”.

EXAMPLE 2.78

Determine a tree and a spanning tree for the connected graph given below (Figure 2.184):

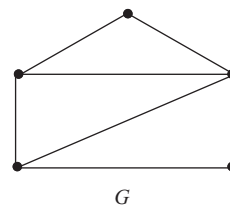


Figure 2.184

Solution. The given graph G contains circuits and we know that removal of the circuits gives a tree. So, we note that the Figure 2.185 is a tree.

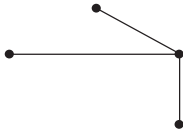


Figure 2.185

And the Figure 2.186 is a spanning tree of the graph G .

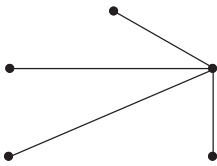


Figure 2.186

EXAMPLE 2.79

Find all spanning trees for the graph G shown below (Figure 2.187).

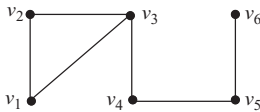


Figure 2.187

Solution. The given graph G has a circuit $v_1 v_2 v_3 v_1$. We know that removal of any edge of the circuit gives a tree. So the spanning trees of G are shown in Figure 2.188.

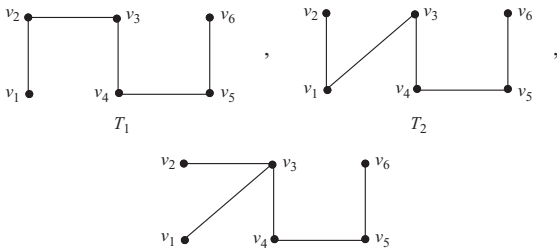


Figure 2.188

Remark 2.16. We know that a tree with n vertices has exactly $n-1$ edges. Therefore if G is a connected graph with n vertices and m edges, a spanning tree of G must have $n-1$ edges. Hence the number of edges that must be removed before a spanning tree is obtained must be

$$m - (n - 1) = m - n + 1.$$

For illustration, in EXAMPLE 2.79, $n=6$, $m=6$, so, we had to remove one edge to obtain a spanning tree.

Definition 2.110. A **branch** of a tree is an edge of the graph that is in the tree.

Definition 2.111. A **chord** (or a **link**) of a tree is an edge of the graph that is not in the tree.

It follows from the above remark that the number of chords in a tree is equal to $m-n+1$, where n is the number of vertices and m is the number of edges in the graph related to the tree.

Definition 2.112. The set of the chords of a tree is called the **complement of the tree**.

EXAMPLE 2.80

Consider the graph discussed in the above example. We note that the edge (v_2, v_3) is a branch of the tree T_1 , whereas (v_1, v_3) is a chord of the tree T_1 .

Theorem 2.28. A graph G has a spanning tree if and only if G is connected.

Proof: Suppose first that a graph G has a spanning tree T . If v and w are vertices of G , then they are also vertices in T and since T is a tree there is a path from v to w in T . This path is also a path in G . Thus every two vertices are connected in G . Hence G is connected.

Conversely, suppose that G is connected. If G is acyclic, then G is its own spanning tree and we are done. So suppose that G contains a cycle C_1 . If we remove an edge from the cycle, the subgraph of G so obtained is also connected. If it is acyclic, then it is a spanning tree and we are done. If not, it has at least one circuit, say C_2 . Removing one edge from C_2 , we get a subgraph of G which is connected. Continuing in this way, we obtain a connected circuit free subgraph T of G . Since T contains all vertices of G , it is a spanning tree of G .

Theorem 2.29. (Cayley's Formula). The number of spanning trees of the complete graph K_n , $n \geq 2$ is n^{n-2} .

(Proof of this formula is out of scope of this book)

EXAMPLE 2.81

Find all the spanning trees of K_4 (Figure 2.189).

Solution. According to Cayley's formula, K_4 has $4^{4-2} = 4^2 = 16$ different spanning trees.

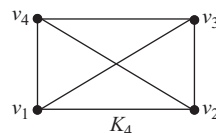


Figure 2.189

Here $n=4$, so the number of edges in any tree should be $n-1=4-1=3$. But here number of edges is equal to 6. So to get a tree, we have to remove three edges of K_4 . The 16 spanning trees so obtained are shown in Figure 2.190.

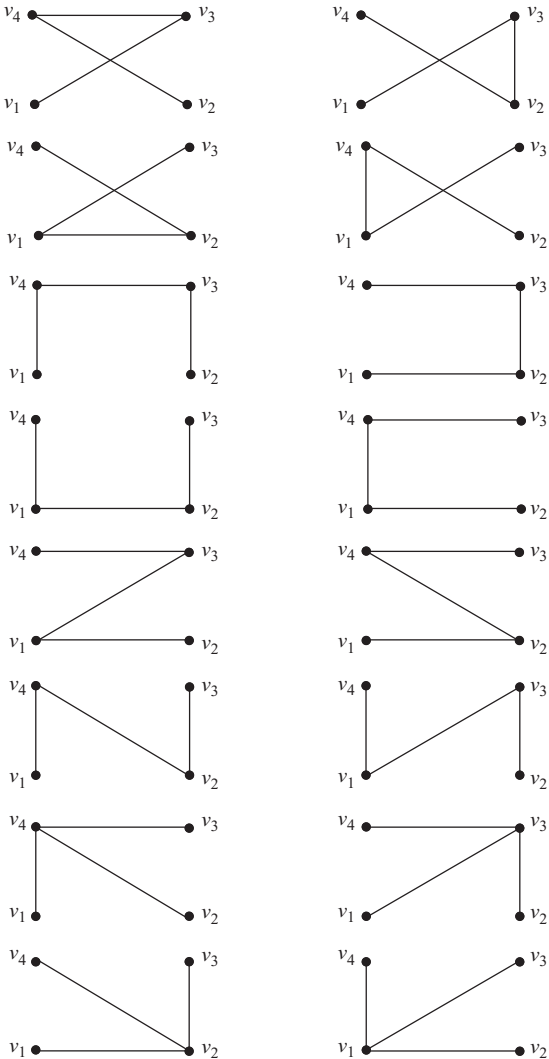


Figure 2.190

2.16 SHORTEST PATH PROBLEM

Let s and t be two vertices of a connected weighted graph G . Shortest path problem is to find a **path** from s to t whose total edge weight is minimum.

We now discuss algorithm due to $E \cdot W$. Dijkstra which efficiently solves the shortest path problem. The idea is to grow a Dijkstra tree, starting at the vertex s , by adding, at each iteration, a frontier edge, whose non-tree endpoint is as close as possible to s . The algorithm involves assigning labels to vertices.

For each tree vertex x , let $\text{dist}[x]$ denote the distance from vertex s to x and for each edge e in the given weighted graph G , let $w(e)$ be its edge-weight.

After each iteration, the vertices in the Dijkstra tree (the labelled vertices) are those to which the shortest paths from s have been found.

Priority of the Frontier Edges: Let e be a frontier edge and let its P value be given by

$$P(e) = \text{dist}[x] + w(e),$$

where x is the labelled endpoint of e and $w(e)$ is the edge-weight of e . Then,

- (i) The edge with the smallest P value is given the **highest priority**.
- (ii) The P value of this highest priority edge e gives the distance from the vertex s to the unlabelled endpoint of e .

We are now in a position to describe Dijkstra's shortest path algorithm.

2.16.1 Dijkstra's Shortest Path Algorithm

Input: A connected weighted graph G with non-negative edge-weights and a vertex s of G .

Output: A spanning tree T of G , rooted at the vertex s , whose path from s to each vertex v is a shortest path from s to v in G and a vertex labelling giving the distance from s to each vertex.

Initialize the Dijkstra tree T as vertex s .

Initialize the set of frontier edges for the tree T as empty.

$$\text{dist}[s] = 0.$$

Write label 0 on vertex s .

While Dijkstra tree T does not yet span G .

For each frontier edge e for T ,

Let x be the labelled endpoint of edge e .
 Let y be the unlabelled endpoint of edge e .
 Set $P(e) = \text{dist}[x] + w(e)$
 Let e be a frontier edge for T that has smallest P value
 Let x be the labelled endpoint of edge e
 Let y be the unlabelled endpoint of edge e
 Add edge e (and vertex y) to tree T

$\text{dist}[y]: P(e)$

Write label $\text{dist}[y]$ on vertex y .

Return Dijkstra tree T and its vertex labels.

EXAMPLE 2.82

Apply Dijkstra algorithm to find shortest path from s to each other vertex in the graph given in Figure 2.191.

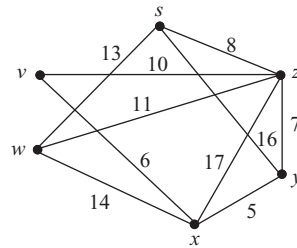
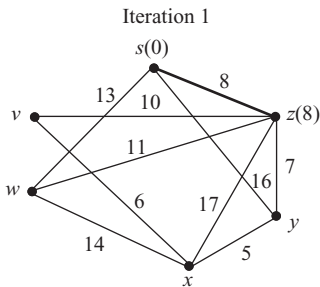


Figure 2.191

If t is the labelled endpoint of edge e , then P values are given by

$$P(e) = \text{dist}[t] + w(e),$$

where $\text{dist}[t]$ = distance from s to t and $w(e)$ is the edge weight of edge e . For each vertex v , $\text{dist}[v]$ appears in the parenthesis. Iteration tree at the end of each iteration is **drawn in dark line**.



$\text{dist}[s] = 0$ $P(sw) = 13$ (minimum)

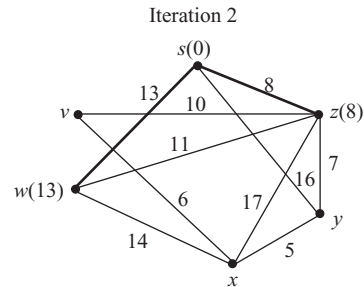
$\text{dist}[z] = 8$ $P(zy) = 8 + 7 = 15$

$P(sy) = 16$

$P(zv) = 8 + 10 = 18$

$P(zw) = 8 + 11 = 19$

$P(zx) = 8 + 17 = 25$



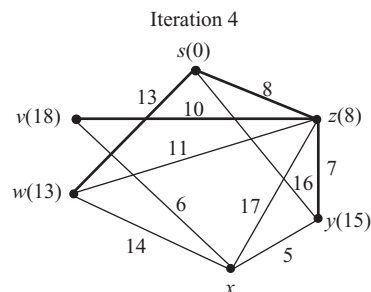
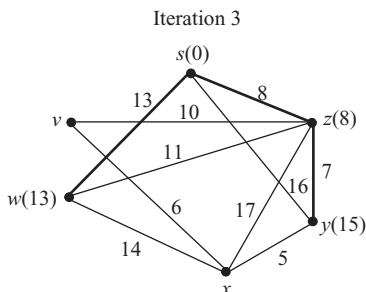
$\text{dist}[s] = 0$ $P(zy) = 8 + 7 = 15$ (minimum)

$\text{dist}[z] = 8$ $P(zx) = 8 + 17 = 25$

$\text{dist}[w] = 13$ $P(zv) = 8 + 10 = 18$

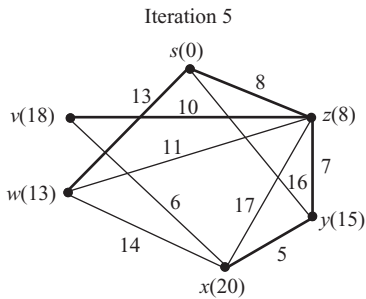
$P(sy) = 16$

$P(wx) = 13 + 14 = 27$



$\text{dist}[s]=0$ $P(zv)=18$ (minimum)
 $\text{dist}[z]=8$ $P(zx)=8+17=25$
 $\text{dist}[w]=13$ $P(wx)=13+14=27$
 $\text{dist}[y]=15$ $P(yx)=15+5=20$

$\text{Dist}[s]=0$ $P(yx)=20$ (minimum)
 $\text{Dist}[z]=8$ $P(zx)=8+17=25$
 $\text{Dist}[w]=13$ $P(vx)=18+6=24$
 $\text{Dist}[y]=15$ $P(wx)=13+14=27$
 $\text{Dist}[v]=18$



$\text{dist}[s]=0$
 $\text{dist}[z]=8$
 $\text{dist}[w]=13$
 $\text{dist}[y]=15$
 $\text{dist}[v]=18$
 $\text{dist}[x]=20$,

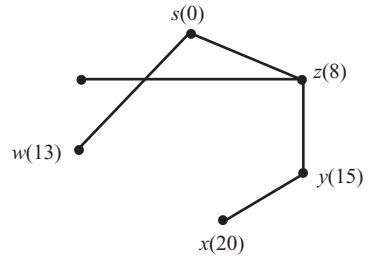


Figure 2.192

EXAMPLE 2.83

Apply Dijkstra algorithm to find the shortest path from s to t in the graph given below (Figure 2.193).

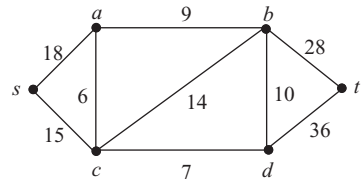
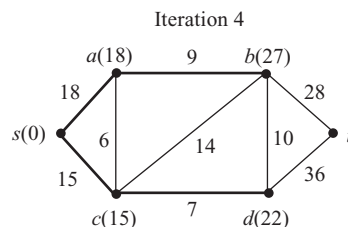
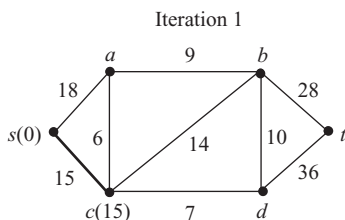


Figure 2.193

which are the required shortest paths from s to any other point. The Dijkstra tree is shown in dark lines in Figure 2.192.

Solution. Let x be the labelled endpoint of the edge e , then P values are given by $P(e)=\text{dist}[x]+w(e)$, where $\text{dist}[x]$ denotes the distance of x from s and $w(e)$ is the edge weight of e .

For each vertex v , $\text{dist}[v]$ appears in the bracket. Iteration tree at the end of each iteration is shown by dark lines:



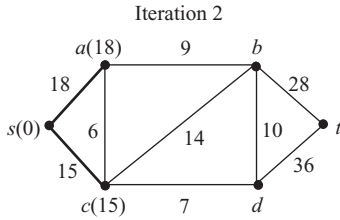
$$\text{dist } [s]=0 \quad P(s \ a)=18 \text{ (minimum)}$$

$$\text{dist } [c]=15 \quad P(c \ a)=15+6=21$$

$$P(c \ b)=15+14=29$$

$$P(c \ d)=15+7=22$$

$$P(c \ t)=\infty$$



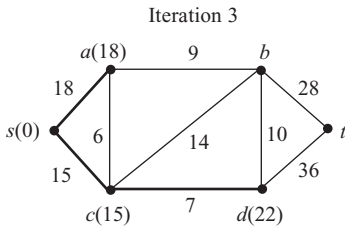
$$\text{dist } [s]=0 \quad P(a \ b)=18+9=27$$

$$\text{dist } [c]=15 \quad P(c \ b)=15+14=29$$

$$\text{dist } [a]=18 \quad P(c \ d)=15+7=22 \text{ (minimum)}$$

$$P(c \ t)=\infty$$

$$P(a \ t)=\infty$$



$$\text{dist } [s]=0 \quad P(c \ b)=15+14=29$$

$$\text{dist } [c]=15 \quad P(a \ b)=18+9=27$$

(minimum)

$$\text{dist } [a]=18 \quad P(d \ b)=22+10=32$$

$$\text{dist } [d]=22 \quad P(d \ t)=22+36=58$$

$$\text{dist } [s]=0 \quad P(d \ t)=22+36=58$$

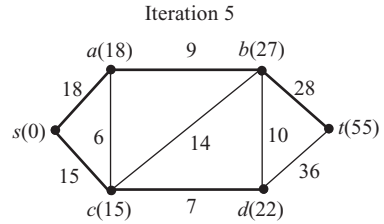
$$\text{dist } [c]=15 \quad P(b \ t)=27+28=55$$

(minimum)

$$\text{dist } [a]=18$$

$$\text{dist } [d]=22$$

$$\text{dist } [b]=27$$



$$\text{dist } [s]=0$$

$$\text{dist } [c]=15$$

$$\text{dist } [a]=18$$

$$\text{dist } [s]=0$$

$$\text{dist } [c]=15$$

$$\text{dist } [a]=18$$

$$\text{dist } [d]=22$$

$$\text{dist } [b]=27$$

$$\text{dist } [t]=55$$

Hence, the length of the shortest path (s, a, b, t) from s to t is 55 and the Dijkstra's tree is shown in Figure 2.194.

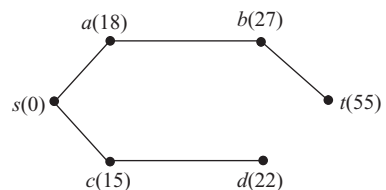


Figure 2.194

EXAMPLE 2.84

Find a shortest path from s to t and its length for the graph given below:

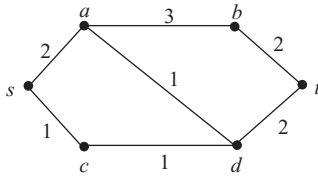


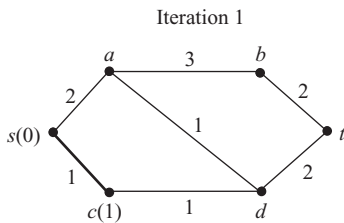
Figure 2.195

Solution. Let x be the labelled endpoint of edge e , then P values are given by

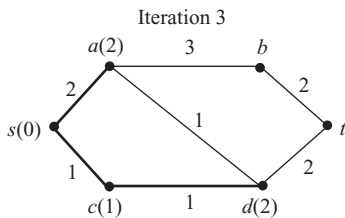
$$P(c) = \text{dist}[s] + w(e),$$

where $\text{dist}[x]$ denotes the distance from s to x and $w(e)$ is the weight of the edge e .

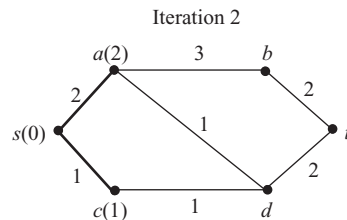
For each vertex v , $\text{dist}[v]$ appears in the bracket. Iteration tree at the end of each iteration is shown in dark lines.



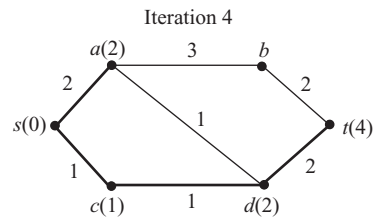
$\text{dist}[s]=0$ $P(c\ d)=2$
 $\text{dist}[c]=1$ $P(s\ a)=2$ (minimum)
 $P(s\ b)=\infty$
 $P(s\ t)=\infty$
 $P(d\ t)=\infty$
 $P(b\ t)=\infty$



$\text{dist}[s]=0$ $P(d\ t)=2$ (minimum)
 $\text{dist}[c]=1$ $P(a\ b)=5$
 $\text{dist}[a]=2$ $P(b\ t)=\infty$
 $\text{dist}[d]=2$

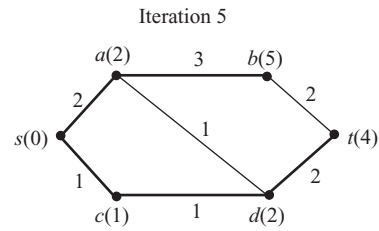


$\text{dist}[s]=0$ $P(a\ b)=2+3=5$
 $\text{dist}[c]=1$ $P(c\ d)=1+1=2$ (minimum)
 $\text{dist}[a]=2$ $P(d\ t)=\infty$
 $P(b\ t)=\infty$
 $P(a\ d)=3$



$\text{dist}[s]=0$ $P(a\ b)=5$ (minimum)
 $\text{dist}[c]=1$ $P(b\ t)=\infty$
 $\text{dist}[a]=2$
 $\text{dist}[d]=2$
 $\text{dist}[t]=4$

$\text{dist}[s] = 0$
 $\text{dist}[c] = 1$
 $\text{dist}[a] = 2$
 $\text{dist}[d] = 2$
 $\text{dist}[t] = 4$
 $\text{dist}[b] = 5$



Thus, the Dijkstra tree is

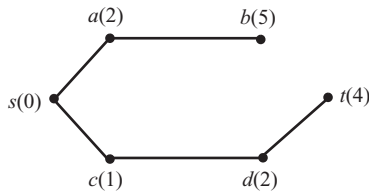


Figure 2.196

Thus the shortest path is $s c d t$ and its length is 4.

2.16.2 Shortest Path if All Edges Have Length 1

If all edges in a connected graph G have length 1, then a **shortest path** $v_1 \rightarrow v_k$ is the path that has the smallest number of edges among all paths $v_1 \rightarrow v_k$ in the given graph G .

Moore's Breadth First Search Algorithm

This method of finding shortest path in a connected graph G from a vertex s to a vertex t is used when all edges have length 1.

Input: Connected graph $G=(V, E)$, in which one vertex is denoted by s and one by t and each edge (v_i, v_j) has length 1.

Initially all vertices are unlabelled.

Output: A shortest path $s \rightarrow t$ in $G=(V, E)$.

1. Label s with 0.
2. Set $v_i = 0$.
3. Find all unlabelled vertices **adjacent to a vertex labelled** v_i .
4. Label the vertices just found with v_{i+1} .
5. If vertex t is labelled, then "**back tracking**" gives the shortest path. If k is label of t (i.e., $t = v_k$), then

Output: $v_k, v_{k-1}, \dots, v_1, 0$.

Else increase i by 1. Go to Step 3.

End Moore.

Remark 2.17. There could be several shortest paths from s to t .

EXAMPLE 2.85

Use BFS algorithm to find shortest path from s to t in the connected graph G given in Figure 2.197.

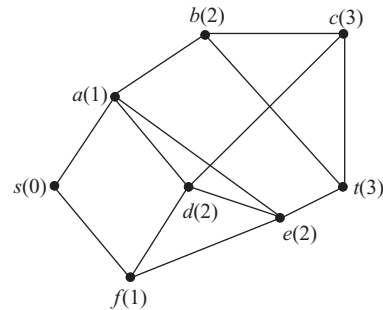


Figure 2.197

Solution. Label s with 0 and then label the adjacent vertices with 1. Thus two vertices have been labelled by 1. Now label the adjacent vertices of all vertices labelled by 1 with label 2. Thus three vertices have been labelled with 2. Label the vertices adjacent to these vertices (labelled by 2) with 3. Thus two vertices have been labelled with 3. We have reached t . Now backtracking yields the following three shortest paths:

- (i) $t(3), e(2), f(1), s(0)$, that is, $s f e t$,
- (ii) $t(3), b(2), a(1), s(0)$, that is, $s a b t$,
- (iii) $t(3), e(2), a(1), s(0)$, that is, $s a e t$.

Thus there are three possible shortest paths of length 3.

EXAMPLE 2.86

Find a shortest path $s \rightarrow t$ in the graph given in Figure 2.198 (using Moore's BFS algorithm).

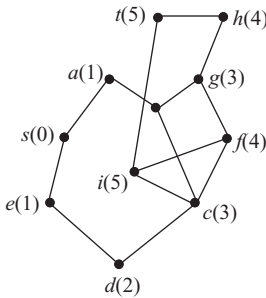


Figure 2.198

Solution. Label s with 0 and then label the adjacent vertices with 1. Thus two vertices have been marked as 1. Label the adjacent vertices of all vertices labelled as 1 with 2. Thus two vertices have been marked as 2. Now label the adjacent vertices of all vertices labelled as 2 with 3. Thus two vertices have been marked as 3. Label the adjacent vertices of all vertices labelled as 3 with 4. Thus three vertices have been marked as 4. Now there is only one vertex t to be labelled. Label it with 5. We have reached at t .

Now backtracking gives the following shortest paths:

$thgbas$ or $sabght$ with length 5,
 $ticdes$ or $sebcit$ of length 5,
 $ticbas$ or $sabcit$ of length 5.

Thus, there are three possible shortest paths of length 5.

2.17 MINIMAL SPANNING TREE

Definition 2.113. Let G be a weighted graph. A spanning tree of G with minimum weight is called **minimal spanning tree** of G .

We discuss two algorithms to find a minimal spanning tree for a weighted graph G .

2.17.1 Prim Algorithm

Prim algorithm builds a minimal spanning tree T by expanding outward in connected links from some vertex. In this algorithm, one edge and one vertex are added at each step. The edge added is the one of least weight that connects the vertices already in T with those not in T .

Input: A connected weighted graph G with n vertices.

Output: The set of edges E in a minimal spanning tree.

1. Choose a vertex v_1 of G . Let $V = \{v_1\}$ and $E = \{\}$.
2. Choose a nearest neighbour v_i of V that is adjacent to v_j , $v_j \in V$ and for which the edge (v_i, v_j) does not form a cycle with members of E . Add v_i to V and add (v_i, v_j) to E .
3. Repeat Step 2 till number of edges in T is $n-1$. Then V contains all n vertices of G and E contains the edges of a minimal spanning tree for G .

Definition 2.114. A **greedy algorithm** is an algorithm that optimizes the choice at each iteration without regard to previous choices.

For example, **Prim algorithm is a greedy algorithm.**

EXAMPLE 2.87

Find a minimal spanning tree for the graph shown in Figure 2.199.

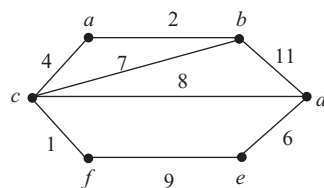


Figure 2.199

Solution. We shall use **Prim algorithm** to find the **required minimal spanning tree**. We note that

number of vertices in this connected weighted graph is 6. Therefore the tree will have five edges.

We start with any vertex, say c . The nearest neighbour of c is f and (c, f) does not form a cycle. Therefore (c, f) is the first edge selected.

Now we consider the set of vertices $V = \{c, f\}$. The vertex a is nearest neighbour to $V = \{c, f\}$ and the edge (c, a) does not form a cycle with the member of set of edges selected so far. Thus,

$$E = \{(c, f), (c, a)\} \quad \text{and} \quad V = \{c, f, a\}.$$

The vertex b is now nearest neighbour to $V = \{c, f, a\}$ and the edge (a, b) do not form a cycle with the member of $E = \{(c, f), (c, a)\}$. Thus,

$$E = \{(c, f), (c, a), (a, b)\} \quad \text{and} \quad V = \{c, f, a, b\}.$$

Now the edge (b, c) cannot be selected because it forms a cycle with the members of E . We note that d is the nearest point to $V = \{c, f, a, b\}$ and (c, d) is the edge which does not form a cycle with members of $E = \{(c, f), (c, a), (a, b)\}$. Thus we get

$$E = \{(c, f), (c, a), (a, b), (c, d)\}, \\ V = \{c, f, a, b, d\}.$$

The nearest vertex to V is now e and (d, e) is the corresponding edge. Thus,

$$E = \{(c, f), (c, a), (a, b), (c, d), (d, e)\}, \\ V = \{c, f, a, b, d, e\}$$

Since number of edges in the Prim tree is 5, the process is complete. The minimal spanning tree is shown in Figure 2.200:

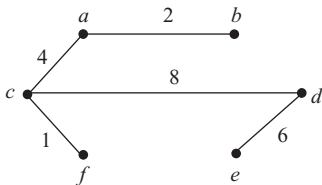


Figure 2.200

The length of the tree is $1 + 4 + 2 + 8 + 6 = 21$

EXAMPLE 2.88

Build the lowest-cost road system that will connect the cities in the graph given in Figure 2.201.

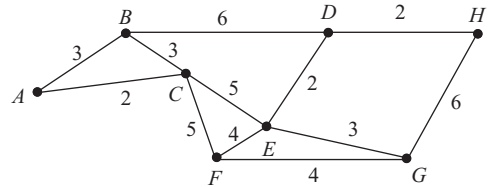


Figure 2.201

Solution. We shall use Prim algorithm to build this system. Here the number of vertices (Cities) is 8. So, the minimal spanning tree would have seven edges.

We start from any vertex, say A and have at the initial stage:

$$V = \{A\}, \quad E = \{\}.$$

The nearest neighbour of $V = \{A\}$ is C and the corresponding edge is (A, C) . So we have

$$E = \{(A, C)\}, \quad V = \{A, C\}.$$

Now B is the nearest neighbour of $V = \{A, C\}$ and edge (C, B) does not form cycle with the edges in $E = \{(A, C)\}$. Here the edge (A, B) can also be chosen. Thus, choosing arbitrarily one out of (A, B) and (C, B) , we have

$$E = \{(A, C), (C, B)\}, \quad V = \{A, C, B\}.$$

Now, E and F are both nearest neighbour to V and (C, E) or (C, F) do not form cycle with the members of $E = \{(A, C), (C, B)\}$. So we may choose any of E or F . Let us choose F and the corresponding edge (C, F) . Then,

$$E = \{(A, C), (C, B), (C, F)\}, \quad V = \{A, C, B, F\}.$$

Now E and G are nearest neighbours of $V = \{A, C, B, F\}$ and both (F, E) and (F, G) do not form cycle with the members of $E = \{(A, C), (C, B), (C, F)\}$. Therefore we may pick up any of these vertices. We pick arbitrarily E .

Then,

$$E = \{(A, C), (C, B), (C, F), (F, E)\},$$

$$V = \{A, C, B, F, E\}.$$

Now D is the nearest neighbour of $V = \{A, C, B, F, E\}$ and (E, D) is the corresponding edge to be added to the tree. So, we have

$$E = \{(A, C), (C, B), (C, F), (F, E), (E, D)\},$$

$$V = \{A, C, B, F, E, D\}.$$

Now H is the nearest neighbour and (D, H) is the corresponding edge and we have

$$E = \{(A, C), (C, B), (C, F), (F, E), (E, D), (D, H)\},$$

$$V = \{A, C, B, F, E, D, H\}.$$

Now G is the nearest neighbour and (E, G) is the corresponding edge. So, we have

$$E = \{(A, C), (C, B), (C, F), (F, E), (E, D),$$

$$(D, H), (E, G)\},$$

$$V = \{A, C, B, F, E, D, H, G\}$$

Since the number of edges in the Prim tree is 7, the process is complete. The minimal spanning tree is shown in Figure 2.202:

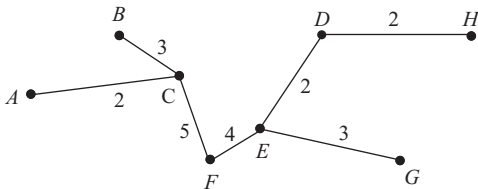


Figure 2.202

The total length of the roads is

$$2+3+5+4+2+2+3=21.$$

EXAMPLE 2.89

Using Prim algorithm, find the minimal spanning tree of the following graph (Figure 2.203):

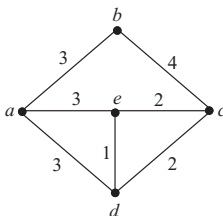


Figure 2.203

Solution. Pick up the vertex a . Then

$$E = \{\} \quad \text{and} \quad V = \{a\}.$$

The nearest neighbour of V is b, d or e and the corresponding edges are (a, b) or (a, d) or (a, e) . We choose arbitrarily (a, b) and have

$$E = \{(a, b)\}, \quad V = \{a, b\}.$$

Now d is the nearest neighbour of $V = \{a, b\}$ and the corresponding edge (a, d) does not form cycle with (a, b) . Thus we get

$$E = \{(a, b), (a, d)\}, \quad V = \{a, b, d\}.$$

Now e is the nearest neighbour of $\{a, b, d\}$ and (d, e) does not form cycle with $\{(a, b), (a, d)\}$.

Hence,

$$E = \{(a, b), (a, d), (d, e)\}, \quad V = \{a, b, d, e\}.$$

Now c is the nearest neighbour of $V = \{a, b, d, e\}$ and the corresponding edges are (e, c) , (d, c) . Thus we have, choosing (e, c) ,

$$E = \{(a, b), (a, d), (d, e), (e, c)\}, \quad V = \{a, b, d, e, c\}$$

Total weight $= 3+3+1+2=9$.

(If we choose (d, c) , then total weight is $3+3+1+2=9$).

The minimal tree is shown in Figure 2.204.

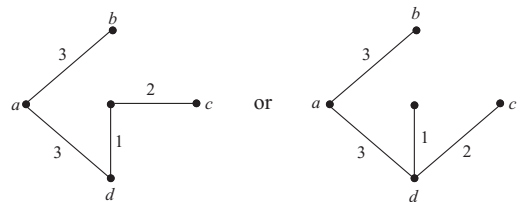


Figure 2.204

2.17.2 Kruskal's Algorithm

In Kruskal's algorithm, the edges of a weighted graph are examined one by one in order of increasing weight. At each stage, an edge with least weight out of edge-set remaining at that stage is added provided this additional edge does not create a circuit with the members of existing edge set at that stage. After $n-1$ edges have been added, these edges together with the n vertices of the connected weighted graph form a minimal tree.

Algorithm

Input: A connected weighted graph G with n vertices and the set $E = \{e_1, e_2, \dots, e_k\}$ of weighted edges of G .

Output: The set of edges in a minimal spanning tree T for G .

Step 1. Initialize T to have all vertices of G and no edges.

Step 2. Choose an edge e_1 in E of least weight. Let

$$E^* = \{e_1\}, \quad E = E - \{e_1\}.$$

Step 3. Select an edge e_i in E of least weight that does not form circuit with members of E^* . Replace

$$E^* \text{ by } E^* \cup \{e_i\} \quad \text{and} \quad E \text{ with } E - \{e_i\}.$$

Step 4. Repeat Step 3 until number of edges in E^* is equal to $n - 1$.

EXAMPLE 2.90

Use Kruskal's algorithm to determine a minimal spanning tree for the connected weighted graph G shown in Figure 2.205.

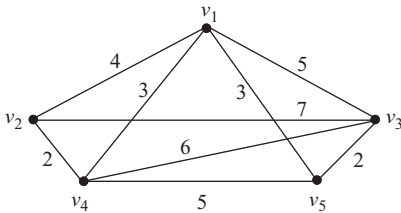


Figure 2.205

Solution. The given weighted graph has five vertices. The minimal spanning tree would, therefore, have four edges.

Let

$$E = \{(v_1, v_2), (v_1, v_4), (v_1, v_5), (v_2, v_3), (v_1, v_3), (v_2, v_4), (v_4, v_5), (v_5, v_3), (v_3, v_4)\}.$$

The edges (v_2, v_4) and (v_3, v_5) have minimum weight. We choose arbitrarily one of these, say (v_2, v_4) .

Thus

$$E^* = \{(v_2, v_4)\}, \\ E = E - \{(v_2, v_4)\}.$$

The edge (v_3, v_5) has minimum weight, so we pick it up. We have thus

$$E^* = \{(v_2, v_4), (v_3, v_5)\}, \\ E = E - \{(v_2, v_4), (v_3, v_5)\}.$$

The edges (v_1, v_4) and (v_1, v_5) have minimum weight in the remaining edge set. We pick (v_1, v_4) say, as it does not form a cycle with E^* . Thus

$$E^* = \{(v_2, v_4), (v_3, v_5), (v_1, v_4)\}, \\ E = E - \{(v_2, v_4), (v_3, v_5), (v_1, v_4)\}.$$

Now the edge (v_1, v_5) has minimum weight in $E - \{(v_1, v_4), (v_3, v_5), (v_1, v_4)\}$ and it does not form a cycle with E^* . So, we have

$$E^* = \{(v_2, v_4), (v_3, v_5), (v_1, v_4), (v_1, v_5)\}$$

and

$$E = E - \{(v_2, v_4), (v_3, v_5), (v_1, v_4), (v_1, v_5)\}.$$

Thus all the four edges have been selected. The minimal tree has the edges.

$$(v_2, v_4), (v_3, v_5), (v_1, v_4), (v_1, v_5)$$

and is shown in the Figure 2.206.

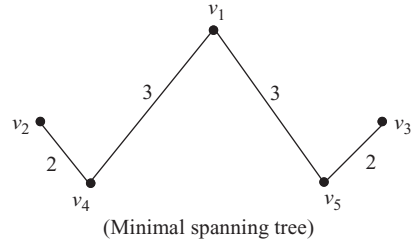


Figure 2.206

EXAMPLE 2.91

Using Kruskal's algorithm, find the minimal spanning tree for the graph given in the Figure 2.207.

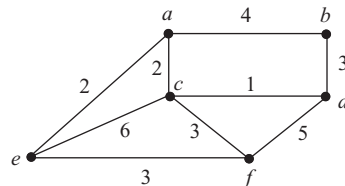


Figure 2.207

Solution. The given graph has six vertices, therefore the minimal spanning tree would have $6 - 1 = 5$ edges

Let

$$E = \{(a, b), (a, c), (b, d), (c, f), (d, f), (c, d), (a, e), (e, f), (e, c)\}.$$

Choose the edge (c, d) first because it has minimum weight. Therefore,

$$E^* = \{(c, d)\}, \quad E = E - \{(c, d)\}.$$

Now choose the edge (a, c) because it has minimum weight. So,

$$E^* = \{(c, d), (a, c)\}, \quad E = E - \{(c, d), (a, c)\}.$$

Now choose the edge (a, e) , since it has also minimum weight in $E - \{(c, d), (a, c)\}$ and it does not form cycle with E^* . So

$$E^* = \{(c, d), (a, c), (a, e)\}, \quad E = E - \{(c, d), (a, c), (a, e)\}.$$

Now choose (b, d) since it has minimum weight and it does not form circuit with E^* . Thus

$$E^* = \{(c, d), (a, c), (a, e), (b, d)\}, \\ E = E - \{(c, d), (a, c), (a, e), (b, d)\}.$$

Now choose (c, f) or (e, f) because they have equal and minimal weight. Let us choose (c, f) . Then

$$E^* = \{(c, d), (a, c), (a, e), (b, d), (c, f)\}, \\ E = E - \{(c, d), (a, c), (a, e), (b, d), (c, f)\}.$$

We have thus obtained all the five required edges. The minimal spanning tree is shown in the Figure 2.208.

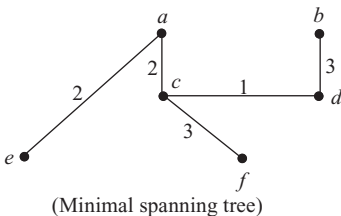


Figure 2.208

Remark 2.18. In the above example, if we had chosen (e, f) in place of (c, f) in the last step, then the minimal spanning tree would have been as shown in the Figure 2.209.

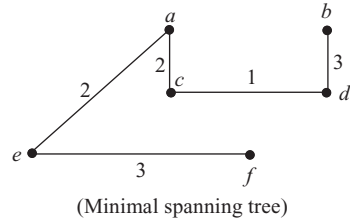


Figure 2.209

EXAMPLE 2.92

Use Kruskal's algorithm to find a shortest spanning tree for the graph G given in the Figure 2.210.

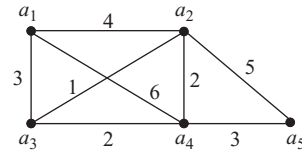


Figure 2.210

Solution. The given graph has five vertices. So the required spanning tree would have $5 - 1 = 4$ edges. Let

$$E = \{(a_1, a_2), (a_1, a_3), (a_3, a_4), (a_2, a_4), (a_2, a_5), (a_4, a_5), (a_2, a_3), (a_1, a_4)\}.$$

Choose the edge (a_2, a_3) since it has minimum weight. Then

$$E^* = \{(a_2, a_3)\}, \\ E = E - \{(a_2, a_3)\} \\ = \{(a_1, a_2), (a_1, a_3), (a_3, a_4), (a_2, a_4), (a_2, a_5), (a_4, a_5), (a_1, a_4)\}.$$

Now choose the edge (a_2, a_4) or (a_3, a_4) having minimum weight in E . Let us take (a_2, a_4) , say. Then

$$E^* = \{(a_2, a_3), (a_2, a_4)\}$$

and

$$E = E - \{(a_2, a_3), (a_2, a_4)\} \\ = \{(a_1, a_2), (a_1, a_3), (a_3, a_4), (a_2, a_5), (a_4, a_5), (a_1, a_4)\}.$$

We cannot choose (a_3, a_4) now because it will form circuit with already chosen edges $(a_2, a_3), (a_2, a_4)$. Therefore we can choose (a_1, a_3) or (a_4, a_5) . Suppose we choose (a_1, a_3) . Then

$$\begin{aligned} E^* &= \{(a_2, a_3), (a_2, a_4), (a_1, a_3)\}, \\ E &= E^* - \{(a_2, a_3), (a_2, a_4), (a_1, a_3)\} \\ &= \{(a_1, a_2), (a_3, a_4), (a_2, a_5), (a_4, a_5), (a_1, a_4)\}. \end{aligned}$$

Now choose (a_4, a_5) as it has minimal weight in E . Then

$$E^* = \{(a_2, a_3), (a_2, a_4), (a_1, a_3), (a_4, a_5)\}$$

and

$$\begin{aligned} E &= E^* - \{(a_2, a_3), (a_2, a_4), (a_1, a_3), (a_4, a_5)\} \\ &= \{(a_1, a_2), (a_3, a_4), (a_2, a_5), (a_1, a_4)\}. \end{aligned}$$

Thus we have obtained four edges and therefore the process is complete.

The Kruskal minimal spanning tree is shown in the Figure 2.211.

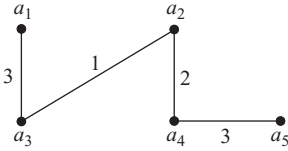


Figure 2.211

Its length is $3 + 1 + 2 + 3 = 9$.

2.18 CUT SETS

Let G be a connected graph. We know that the distance between two vertices v_1 and v_2 , denoted by $d(v_1, v_2)$, is the **length of the shortest path**.

Definition 2.115. The **diameter** of a connected graph G , denoted by $\text{diam}(G)$, is the maximum distance between any two vertices in G .

For example, in graph G shown in the Figure 2.212, we have

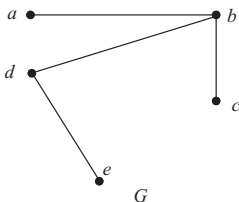


Figure 2.212

$d(a, e) = 3, d(a, c) = 2, d(b, e) = 2$ and $\text{diam}(G) = 3$.

Definition 2.116. A vertex v in a connected graph G is called a **cut point** if $G - v$ is disconnected, where $G - v$ is the graph obtained from G by deleting v and all edges containing v .

For example, in the above graph, d is a cut point.

Definition 2.117. An edge e of a connected graph G is called a **bridge** (or **cut edge**) if $G - e$ is disconnected, where $G - e$ is the graph obtained by deleting the edge e .

For example, consider the graph G shown in the Figure 2.213.

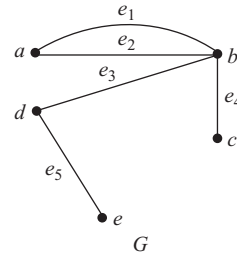


Figure 2.213

We observe that $G - e_3$ is disconnected. Hence the edge e_3 is a bridge.

Definition 2.118. A minimal set C of edges in a connected graph G is said to be a **cut set** (or **minimal edge-cut**) if the subgraph $G - C$ has more connected components than G has.

For example, in the above graph, if we delete the edge $(b, d) = e_3$, the resulting subgraph $G - e_3$ has two connected components shown in the Figure 2.214.

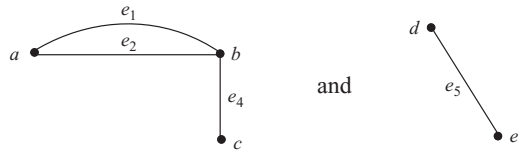


Figure 2.214

So, in this example, the cut set consists of single edge $(b, d) = e_3$, which is called cut edge or bridge.

EXAMPLE 2.93

Find a cut set for the graph given below in the Figure 2.215.

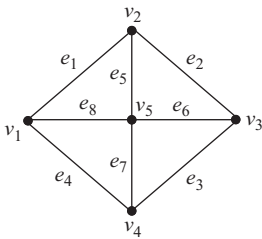


Figure 2.215

Solution. The given graph is connected. It is sufficient to reduce the graph into two connected components. To do so we have to remove the edges e_1, e_4 ,

e_5, e_6, e_7 . The two connected components are shown in the Figure 2.216.

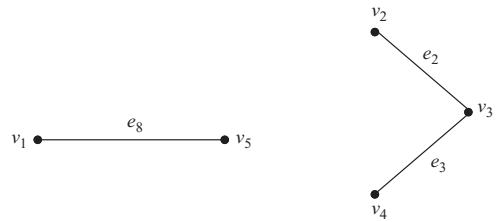
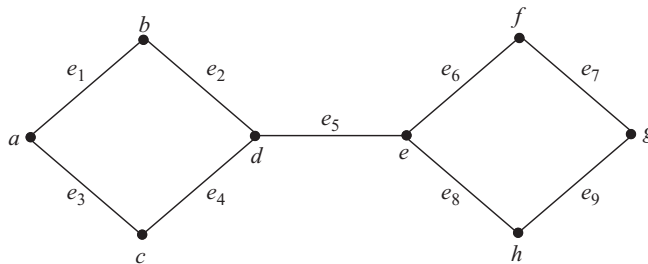


Figure 2.216

But, if we remove any proper subset of $\{e_1, e_4, e_5, e_6, e_7\}$, then there is no increase in connected components of G . Hence $\{e_1, e_4, e_5, e_6, e_7\}$ is a cut set.

EXAMPLE 2.94

Find a cut set for the graph given in the Figure 2.217.



G

Figure 2.217

The given graph G is connected. Clearly $G - e_5$ has two connected components given in the Figure 2.218.

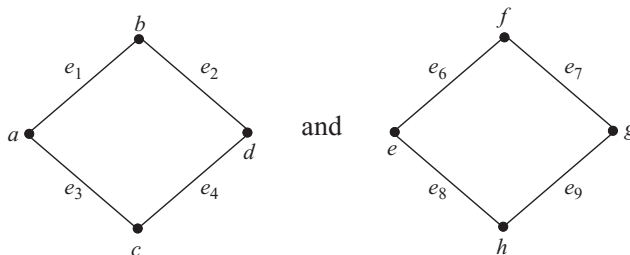


Figure 2.218

Hence a cut set for this graph is $\{e_5\}$. Some other cut sets for G are $\{e_1, e_3\}$, $\{e_3, e_2\}$, $\{e_6, e_9\}$, $\{e_6, e_8\}$.

EXAMPLE 2.95

Find a cut set for the graph represented by the Figure 2.219.

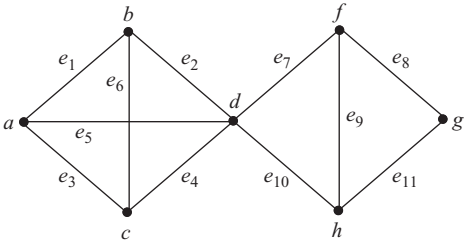


Figure 2.219

Solution. The given graph is a connected graph. We note that removal of the edges e_7 and e_{10} creates two connected components of G shown in Figure 2.220.

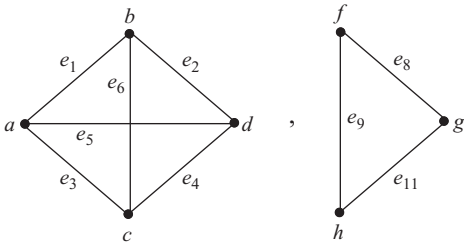


Figure 2.220

Hence the set $\{e_7, e_{10}\}$ is a cut set for the given graph G .

EXAMPLE 2.96

Find a cut set (minimal edge cut) for the following graph (Figure 2.221).

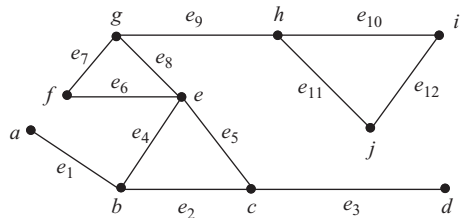


Figure 2.221

Solution. We note that $\{e_4, e_5\}$ is a cut set (minimal edge cut) for the given graph because removal of e_4 and e_5 gives rise to two components shown in the Figure 2.222.

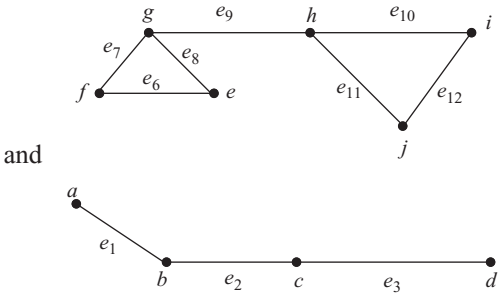


Figure 2.222

Similarly, $\{e_9\}$ forms a cut set (bridge) for this graph.

Theorem 2.30. Let G be a connected graph with n vertices. Then G is a tree if and only if every edge of G is a bridge (cut edge).

(This theorem asserts that every edge in a tree is a bridge).

Proof: Let G be a tree. Then it is connected and has $n-1$ edges (proved already). Let e be an arbitrary edge of G . Since $G-e$ has $n-2$ edges, and also we know that a graph G with n vertices has at least $n-C(G)$ edges, it follows that $n-2 \geq n-C(G-e)$. Thus, $G-e$ has at least two components. Thus removal of the edge e created more components than in the graph G . Hence e is a cut edge. This proves that every edge in a tree is a bridge.

Conversely, suppose that G is connected and every edge of G is a bridge. We have to show that G is a tree. To prove it, we have only to show that G is circuit-free. Suppose on the contrary that there exists a cycle between two points x and y in G (Figure 2.223). Then any edge on this cycle is not a cut edge which contradicts the fact that every edge of G is a cut edge. Hence G has no cycle. Thus G is connected and acyclic and so is a tree.



Figure 2.223

2.18.1 Relation Between Spanning Trees, Circuits and Cut Sets

A spanning tree contains a unique path between any two vertices in the graph. Therefore, addition of a chord to the spanning tree yields a subgraph that contains exactly one circuit. For example, consider the graph G shown below in the Figure 2.224.

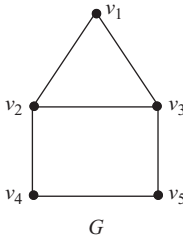


Figure 2.224

For this graph, the Figure 2.225 is a spanning tree.

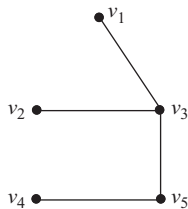


Figure 2.225 (Spanning tree)

The chords of this tree are (v_1, v_2) and (v_2, v_4) . If we add (v_1, v_2) to this spanning tree, we get a circuit $v_1 v_2 v_3 v_1$. Similarly addition of (v_2, v_4) gives one more circuit $v_2 v_3 v_5 v_4 v_2$. If there are v vertices and e edges in a graph, then there are $e - v + 1$ chords in a spanning tree. Therefore, if we add all the chords to the spanning tree, there will be $e - v + 1$ circuits in the graph.

Definition 2.119. Let v be the number of vertices and e be the number of edges in a graph G .

Then the set of $e - v + 1$ circuits obtained by adding $e - v + 1$ chords to a spanning tree of G is called the **fundamental system of circuits relative to the spanning tree**.

A circuit in the fundamental system is called a **fundamental circuit**.

For example, $\{v_1, v_2, v_3, v_1\}$ is the fundamental circuit corresponding to the chord (v_1, v_2) .

On the other hand, since each branch of a tree is cut edge, removal of any branch from a spanning tree breaks the spanning tree into two trees. For example, if we remove (v_1, v_3) from the above figured spanning tree, the resulting components are shown in the Figure 2.226.

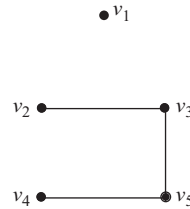


Figure 2.226

Thus, to every branch in a spanning tree, there is a corresponding cut set. But, in a spanning tree, there are $v - 1$ branches. Therefore, there are $v - 1$ cut sets corresponding to $v - 1$ branches.

Definition 2.120. The set of $v - 1$ cut sets corresponding to $v - 1$ branches in a spanning tree of a graph with v vertices is called the **fundamental system of cut sets relative to the spanning tree**.

A cut set in the fundamental system of cut sets is called a **fundamental cut set**.

For example, the fundamental cut sets in the spanning tree (figured above) are

$$\{(v_1, v_2), (v_1, v_3)\}, \{(v_1, v_3), (v_2, v_3), (v_3, v_4)\}, \\ \{(v_3, v_5), (v_4, v_5)\}, \{(v_2, v_4), (v_4, v_5)\}.$$

Theorem 2.31. A circuit and the complement of any spanning tree must have at least one edge in common.

Proof: We recall that the set of all chords of a tree is called the complement of the tree. Suppose on the contrary that a circuit has no common edge with the complement of a spanning tree. This means the circuit is wholly contained in the spanning tree. This contradicts the fact that a tree is acyclic (circuit-free). Hence a circuit has at least one edge in common with complement of a spanning tree.

Theorem 2.32. A cut set and any spanning tree must have at least one edge in common.

Proof: Suppose on the contrary that there is a cut set which does not have a common edge with a spanning tree. Then removal of cut set has no effect on the tree, that is, the cut set will not separate the graph into two components. But this contradicts the definition of a cut set. Hence the result.

Theorem 2.33. Every circuit has an even number of edges in common with every cut set.

Proof: We know that a cut set divides the vertices of the graph into two subsets each being set of vertices in one of the two components. Therefore a path connecting two vertices in one subset must traverse the edges in the cut set an even number of times. Since a circuit is a path from some vertex to itself, it has an even number of edges in common with every cut set.

2.19 TREE SEARCHING

Let T be a binary tree of height $h \geq 1$ and root v . Since $h \geq 1$, v has at least one child : v_L and/or v_R . Now v_L and v_R are the roots of the left and right subtrees of v called T_L and T_R , respectively.

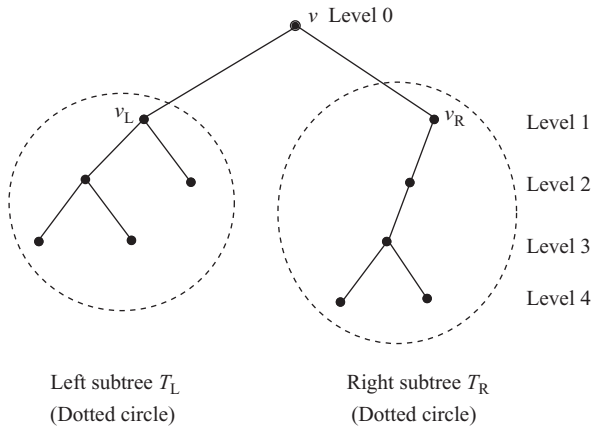


Figure 2.227

Definition 2.121. Performing appropriate tasks at a vertex is called **visiting the vertex**.

Definition 2.122. The process of visiting each

vertex of a tree in some specified order is called **searching the tree** or **walking** or **traversing the tree**.

We now discuss methods of searching a tree.

1. Pre-order Search Method

Input: The root v of a binary tree.

Output: Vertices of a binary tree using preorder traversal

1. **Visit** v
2. If v_L (left child of v) exists, then apply the algorithm to $(T(v_L), v_L)$
3. If v_R (right child of v) exists, then apply this algorithm to $(T(v_R), v_R)$

End of algorithm pre-order.

In other words, pre-order search of a tree consists of the following steps:

Step 1. Visit the root.

Step 2. Search the left subtree if it exists.

Step 3. Search the right subtree if it exists.

EXAMPLE 2.97

Find the order of the vertices of the following tree T processed using pre-order traversal.

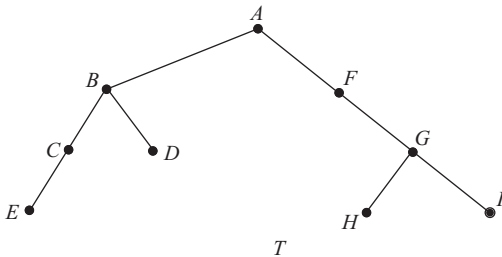


Figure 2.228

Solution. The root of the given tree is A . It has two children B and F . The left subtree and right subtree are shown in Figure 2.229.

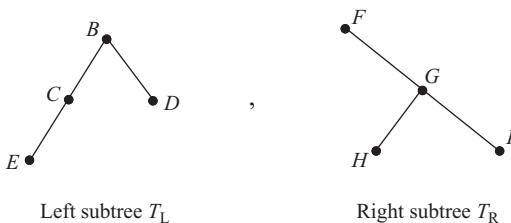


Figure 2.229

If we apply pre-order algorithm to the given tree T , we visit the root and print A . Using pre-order to left subtree, we visit the root B and print B and then search T_L printing C and E and then D . Up to this point, the search has yielded the string $A B C E D$. Thus the search of T_L is complete. We now go to subtree T_R . Using the same procedure, we get the string $F G H I$. Thus the complete search of the tree T is

$A B C E D F G H I$.

EXAMPLE 2.98

Find binary tree representation of the expression

$$(a-b) \times (c+(d \div e))$$

and represent the expression in string form using pre-order traversal.

Solution. In the given expression, $\times$ is the central operator and therefore shall be the root of the binary tree. Then the operator $-$ acts as v_L and the operator $+$ acts as v_R . Thus the tree representation of the given expression is as shown in the Figure 2.230.

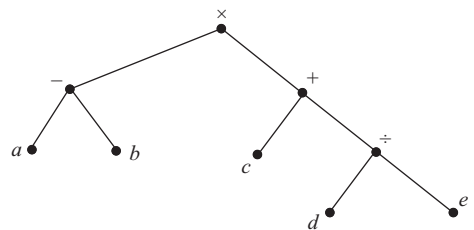


Figure 2.230

The result of the pre-order traversal to this binary tree is the string

$$\times - a b + c \div d e.$$

This form of the expression is called **prefix form** or **polish form** of the expression

$$(a-b) \times (c+(d \div e)).$$

In a polish form, the variables $a, b, c, \dots$ are called **operands** and $-, +, \times, \div$ are called **operators**. We observe that, **in polish form, the operands follow the operator.**

EXAMPLE 2.99

Represent the expression $(A+B) * (C-D)$ as a binary tree and write the prefix form of the expression.

Solution. Here $*$ is the central operator. Further $+$ and $-$ operators are v_L and v_R . Hence the binary tree for the given expression is as shown in the Figure 2.231.

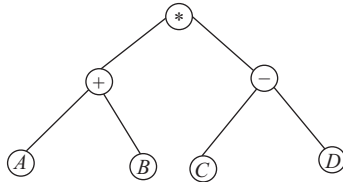


Figure 2.231

Using pre-order traversal, the prefix expression for it is

$$* + A B - C D.$$

2.19.1 Procedure to Evaluate an Expression Given in Polish Form

To find the value of a polish form, we proceed as follows:

Move from left to right until we find a string of the form $K x y$, where K is operator and x, y are operands.

Evaluate $x K y$ and substitute the answer for the string $K x y$. Continue this procedure until only one number remains.

EXAMPLE 2.100

Find parenthesized form of the polish expression

$$- + A B C.$$

Solution. The parenthesized form of the given polish expression is derived as follows:

$$\begin{aligned} &-(A+B) C, \\ &(A+B) - C. \end{aligned}$$

The corresponding binary tree is shown in the Figure 2.232.

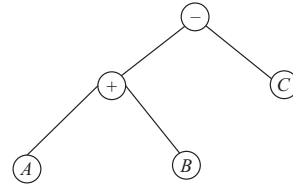


Figure 2.232

EXAMPLE 2.101

Evaluate the polish form

$$\times - 64 + 5 \div 22$$

Solution. We have the following steps in this regard:

1. $\times (6-4) + 5 \div 22$
2. $\times 2 + 5 \div 22$
3. $\times 2 + 5 (2 \div 2)$
4. $\times 2 + 51$
5. $\times 2 (5+1)$
6. $\times 26$
7. 2×6
8. 12, which is the required value of the expression.

2. Post-order Search Method

Algorithm

Step 1. Search the left subtree if it exists

Step 2. Search the right subtree if it exists

Step 3. Visit the root

End of algorithm

EXAMPLE 2.102

Represent the expression

$$(A+B) * (C-D)$$

as a binary tree and write the result of post-order search for that tree.

Solution. The binary tree expression (as shown earlier) of the given algebraic expression is given in Figure 2.233.

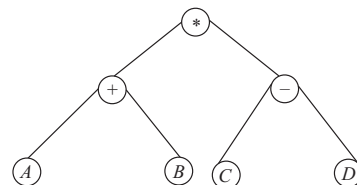


Figure 2.233

The result of post-order search of this tree is

$$A B + C D - *$$

This form of the expression is called **postfix form** of the expression or **reverse polish form** of the expression.

In postfix form, the operator follows its operands.

EXAMPLE 2.103

Find the parenthesized form of the postfix form

$$A B C ** C D E + / - .$$

Solution. We have

1. $A B C ** C D E + / -$
2. $A (B * C) * C (D + E) / -$
3. $(A * (B * C)) (C / (D + E)) -$
4. $(A * (B * C)) - (C / (D + E))$.

The corresponding binary tree is shown in the Figure 2.234.

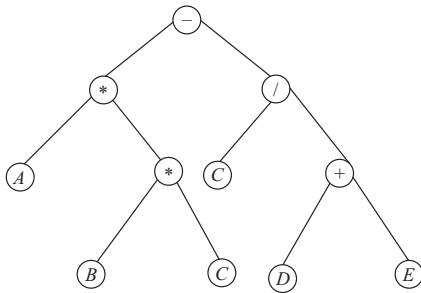


Figure 2.234

EXAMPLE 2.104

Evaluate the postfix form

$$21 - 342 \div + \times .$$

Solution. We have

$$\begin{aligned} 21 - 342 \div + \times &= (2 - 1) 342 \div + \times \\ &= 13 (4 \div 2) + \times \\ &= 13 \div + \times \\ &= 1 (3 \div 2) \times \\ &= 1 \times \\ &= 1 \times 5 = 5. \end{aligned}$$

2.20 TRANSPORT NETWORKS

An important application of weighted directed graph is to model transport networks like oil pipeline, water supply pipeline, communication network, electric power distribution system and highway system. The purpose of a network is to implement the flow of oil, water, electricity, messages, traffic, etc and we desire that the flow through the network should be largest possible value. Such a flow will be called maximum flow.

Definition 2.123. A *transport network* or simply a *network* is a simple weighted directed graph with the following properties:

- (i) There is a designated vertex (node), called the **source**, that has no incoming edge.
- (ii) There is a designated vertex, called the **sink**, that has no outgoing edge.
- (iii) There is a non-negative weight c_{ij} on the directed edge (i, j) , called the **capacity** of the edge (i, j) .

Since the graph is simple, it follows that if the edge (i, j) is in the network, then (j, i) is not there.

Definition 2.124. A *flow* in a network is a function that assigns to each (i, j) of the network a non-negative number f_{ij} such that

- (i) $0 \leq f_{ij} \leq c_{ij}$, where c_{ij} is the capacity of the edge (i, j)
- (ii) For each vertex j , which is neither the source nor the sink,

$$\sum_i f_{ij} = \sum_k f_{jk}.$$

The non-negative number f_{ij} is called the **flow** in the edge (i, j) . For any vertex (distinct from source and sink) j , the sum $\sum_i f_{ij}$ is called the **flow into** j and the sum $\sum_k f_{jk}$ is called the **flow out of** j . Thus the condition (ii) implies that material cannot accumulate, be created, dissipate or lost at any vertex other than the source or the sink. Hence the equation

$\sum_i f_{ij} = \sum_k f_{jk}$ is called **conservation of flow**. As a consequence of conservation, it follows that the **sum of flows leaving the source must be equal to the sum of the flows entering the sink**. This sum is called the

value of the flow and is denoted by value (F). A flow F in a network is represented by labelling each edge (i, j) with the pair (c_{ij}, f_{ij}) .

For example consider the oil pipeline network shown in Figure 2.235.

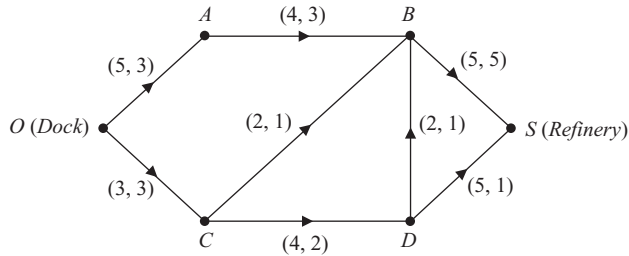


Figure 2.235

In this network, the crude oil unloaded at the dock O (source) is being pumped to the refinery S (sink). The capacities of the flow are

$$c_{OA} = 5, \quad c_{AB} = 4, \quad c_{BS} = 5, \quad c_{OC} = 3, \\ c_{CB} = 2, \quad c_{CD} = 4, \quad c_{DB} = 2, \quad c_{DS} = 5.$$

The flows in the edges are

$$f_{OA} = 3, \quad f_{AB} = 3, \quad f_{BS} = 5, \quad f_{OC} = 3, \\ f_{CB} = 1, \quad f_{CD} = 2, \quad f_{DB} = 1, \quad C_{DS} = 1.$$

We note that

$$\text{Flow into the vertex } B = 3 + 1 + 1 = 5$$

Flow out of the vertex $B = 5$.

Therefore, **conservation of flow** is satisfied. Further, value (F) = 3 + 3 = 5 + 1 = 6.

EXAMPLE 2.105

Consider the water pipeline network shown in Figure 2.236 for two towns A and B in which water is supplied from four borewells w_1, w_2, w_3, w_4 and a, b, c, d represent intermediate pumping stations. Form an equivalent network with unique source and unique sink and determine the value of the flow.

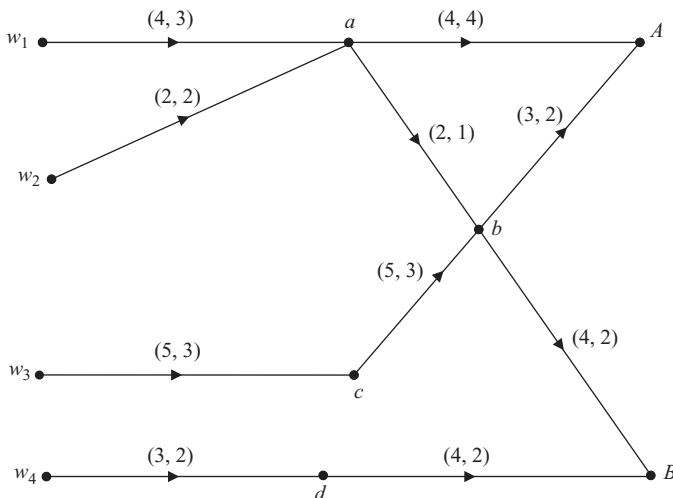


Figure 2.236

Solution. To obtain the required equivalent network with designated source and designated sink, we tie together the given sources w_1, w_2, w_3 and w_4 into a

super source O and the towns A and B into a *super sink* S as shown in Figure 2.237.

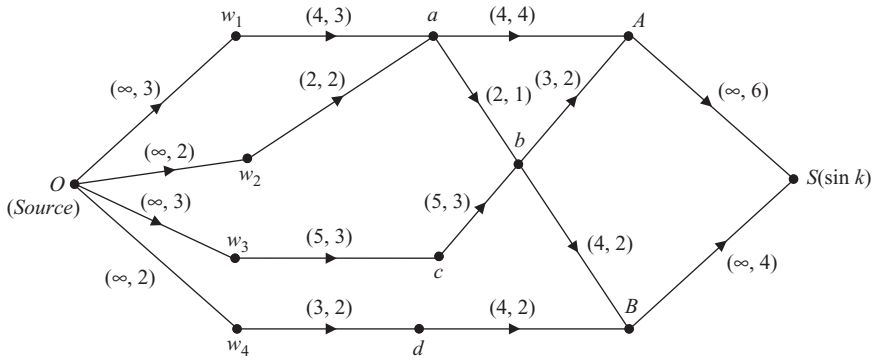


Figure 2.237

The value of the flow is

$$\text{Value } (F) = 3 + 2 + 3 + 2 = 6 + 4 = 10.$$

Definition 2.125. A *maximal flow* in a network N is a flow with maximum value.

In general, there will be several flows having the same maximum value. For example, consider the network shown in Figure 2.238.

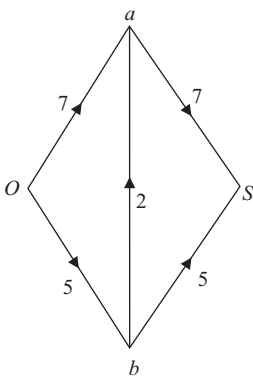


Figure 2.238

Consider the following two flows for this network:

The value of the flow in Figure 2.239(a) is 10 and three of the five edges in this network attain their maximum capacity. But for the same network, the value of the flow in Figure 2.239(b) is 12 and four

out of five edges are carrying maximum capacity. Therefore, the flow function in the second case is better than the first one.

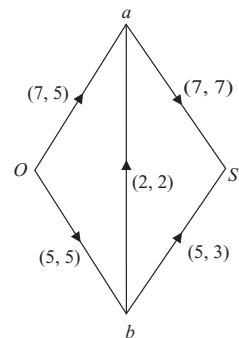


Figure 2.239(a)

and

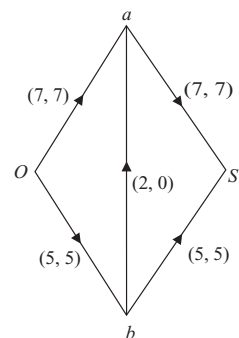


Figure 2.239(b)

EXAMPLE 2.106

Find the maximum possible increase in the following network (Figure 2.240).

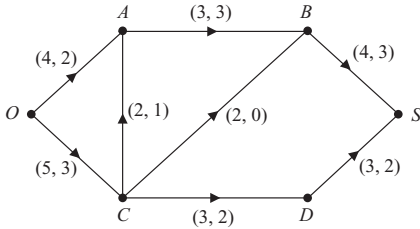


Figure 2.240

Solution. We cannot increase the flow in the path $OABS$ because the capacity and the flow of the edge AB are equal. But if increase flow in the path $OCDS$ by 1, we get the network shown in Figure 2.241.

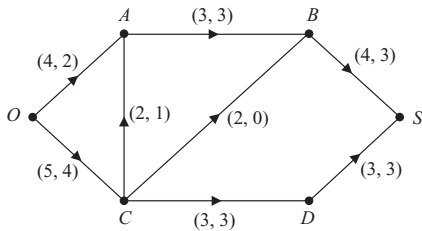


Figure 2.241

Further, if we increase the flow in the path $OCBS$ by 1, we get the network shown in Figure 2.242.

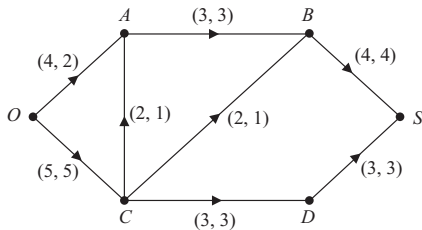


Figure 2.242

No further increase in flow is possible since the capacities of all the edges leading to the sink have been exhausted. The value of the flow is 7.

EXAMPLE 2.107

Find a maximal flow in the network shown in Figure 2.243.

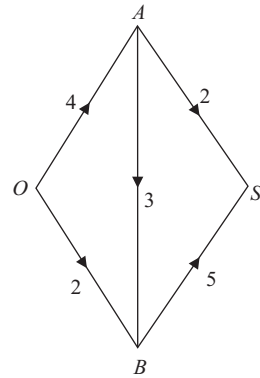


Figure 2.243

Solution. The capacity of each edge is shown in Figure 2.243. Let the initial flow be 0 in each edge. Thus we have the network as shown in Figure 2.244(a).

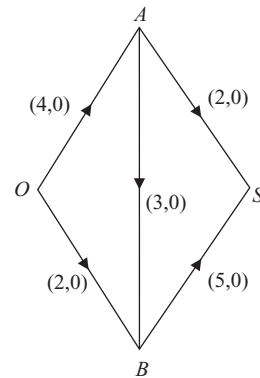


Figure 2.244(a)

Increase the flow by 2 in the path OAS to get the network shown in the Figure 2.244(b).

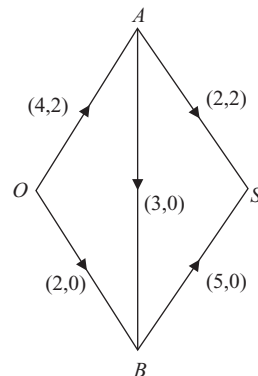


Figure 2.244(b)

Now increase the flow by 2 in the path OBS to get the network shown in Figure 2.244(c).

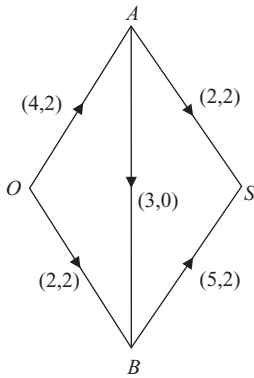


Figure 2.244(c)

Now increase the flow by 2 in the path $OABS$ to get the network shown in Figure 2.244(d).

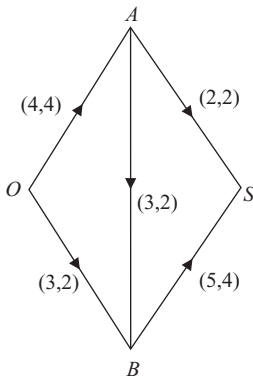


Figure 2.244(d)

Any further increase in the flow is not possible. Thus maximal flow in the network has been reached. Thus, value $(F)=2+4=6$.

EXAMPLE 2.108

Find a maximal flow in the network shown in Figure 2.245.

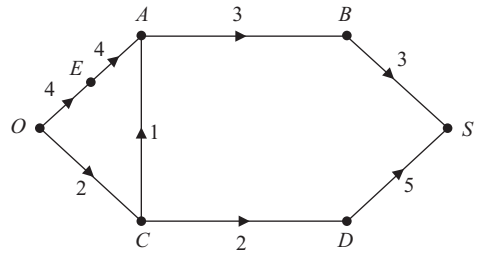


Figure 2.245

Solution. The initial labelling yields the network shown in Figure 2.246(a).

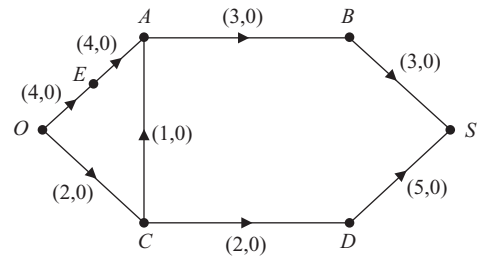


Figure 2.246(a)

Increase the flow by 2 in the path $OCDS$ by to get the network shown in Figure 2.246(b).

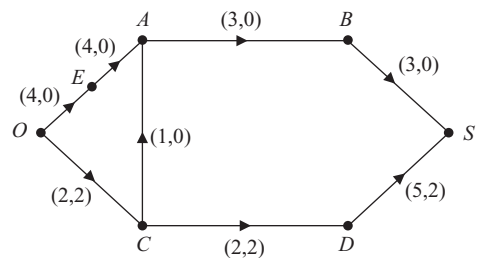


Figure 2.246(b)

Now increase the flow by 3 in the path $OEABS$ to get the network shown in Figure 2.246(c).

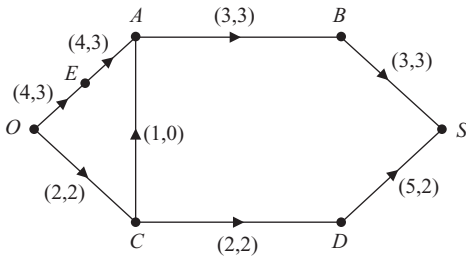


Figure 2.246(c)

No further increase in flow is possible since the capacities in the edges (A, B) and (C, D) has been exhausted. Hence value $(F)=5$.

Definition 2.126. Let N be a network. A **cut** in N is a partition $(P, \bar{P})$ of the set of vertices in N such that the source $O \in P$ and the sink $S \in \bar{P}$.

Thus, a cut $(P, \bar{P})$ in a network N is a set K of edges $(v, w), v \in P, w \in \bar{P}$ such that every path from source to sink contains at least one edge from K . In fact, a cut does cut a directed graph into two pieces, one containing the source and the other containing the sink. **Nothing can flow from source to sink if edges of a cut are removed.** We indicate a cut by drawing a dashed line to partition the vertex set. As an illustration, consider the network of EXAMPLE 2.106. The dashed line divides the vertex set into the sets $P = \{O, A\}$ and $\bar{P} = \{C, B, D, S\}$.

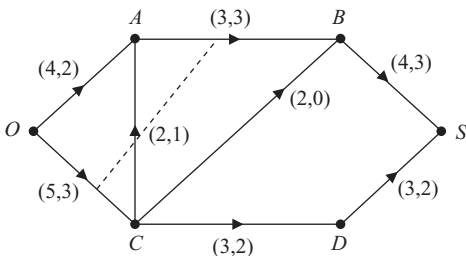


Figure 2.247

Definition 2.127. The **capacity of a cut** $K = (P, \bar{P})$ is the sum of the capacities of all edges in K and is denoted by $C(K)$ or $C(P, \bar{P})$.

For example, the capacity of the cut in Figure 2.247 is

$$C(P, \bar{P}) = C_{OC} + C_{AC} + C_{AB} = 5 + 2 + 3 = 10.$$

EXAMPLE 2.109

Find the capacity of the cut, shown by dashed line, in the network shown in Figure 2.248.

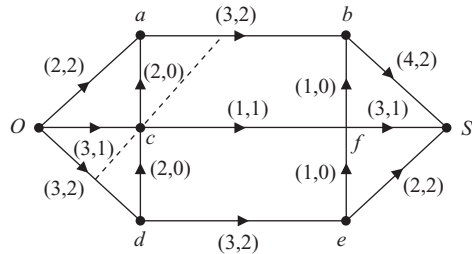


Figure 2.248

Solution. The cut $(P, \bar{P})$ is given by

$$P = \{O, a, c\}, \quad \bar{P} = \{d, e, f, b, S\}.$$

Therefore the capacity of the given cut is

$$C(P, \bar{P}) = c_{Od} + c_{cf} + C_{ab} = 3 + 1 + 3 = 7.$$

Definition 2.128. A **minimal cut** is a cut having minimum capacity.

Theorem 2.34. (The Max Flow Min Cut Theorem) A maximum flow F in a network has value equal to the capacity of a minimum cut of the network.

Proof: Let F be any flow and K be any cut in a network. Since all parts of F must pass through the edges of the cut K and since $C(K)$ is the maximum amount that can pass through the edges of K , it follows that **value** $(F) \leq C(K)$. The equality will hold if the flow F uses the full capacity of all edges in the cut K . Thus equality holds if the flow is maximum. Further, K must be a minimum capacity cut since every cut must have capacity at least equal to **value** (F) . This completes the proof of the theorem.

EXAMPLE 2.110

In the network shown in Figure 2.249, find a maximum flow, give its value and prove that it is

maximum by appealing to the max flow min cut theorem.

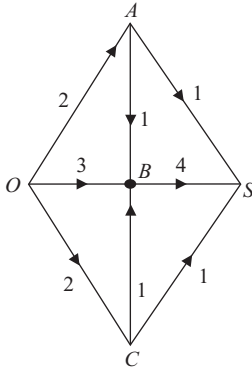


Figure 2.249

Now increase the flow by 1 in the path OCS to get the network shown in Figure 2.250(c).

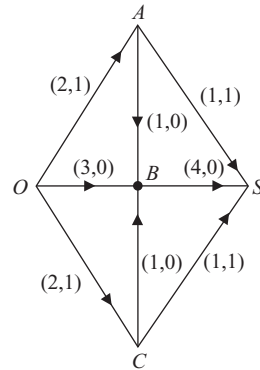


Figure 2.250(c)

Solution. The initial labelling yields the following network shown in Figure 2.250(a):

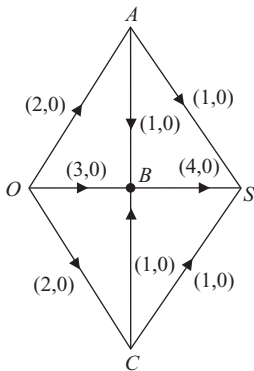


Figure 2.250(a)

Further, increase in flow by 1 in the path $OABS$ to get the network of Figure 2.250(d).

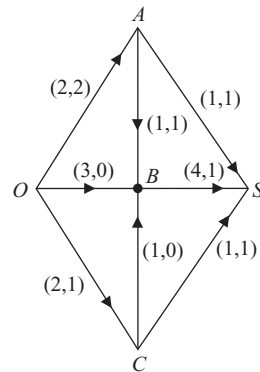


Figure 2.250(d)

Increase the flow by 1 in the path OAS to get the network shown in Figure 2.250(b).

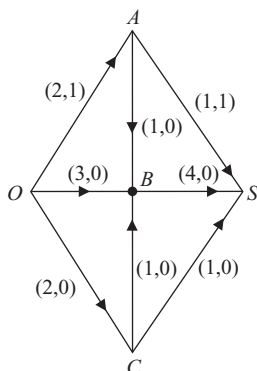


Figure 2.250(b)

Now increase the flow by 1 in the path $OCBS$ and get the network shown in Figure 2.250(e).

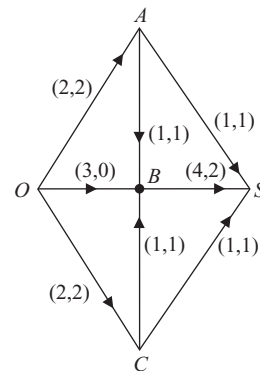


Figure 2.250(e)

Lastly, we increase the flow by 2 in the path OBS and get the network with maximum flow as shown in Figure 2.250(f). In fact the capacities in the edges leading to the sink have been completely exhausted.

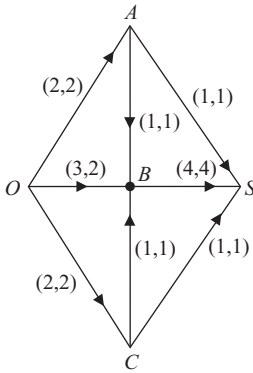


Figure 2.250(f)

We note that value ($F=2+2+2=1+4+1=6$). To check whether it is maximum flow, we appeal to max flow min cut theorem. We consider the following cuts:

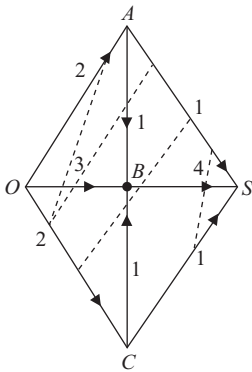


Figure 2.250(g)

- (i) $K_1 = \{OC, OB, OA\}$
with cap
(K_1) = $C_{OC} + C_{OB} + C_{OA} = 2 + 3 + 2 = 7$,
- (ii) $K_2 = \{OC, BC, BS, AS\}$
with cap
(K_2) = $C_{OC} + C_{BC} + C_{BS} + C_{AS}$
 $= 2 + 1 + 4 + 1 = 8$,

- (iii) $K_3 = \{CS, BS, AS\}$
with cap(K_3) = $C_{CS} + C_{BS} + C_{AS} = 1 + 4 + 1 = 6$,
- (iv) $K_4 = \{OC, OB, AS, AB\}$
with cap (K_4) = $C_{OC} + C_{OB} + C_{AS} + C_{AB}$
 $= 2 + 3 + 1 + 1 = 7$.

We note that capacity of the minimum cut in 6. Hence, the maximum flow is 6.

EXERCISES

1. Is there a non-empty simple graph with twice as many edges as vertices?

Ans. Yes. In fact if n = number of vertices
 $2n$ = number of edges, then using the formula for number of edges in a complete graph:

$$\frac{n(n-1)}{2}, \text{ we have } 2n = \frac{n(n-1)}{2} \text{ which}$$

yields $n = 5$. But in a complete graph K_5 the number of edges is 10. Thus the complete graph K_5 , which is simple by definition, has five vertices and ten edges.

2. Is the graph given below a bipartite graph?

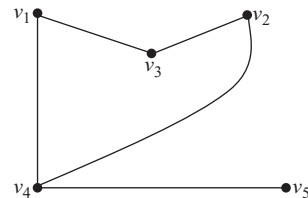


Figure 2.251

Ans. Yes, it is a bipartite graph, where the set of vertices V has been partitioned into two subsets, $V_1 = \{v_3, v_4\}$, $V_2 = \{v_1, v_2, v_5\}$. But it is not a complete bipartite graph.

3. Find a formula for the number of edges in complete bipartite graph K_{mn} .

Ans. The $m + n$ vertices having been divided into two subsets V_1 and V_2 containing m vertices and n vertices, respectively. Every vertex in V_1 is joined to all the n vertices in V_2 . Thus the task of joining each vertex of V_1 to each vertex in V_2 can be performed in mn ways. Hence the number of edges in K_{mn} is mn .

4. A graph G has vertices of degrees 1, 4, 3, 7, 3 and 2. Find the number of edges in the graph.

Ans. Total degree of G is $1 + 4 + 3 + 7 + 3 + 2 = 20$ which should be twice the number of edges. Hence number of edges in G is 10.

5. How many edges are there in an n -cube?

Ans. Degree of each vertex in an n -cube is n . The number of vertices in an n -cube is 2^n . Therefore total degree of n -cube is $n \cdot 2^n$. Hence the number of edges in an n -cube is

$$\frac{n \cdot 2^n}{2} = n \cdot 2^{n-1}.$$

6. When does the complete graph K_n possess an Euler circuit?

Ans. Complete graph is a connected graph. So it will have an Euler cycle if degree of each vertex is even.

7. Does the graph given below have an Euler circuit?

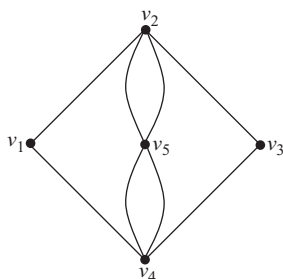


Figure 2.252

Ans. Yes. The graph is connected and degree of each vertex is even. Hence the graph possesses an Euler circuit. For example, $v_1 v_2 v_3 v_4 v_5 v_2 v_3 v_4 v_1$ is an Euler cycle.

8. When does the complete bipartite graph K_{mn} contain Euler cycle?

Ans. When both m and n are even. In such a case the degree of each vertex will be even.

9. In seven bridges problem, was it possible for a citizen of Königsberg to make a tour of the city and cross each bridge exactly twice? Give reasons.

Ans. Yes, because in such a case degree of each vertex of the graph of seven bridges problem shall be even.

10. Show that the graph given below does not contain a Hamiltonian cycle.

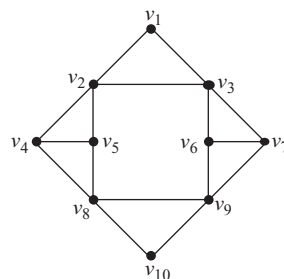
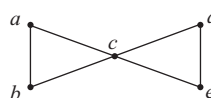


Figure 2.253

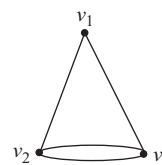
11. Give example of a graph that has Euler circuit but not Hamiltonian circuit.

Ans.



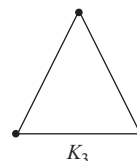
12. Give example of a graph that has an Hamiltonian circuit but not an Euler circuit.

Ans.



13. Give example of a graph that has both an Euler circuit and an Hamiltonian circuit.

Ans.



14. Use Dijkstra's shortest path algorithm to find the shortest path from a to f in the graph given below:

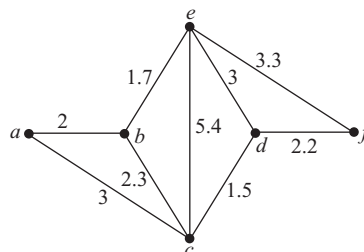


Figure 2.254

Ans. 6.7

15. Find the adjacency matrix of the graph.

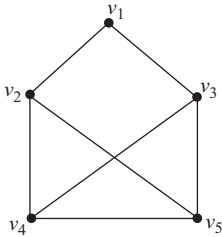


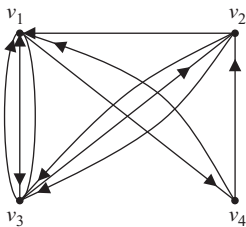
Figure 2.255

Ans.
$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

16. Find directed graph that have the following adjacency matrix.

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 2 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Ans.

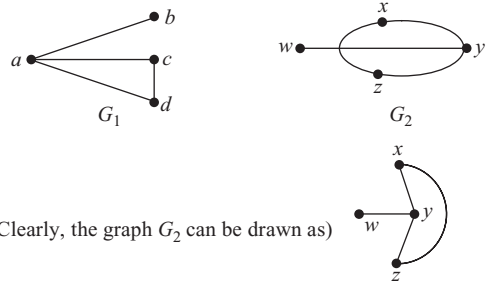


17. A graph G has the adjacency matrix given below. Verify whether it is connected.

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Ans. The graph G is connected because $B = A + A^2 + A^3 + A^4$ has non-zero entries off the main diagonal.

18. Defining an isomorphism. Show that the graphs shown below are isomorphic.



(Clearly, the graph G_2 can be drawn as)

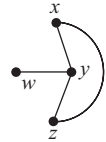


Figure 2.256

Ans. Vertex set of G_1 is $V(G_1) = \{a, b, c, d\}$
vertex set of G_2 is $V(G_2) = \{x, y, z, w\}$

Define $f: V(G_1) \rightarrow V(G_2)$ by

$$f(a) = y, f(b) = w, f(c) = x, f(d) = z.$$

Then f is bijective.

19. Show that the graphs shown below are not isomorphic. Give reasons.

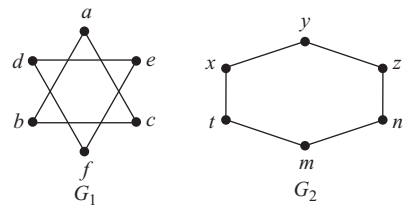


Figure 2.257

Ans. Graph G_2 is connected whereas G_1 is not connected. The connected components of G_1 are



20. Show that the graphs shown below are not isomorphic.

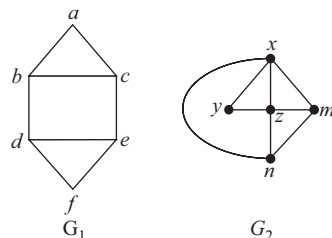


Figure 2.258

Ans. The graphs do not have same number of vertices and so are not isomorphic.

21. Find the complement of the graph shown below.

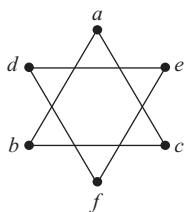
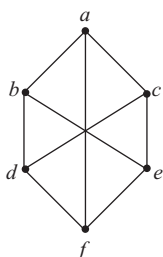


Figure 2.259

Ans. The complement of the given graph is



22. Let G be a graph and V be its vertex set. Then

(i) Eccentricity of $v \in V$ is defined as

$$e(v) = \max \{d(u, v) : u \in V, u \neq v\}.$$

(ii) The radius of the graph is defined by

$$\text{rad}(G) = \min \{e(v) : v \in V\}.$$

(iii) The diameter of G is defined by

$$\begin{aligned} \text{diam}(G) &= \max \{e(v) : v \in V\} \\ &= \max [d(u, v) : u \in V, \\ &\quad v \in V, u \neq v]. \end{aligned}$$

(iv) v is a central point if

$$e(v) = \text{rad}(G).$$

(v) The centre of G is the set of all central points.

Find the centre of the graph given below:

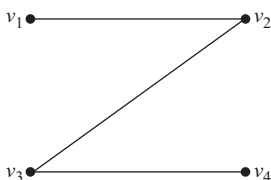


Figure 2.260

Ans. We note that

(i) $d(v_1, v_2) = 1$, $d(v_1, v_3) = 2$, $d(v_1, v_4) = 3$ and so $e(v_1) = 3$

(ii) $d(v_2, v_1) = 1$, $d(v_2, v_3) = 1$, $d(v_2, v_4) = 2$ and so $e(v_2) = 2$

(iii) $d(v_3, v_1) = 2$, $d(v_3, v_2) = 1$, $d(v_3, v_4) = 1$ and so $e(v_3) = 2$

(iv) $d(v_4, v_1) = 3$, $d(v_4, v_2) = 2$, $d(v_4, v_3) = 1$ and so $e(v_4) = 3$

Hence $\text{rad}(G) = \min \{3, 2, 2, 3\} = 2$. The centre is $\{v_2, v_3\}$.

23. Show that any graph having n vertices, $n = 1, 2, 3, 4$ is planar.

24. Show that the colouring of map given below requires at least three colours

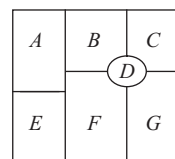


Figure 2.261

Ans. Give one colour to A and D , give second colour to B, E and G , and third colour to C and F . Thus all the adjoining boundaries have different colours.

25. How many colours are required to paint the graph given below?

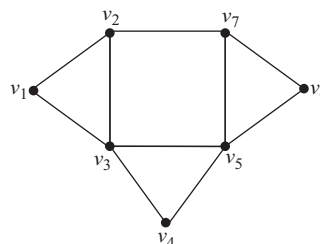


Figure 2.262

Ans. Three. We can paint v_1, v_2, v_3 with colours C_1, C_2, C_3 . Then paint v_4 by C_1 , v_5 by C_2 , v_6 by C_3 and v_7 by C_1 .

26. For which values of m and n is the complete bipartite graph K_{mn} a tree?

Ans. Number of edges in $K_{mn} = mn$. Total number of vertices in $K_{mn} = m + n$. If K_{mn} is a tree, then number of edges in K_{mn} should be $(m + n) - 1$. Hence, to be a tree K_{mn} satisfies the condition

$$mn = (m + 1) - 1 \quad \text{or} \quad m + n = mn + 1,$$

which is satisfied if either $m = 1$ or $n = 1$.

27. Let G be a graph with n vertices and $n - 2$ or fewer edges. Show that G is not connected.

Ans. Suppose G is connected. The graph itself will be a tree or it can be made a tree by eliminating some edges. In either case it must have $n - 1$ edges. But in the questions, number of edges is less than or equal to $n - 2$. Hence G cannot be connected.

28. Represent the weighted graph given below in the matrix form.

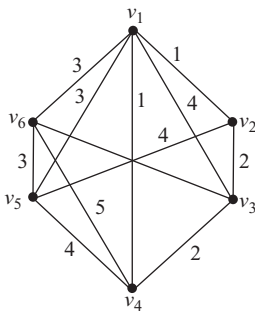


Figure 2.263

Ans.

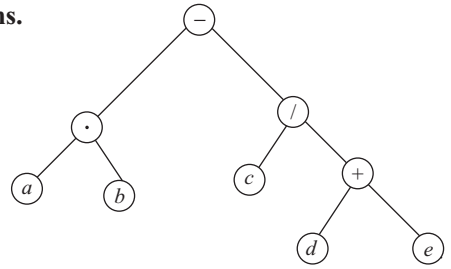
$$\begin{bmatrix} \infty & 1 & 4 & 1 & 3 & 3 \\ 1 & \infty & 2 & \infty & 4 & \infty \\ 4 & 2 & \infty & 2 & \infty & \infty \\ 1 & \infty & 2 & \infty & 4 & 5 \\ 3 & 4 & \infty & 4 & \infty & 3 \\ 3 & \infty & \infty & 5 & 3 & \infty \end{bmatrix}$$

The diagonal is $\infty \infty \dots \infty$ and the matrix is symmetric with respect to the diagonal.

29. Represent the expression given below by a binary tree and obtain the prefix form and postfix form of the expression.

$$a \cdot b - (c / (d + e)).$$

Ans.



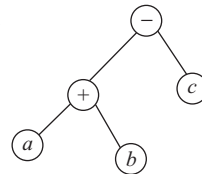
Prefix form: $- \cdot ab - c + de$

Postfix form: $ab \cdot de + c / -$

30. Convert the following postfix form of a binary tree into prefix form and parenthesized infix form and usual infix form

$$ab + c -.$$

Ans.



Prefix form: $- + abc$

Parenthesized form: $(a + b) - c$

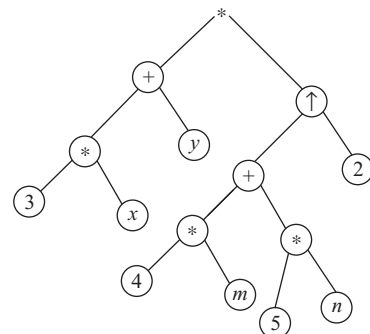
Usual infix form: $a + b - c$

31. Draw a tree for the algebraic expression

$$(3x + y)(4m + 5n)^2$$

and find its prefix polish form.

Ans. Using $\uparrow$ for exponentiation and $*$ for multiplication, the binary tree representing the algebraic expression is



The prefix form is

$$\begin{aligned} & * + * 3xy \uparrow + * 4m * 5n \\ 232 * \uparrow + 233 - 24 &= * \uparrow (2 + 3) 3 (2 - 4) \\ &= * (2 + 3)^3 (2 - 4) \\ &= (2 + 3)^3 * (2 - 4) \\ &= 5^3 * (-2) \\ &= -250 \end{aligned}$$

32. If $\uparrow$ denotes exponentiation, evaluate the polish form

$$* \uparrow + 233 - 24.$$

33. Find all cut sets for the graph G given below.

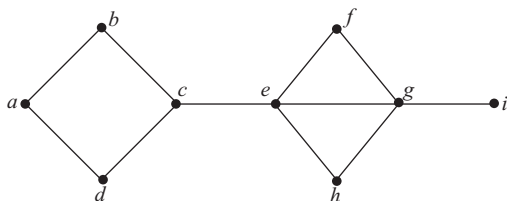


Figure 2.264

Ans. $\{c, e\}$; $\{\{a, b\}, \{a, d\}\}$; $\{\{a, d\}, \{b, c\}\}$; $\{\{e, f\}, \{e, g\}\}$; $\{h, g\}$; $\{g, i\}$; $\{\{e, h\}, \{h, g\}\}$. Try for other cut edges.

34. Find a minimal spanning tree for the graph shown below:

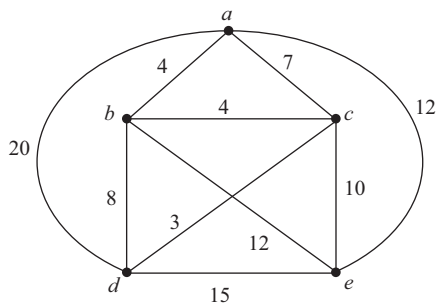
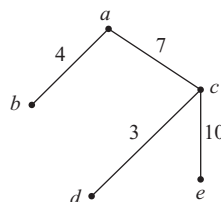


Figure 2.265

Ans.



35. Find a minimal tree for the graph of Exercise 34 by deleting one by one those costliest edges whose deletion does not disconnect the graph.

Ans. First remove $\{a, d\}$, then $\{d, c\}$, $\{a, e\}$, $\{b, e\}$. We cannot remove $\{c, e\}$ now because the graph will be disconnected by doing so. But we can delete $\{b, d\}$. Now delete any of $\{a, c\}$ or $\{b, c\}$. We will get two minimal trees.

36. Find all spanning trees of the graph shown below:

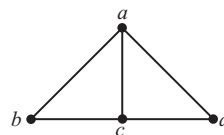


Figure 2.266

Ans. Each spanning tree must have $n-1 = 4-1 = 3$ edges. There are eight such trees.

37. Find the value of the maximum flow in the network of EXAMPLE 2.109 and verify the value obtained by max flow min cut theorem.

Ans. Value (F) for max flow = 7, minimum cut (k) is shown in Figure 8.248.

38. Find the value of the maximum flow in the network shown in Figure 2.267 and verify your answer using max flow min cut theorem.

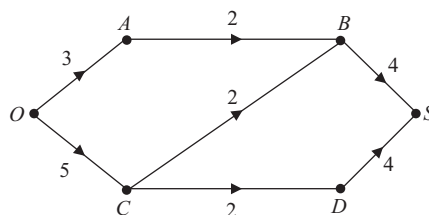


Figure 2.267

Ans. Max flow = 6